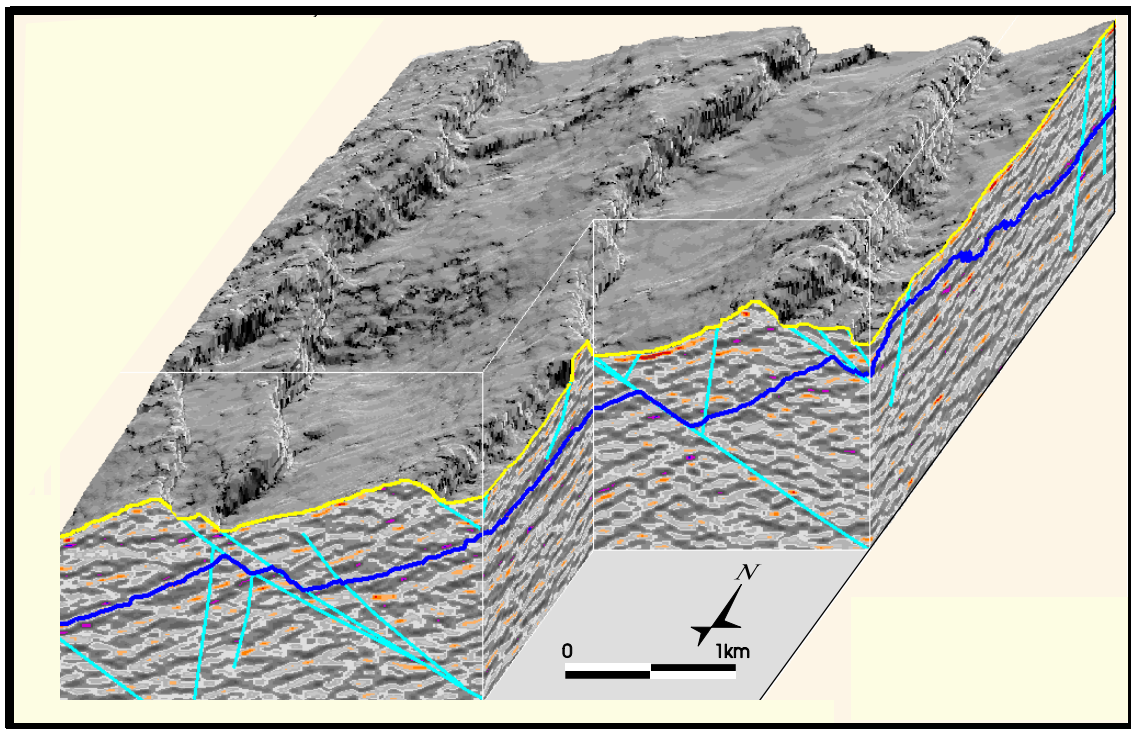


Seismic Data Acquisition and Imaging



Martin Landrø

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1 Introduction

1.1 Background and history

The purpose of this report is to give an introduction to acquisition and processing of seismic reflection data. During the past two decades, the most important step in the seismic exploration industry has been the transfer from 2-dimensional seismic data to 3-dimensional seismic data. A simultaneous growth in acquisition and processing capabilities is the major explanation for this major break through. In this work I will put the main emphasis on marine seismic acquisition. A typical marine seismic vessel is today capable of towing of the order of 10-20 cables rather than 1-2 which was typical 10 years ago. This effectiveness in acquisition combined with a tremendous increase in computer capacity has lead to the success of 3D seismic. Today, most new wells are drilled on the bases of 3D seismic data.

Another important improvement in seismic acquisition is the increased navigation accuracy offered by the new satellite position system being used on most seismic vessels today. In seismic processing it is important to know where the acoustic source is and where the seismic receivers are as accurate as possible. This accuracy is even more important for proper processing of time lapse seismic data.

1.2 Seismic wave propagation

A seismic wave consist of an organized particle motion caused by a seismic source such as for instance an air gun explosion in water, or an earthquake. For reflection seismic purposes, there are two types of seismic waves that are exploited for imaging of the subsurface. The most common one is the compressional seismic wave, often referred to as P-waves. This wave is longitudinal, which means that the local particle motion is in the same direction as the wave propagates. The velocity of a P-wave is related to the bulk modulus (K) and the shear modulus (μ) in the following way

$$v_p = \sqrt{\frac{K + \frac{4}{3}\mu}{\rho}} = \sqrt{\frac{\lambda + 2\mu}{\rho}} \quad (1.1)$$

where ρ is the density and λ is Lames first constant. Another important wave mode for reflection seismic experiments is the shear wave, often referred to as S-wave, and characterized by its particle motion being perpendicular to the wave propagation direction. For an anisotropic medium, this shear wave mode will split into two modes, having different shear wave velocities. This phenomenon is called shear wave splitting, and can be used to quantify the degree of

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anisotropy of certain layers in the subsurface. For an isotropic medium the shear wave velocity is simply given by

$$v_s = \sqrt{\frac{\mu}{\rho}} \quad (1.2)$$

For water, the shear modulus is zero, and hence shear wave propagation in water is impossible. This means that for conventional marine streamer surveys, shear waves do not play an important role, since they can not be measured directly. However, for land seismic surveys and sea bed seismic experiments, such waves can be measured directly, since the recording equipment then is placed within a solid medium, with shear wave modulus being non-zero. In addition to these main wave modes (P and S), special wave modes are generated at the interfaces between subsurface layers, these waves are called interface waves. The most well known interface wave is the Rayleigh wave occurring at a free surface, as for instance between air and ground. Ground roll is also often used as a term for such wave phenomena.

Another frequently used parameter used to describe the properties of acoustic rocks is the Poissons ratio

$$\sigma = \frac{3K - 2\mu}{6K + 2\mu} \quad (1.3)$$

Another frequently used parameter in seismic analysis is the product of seismic velocities and the mass density, called impedances. The P-wave acoustic impedance is the product of P-wave velocity and density, often referred to as acoustic impedance, and the shear wave impedance is the S-wave velocity multiplied with the density. As an illustration of the potential use of knowing both the P and S wave velocities of a medium, a crossplot between measured P and S wave velocities in a North Sea well is shown in Figure 1.1. In seismic exploration the major task is to obtain high quality images of the subsurface structures. Another important goal is to discriminate between different lithologies. A classical problem is to discriminate between sand and shale. Empirical investigations shows that shales generally have higher V_p/V_s -ratios than sands. Such empirical relations might of course vary from one area to another. We observe that

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in such a plot the different lithologies may be discriminated, if one is able to estimate both P and S wave velocities directly from seismic data.

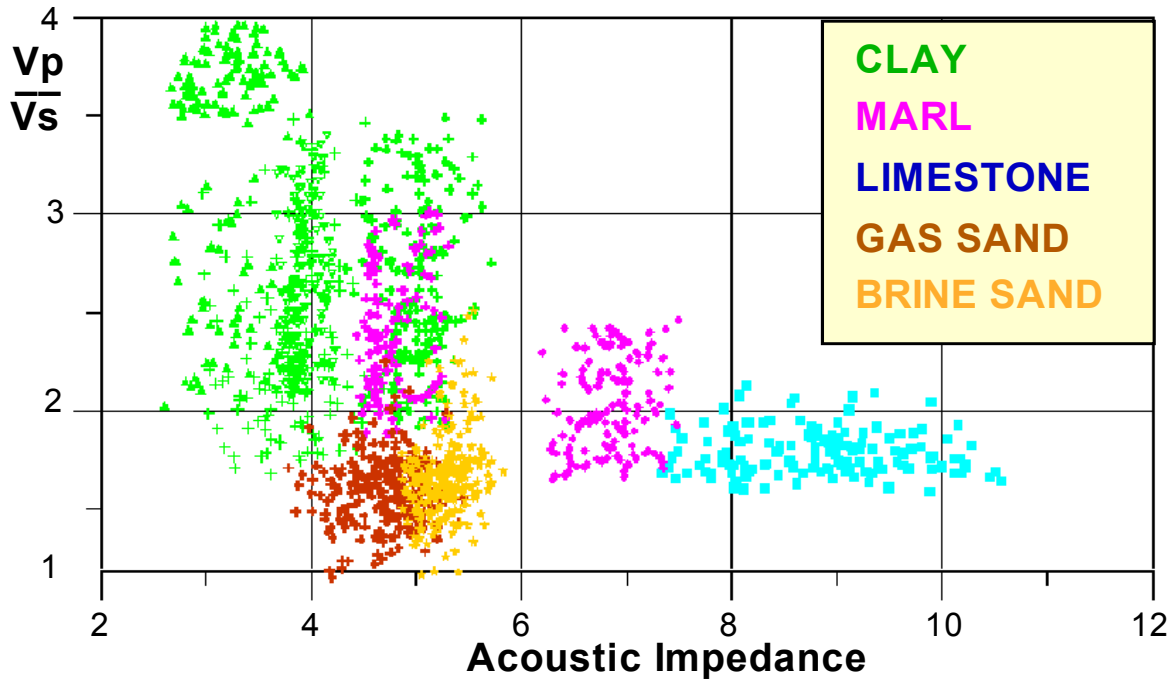


Figure 1.1 Crossplot of V_p/V_s -ratio and acoustic impedance, showing that different lithologies like for instance shale and sand have different V_p/V_s -ratios. Also observe that gas filled and brine filled sand, differs only by the acoustic impedance. (Figure is reprinted from Buland et al (1996)).

1.3 The role of seismic data acquisition and processing

The typical scenario for a normal seismic survey today is that the 3D data is acquired by a commercial contractor. The survey planning is normally being done by the oil company. The operating oil company being responsible for the acquisition hires one person that is present during the acquisition, to assure and control the quality of the acquisition. The processing of the data is also being done by a contractor (not necessarily by the same contractor as acquired the data). The processing is followed up by representatives from the oil company, in order to assure that the oil company gets the product they need for the requested seismic study. During recent years so-called onboard processing, meaning that most of the processing of the seismic data is being done onboard the vessel, and ready to be shipped to the oil company when the seismic vessel returns after the survey. This might be very time efficient, and sometimes it might also be very useful for the oil company to receive processed seismic data directly from the vessel during the acquisition phase. Another frequently used processing work flow is that the oil company invites

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the contractor to process the data inhouse, in close cooperation with the seismic interpreters. Such a solution ensures that the oil company get a final product close to what their interpreters request, and might lead to a more optimal processing sequence.

The next step in seismic exploration, the interpretation, is traditionally being done by the oil companies. An exception from this is the so-called speculative surveys, being acquired on a contractors own risk, and often being interpreted by a contractor company. Seismic interpretation is regarded as a core business for the oil companies, since it is essential for risk assessment in exploration drilling, and for reservoir description and increased reservoir knowledge for a producing oil field.

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2 Acquisition

2.1 Acquisition of land seismic data

The two most common seismic sources used in land acquisition are explosive sources (dynamite) and vibrators. On land, explosive sources are detonated in shallow holes. The source signal emitted by explosives are high frequent. One major shortcoming of explosive sources is the lack of repeatability. It also slows down the acquisition due to the drilling of these shallow holes used for detonation. In the early days of marine seismic exploration, explosive sources were used, but today marine seismic acquisition is totally dominated by the air gun being used as the generator for the acoustic signal. Land vibrators are mounted on trucks for operational efficiency, and generate a low amplitude signal of varying frequency, often referred to as a sweep. Such a sweep might last for several seconds (typically 10-30 seconds). Sensors placed at the baseplate of the vibrator are used to register the assumed output signal from the vibrator. A mechanical model for a typical seismic vibrator is shown in Figure 2.1. According to Sallas (1984) the following Newtonian equations can be set up for this mechanical model of the vibrator (annotation is explained in Figure 2.1):

$$F_a + F_t - F_g = M_b \ddot{X}_b + -D_a(\dot{X}_r - \dot{X}_b) - K_a(X_r - X_b) \quad (2.1)$$

$$-F_a = M_r \ddot{X}_r + D_a(\dot{X}_r - \dot{X}_b) + K_a(X_r - X_b) \quad (2.2)$$

Combining these two equations yields:

$$F_t - F_g = M_r \ddot{X}_r + M_b \ddot{X}_b \quad (2.3)$$

Most seismic vibrators are vibrationally isolated from the baseplate for frequencies above 3 Hz, which means that F_t can be neglected in the above equation. Therefore, a reasonable method for determining the signal being transmitted into the subsurface is to put accelerometers at the baseplate and at the reaction mass.

In the processing step, this recorded output signal is crosscorrelated with the signal recorded by the seismic receivers. This crosscorrelation technique shortens the seismic signal, and enhances the reflection image. A vibrator placed on a free surface emits both P and S wave energy into the subsurface. In fact, only approximately 10% of the energy is P-wave, 25% is S-wave, and

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the remaining 65% is surface waves or ground roll. Due to this fact, the offset to the nearest geophone group is often several hundred meters.

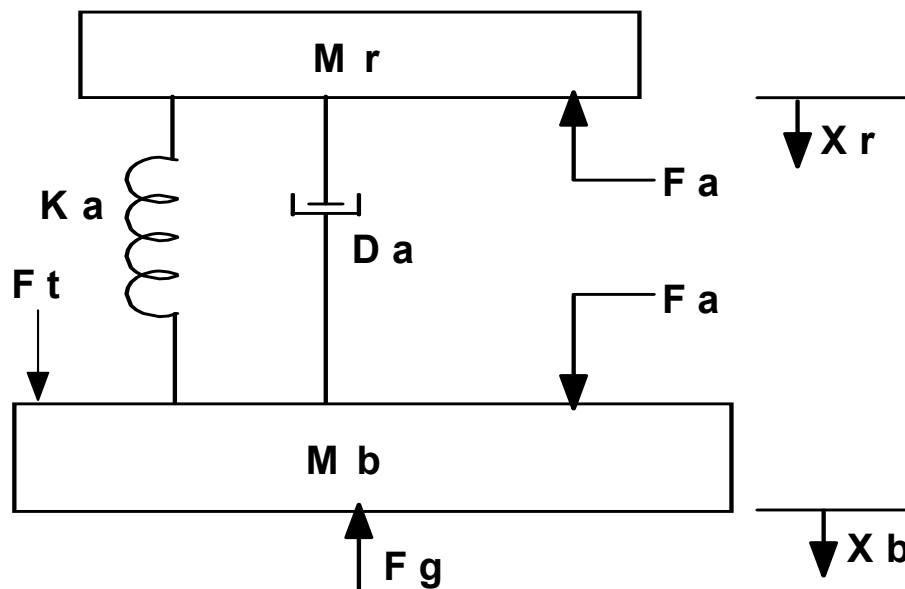


Figure 2.1: Mechanical model of a seismic vibrator (copied from Sallas, 1984).

X_r = Reaction mass displacement

X_b = Baseplate displacement

F_a = Actuator force, force from the hydraulic fluid upon the piston

F_t = Dynamic force from vehicle upon the baseplate

F_g = Force from earth upon baseplate

M_r = Reaction mass

M_b = Baseplate mass

K_a = Spring constant due to oil compressibility

D_a = Dashpot constant accounting for losses in oil column due to leakage

The seismic signal is recorded by geophones planted into the ground. The geophone convert the particle motion generated by the seismic wave into an electrical signal that is registered by a digital recording system. In order to reduce noise, geophones are often planted in groups of 10-20. A conventional land survey geometry is swath shooting, which means that the shooting line is

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perpendicular to several receiver lines. In contradiction to marine seismic surveys, each shot is rather expensive in land seismics.

One major difference between marine seismic acquisition and land acquisition is that the medium where the sources and receivers are located is varying. For a land survey the acoustic properties of the uppermost stratigraphic layer might vary considerably from one receiver location to the next. For marine acquisition however, the sea layer has practically constant acoustic properties, both laterally and vertically. The temperature of the seawater will of course vary, and hence also the sound propagation velocity, but these effects can be treated as second order effects for most marine surveys. For land surveys, however, one has to take this variation into account in the processing of the seismic data, a process referred to as static correction. Also for seabed surveys, one must expect velocity variations, especially for shearwaves, in the uppermost stratigraphic layers (close to the sea bottom).

2.2 Marine seismic data acquisition

The major seismic source used in marine surveys is the air gun. Figure 2.2 shows how the air bubble generated by an air gun expands within the time frame of milliseconds after the gun has been fired. A photo and a schematic view of an air gun are shown in Figure 2.3.

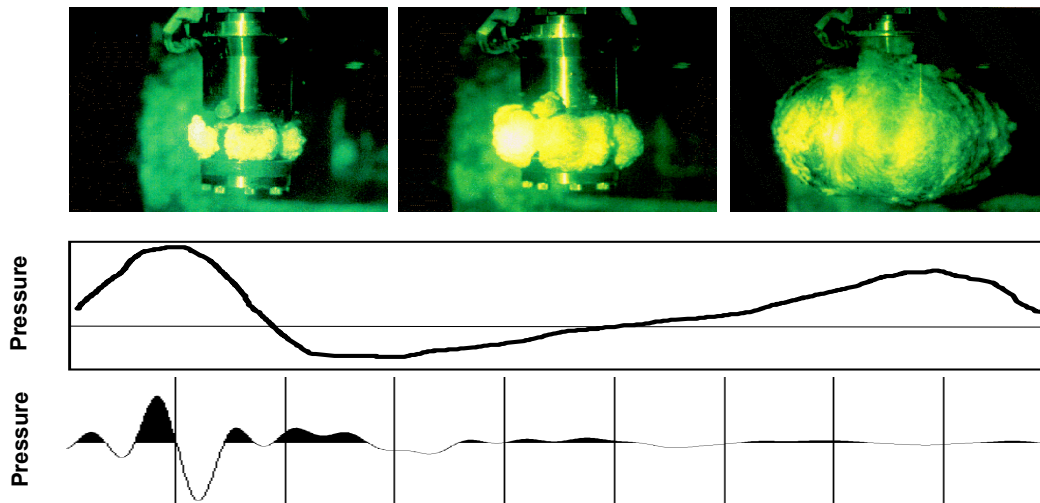


Figure 2.2: The bubble generated by a seismic air gun at different stages after the gun has fired, and corresponding recording of the acoustic pressure signal recorded close to the gun (middle) and a typical far-field signature (bottom). Notice that the time scale is different on the two signature plots.

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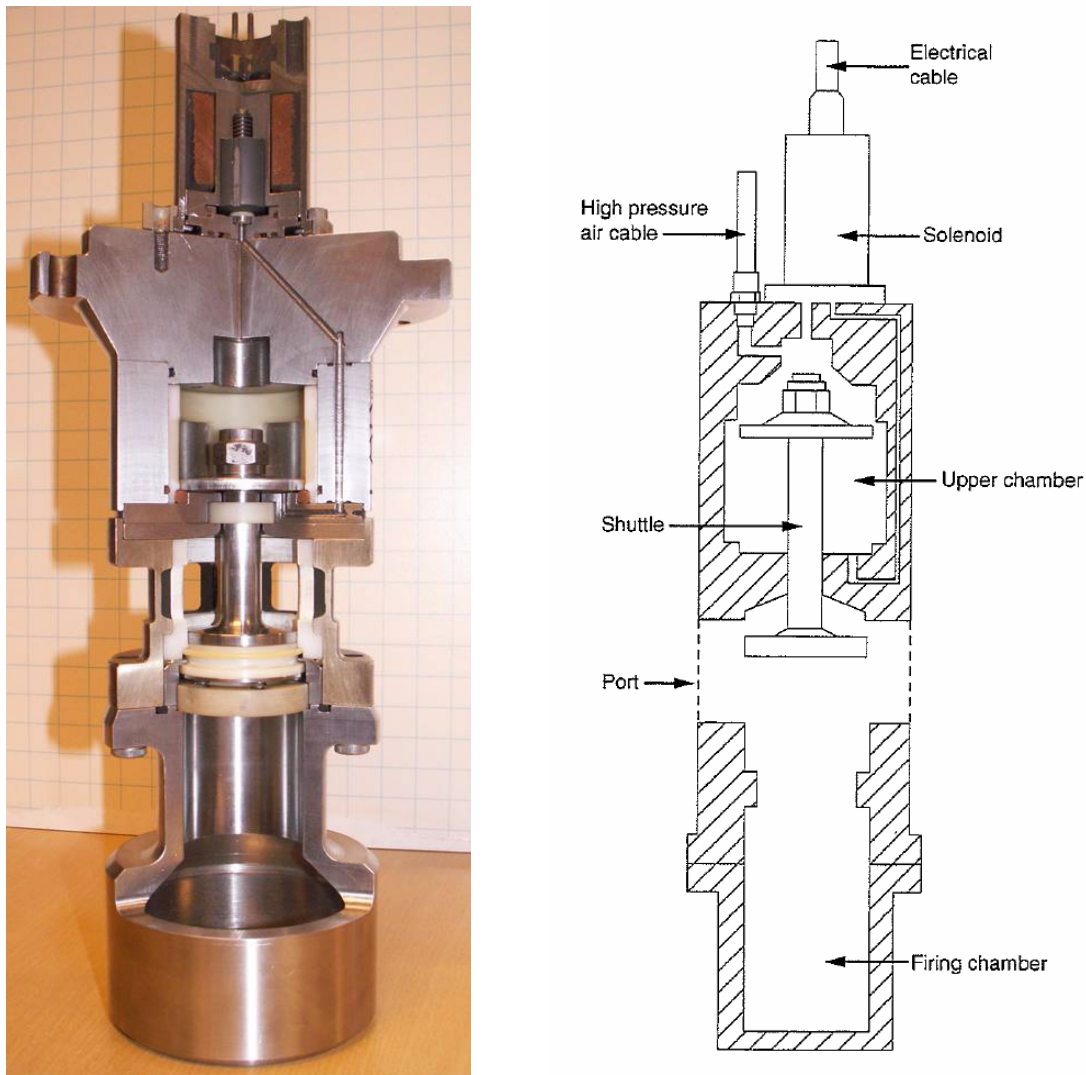


Figure 2.3: Photo (left) of a BOLT airgun and a schematic cross-section of an airgun. Notice that the area of the upper shuttle disk is somewhat larger than the lower shuttle disk, ensuring that the shuttle moves upwards when the solenoid valve is activated. The firing chamber can be replaced with various sizes.

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Rayleigh (1917) derived an equation for a cavity filled with water vapour or gas. Assuming that the cavity or bubble is perfectly spherical, the kinetic energy of a spherical shell of water with thickness dr and mass m is

$$\frac{1}{2}mu^2 = \frac{1}{2}\rho 4\pi r^2 dr u^2 \quad (2.4)$$

where u is the velocity at the distance r from the cavity centre and ρ is the density of water. Assuming that the water is incompressible, the bubble wall velocity U and the bubble radius R must be related to the particle velocity in the water (u) and the distance to an arbitrary point in the water r in the following way:

$$UR^2 = ur^2 \quad (2.5)$$

The total kinetic energy of the water motion is now given by integrating the local kinetic energy from the bubble wall (R) to infinity:

$$E_k = \int_R^\infty \frac{1}{2}mu^2 = 2\rho\pi U^2 R^4 \int_R^\infty \frac{dr}{r^2} = 2\rho\pi U^2 R^3 \quad (2.6)$$

Assuming that the pressure inside the bubble (or cavity) is uniform (varying as a function of time, but not in space) and equal to p , the work done by the expanding bubble is given by

$$W = (p_0 - p_\infty) \frac{4}{3}\pi R_0^3 - (p - p_\infty) \frac{4}{3}\pi R^3 \quad (2.7)$$

where p_∞ is the hydrostatic pressure (water pressure at infinity), and p_0 and R_0 are initial bubble pressure and bubble radius, respectively. The total energy of the system is now equal to $E_{TOT} = E_k + W$. Assuming that the total energy is conserved, differentiation with respect to time of the energy equation leads to the well known Rayleigh equation for the bubble motion:

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{p - p_\infty}{\rho} \quad (2.8)$$

where dot means differentiation with respect to time. We have also neglected terms involving time derivative of the bubble pressure ($\dot{p} \approx 0$). In order to solve the above given equation one needs an equation of state for the gas inside the bubble (for instance the ideal gas law), and an assumption for how the gas bubble expands (for instance adiabatic conditions etc.). A typical solution of this equation is shown in Figure 2.4. Notice that the bubble wall velocity is zero for maximum bubble radius, and that the velocity changes rapidly around the minimum bubble radius. Furthermore, we notice that there is no damping, since the maximum bubble radius remains constant for multiple bubble oscillations. A more realistic equation for the motion of an air bubble created by an air gun, was derived by Kirkwood and Bethe (1942):

$$R\ddot{R}\left(1 - \frac{\dot{R}}{C}\right) + \frac{3}{2}\dot{R}^2\left(1 - \frac{\dot{R}}{3C}\right) = H\left(1 + \frac{\dot{R}}{C}\right) + R\frac{\dot{H}}{C}\left(1 - \frac{\dot{R}}{C}\right) \quad (2.9)$$

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where H denotes the enthalpy at the bubble wall, $H = \int \frac{dp}{\rho}$. The enthalpy at the bubble wall can

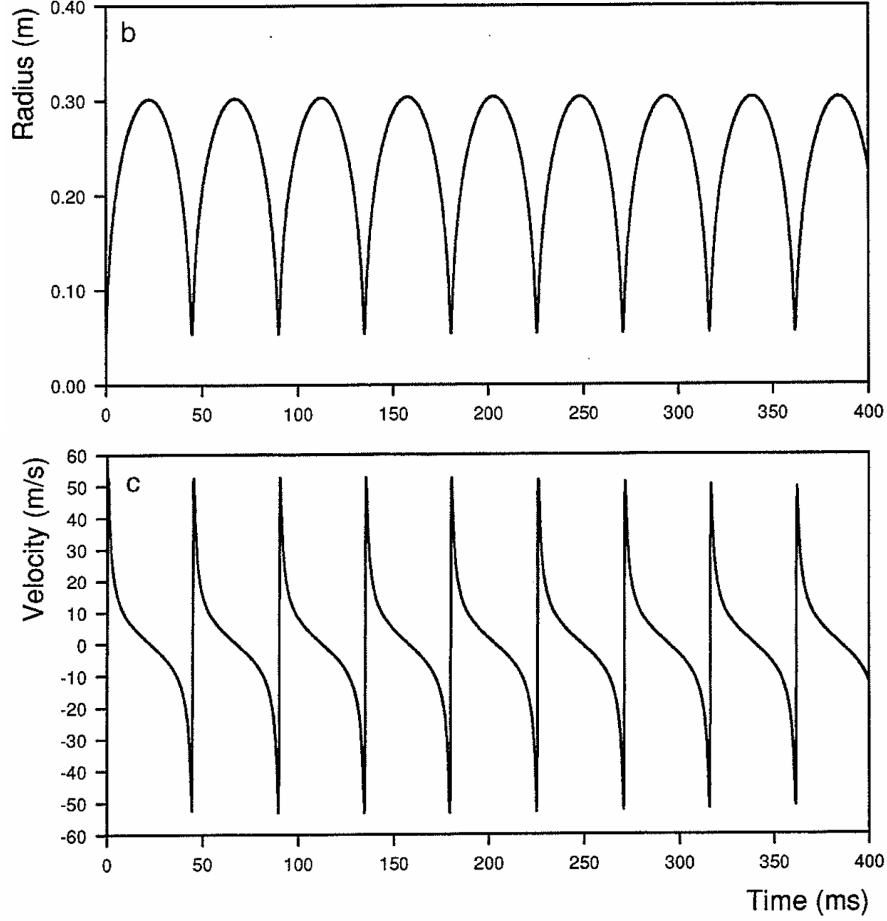


Figure 2.4: Bubble radius and bubble wall velocity calculated using the Rayleigh equation (2.8). Initial volume: 40 cubic inches. Firing pressure: 140 bar. Water depth: 7.5 m.

for instance be evaluated from an equation of state for the water, the most commonly used equation is the Tait equation. Assuming that we solve either the Rayleigh equation or the Kirkwood-Bethe equation, Gilmore derived an equation for the acoustic pressure at any distance from the bubble centre:

$$p - p_{\infty} = \rho_{\infty} \frac{R}{r} \left(H + \frac{\dot{R}^2}{2} \right) \quad (2.10)$$

where higher order terms in $\frac{1}{r}$ have been neglected. This equation can then be used to predict the pressure at any distance from a bubble generated by an air gun.

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2.3 Bubble time period and primary to bubble ratio

Two terms used to characterize source signatures are the bubble time period, and the primary to bubble ratio. The bubble time period is measured as the time interval between the primary peak and the secondary peak in the signature, as shown in Figure 2.5. Notice that the primary to bubble ratio is defined as the total amplitude of the primary signal (max positive minus max negative) divided by the total amplitude of the bubble signal. The primary to bubble ratio is strongly dependent on frequency content. This means that when comparing signatures from different arrays, one always have to make sure that the frequency content of the two signatures are comparable.

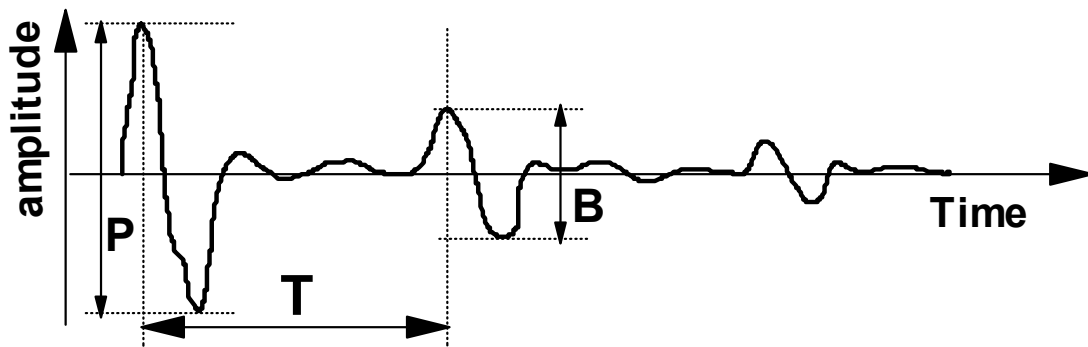


Figure 2.5: Typical far-field source signature showing how the primary (P) signal amplitude and the bubble (B) signal are defined, together with the bubble period (T).

2.4 Methods for damping of bubble oscillations of an air gun signature

As can be seen from Figure 2.2 (upper signature) the pressure signal associated with the collapse of the bubble generated by an air gun is nearly as strong as the primary peak. The ideal seismic signal is a sharp, highfrequency peak only, and the second, third bubble oscillation signal is undesired. Several methods have been developed in order to attenuate this unwanted tail of the seismic signal generated by air guns. These are:

1) **Combination of air guns with different gun chamber volumes.** Air guns with different volumes will have different bubble time periods, and hence the secondary bubble pulse will occur at different times, and in contradiction to the primary signals (which will add up constructively), the secondary bubble oscillations will cancel each other. The resulting source signature for the whole air gun array will thus be a strong primary signal, and a strongly attenuated bubble oscillation signal. This can be seen on the bottom signature in Figure 2.2.

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2) **Clustering of air guns.** If two or more air guns are placed extremely close to each other (of the order of 0.5 m), interaction effects between the guns will be severe. Interaction will give some reduction in the primary amplitude, however, the bubble amplitude will be even more attenuated. Therefore, it is quite common in modern air gun arrays to put one or two clusters of guns in front of each sub-string in the total gun array. In this way, the primary to bubble ratio is improved, compared to conventional arrays.

3) **Wave shape kit.** By placing a steel plate with a small hole in the middle, the release of air from the gun chamber into the bubble is delayed. This delay causes attenuation of the primary signal, but the bubble is attenuated even more. Wave shape kits were commonly used in the 1980's, but is not being used today.

2.5 Source signature estimation

In experimental physics disciplines like particle physics, optics etc. it is essential to have a detailed knowledge of the signal being used to investigate or probe a target. For the seismic experiment the interest for an exact determination of the input signal has not been addressed to the same extent. There are two major reasons for this lack of interest for accurate determination of the seismic input signal:

- The input signal is strongly altered both with respect to amplitude and frequency on its way through the earth down to the target level.
- It has not been sufficiently demonstrated that it is necessary to determine a directional source signature on a shot by shot basis.

In a deterministic perspective, we would like to understand our seismic data in the following way: First determine the source signal before it enters into the subsurface, then add absorption and scattering effects due to changes in the medium parameters as the wavefield propagates through the earth, and in this way obtain a best possible fit with measured seismic data. Three recent developments that might cause an increased interest for source signature determination are:

- Time lapse seismic: In order to exclude source variations between time lapse seismic surveys an exact knowledge of the signatures is required.
- 4-C seimics: For seabed seismic data a directional source signature might be useful in order to determine angle ranges for maximum conversion from P to S energy at the sea bed.
- Multiple attenuation: The new generation of surface related multiple attenuation

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schemes, often requires that the source wavelet in the water layer is known.

The next chapter will therefore shortly describe two different methods for determination of the source signature as being measured at a far-field location in the water layer.

2.5.1 The notional-source method

The notional-source method was proposed and demonstrated by Ziolkowski et al. in 1982 and was refined by Parkes et al. in 1984. It is based upon the assumption that the pressure wavefield of an array of airguns can be expressed as the superposition of a set of monopole wavefields located at the position of the individual oscillating bubbles. This means that the total wavefield can be determined from measurements of pressure made by near-field hydrophones placed approximately 1 m from each air gun. The configuration of near-field hydrophones relative to

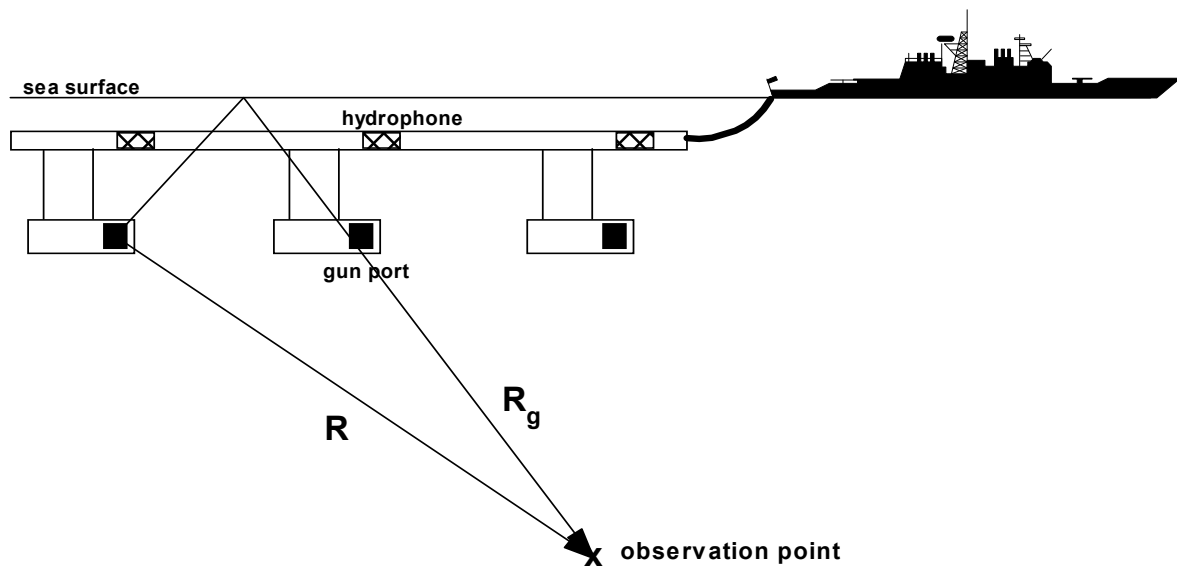


Figure 2.6: Schematic view of the experimental setup used for the notional-source method. A near-field hydrophone is placed close to each gun (70 cm is a typical choice), and the far-field signature is extrapolated on the basis of the near-field hydrophone recordings.

the air guns in an array is sketched in Figure 2.6. Each source element is described by a single monopole time-varying function referred to as the notional-source signature. The effect of interaction between the air guns (the fact that the radiated pressure from one air gun impacts the behaviour of the neighbouring air guns) is included in the notional source signature, which means that the total wavefield of the air gun array can be found as a direct superposition of the notional source signatures:

$$p(R, t) = \sum_j \frac{S_j\left(t - \frac{R_j}{c}\right)}{R_j} + \beta \frac{S_j\left(t - \frac{R_j^g}{c}\right)}{R_j^g} \quad (2.11)$$

where R_j is the distance from bubble j to the far-field point (or observation point) and R_j^g is the corresponding distance for the ghost reflection (surface reflected wavefield), and c is the sound velocity in water. Notice that the bubble position will vary as a function of time, since the bubble rises. Also the hydrophones will move with respect to the bubble centre, with a horizontal velocity equal to the vessel speed. The reflection coefficient for the water-air interface is denoted β in this equation. Parkes et al (1984) derived a recursive solution for the notional-source signatures, which means that the notional-source signature at one time sample can be written as a function of the near-field measurements and delayed notional-source signatures. Once the notional-source signatures are estimated, equation 2.11 is used to compute the far-field signature.

In a recent field test the average residual error energy between estimated and measured far-field signatures were found to be 3.1%. This test included various array configurations (with and without clustered air guns) and various far-field angles, totally 20 comparisons.

2.5.2 The ministreamer inversion method

To use data recorded on a separate signature ministreamer instead of near-field hydrophones and invert for the notional-source signatures was proposed by Landrø and Sollie in 1992. Hargreaves presented a wave-field extrapolation method limited to 2-D also using ministreamer data in 1984. The advantage with using an inversion algorithm to estimate notional-source signatures is that it can be used in 3-D and that it is not limited by finite aperture effects. A schematic view of this method is shown in Figure 2.7.

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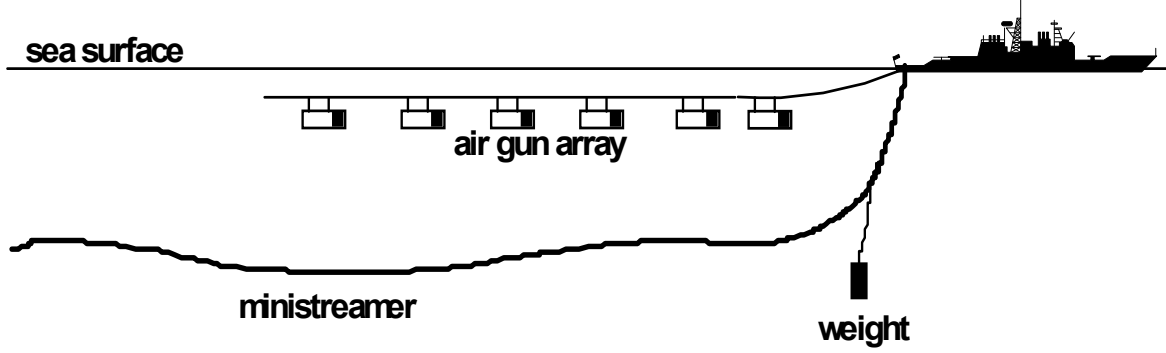


Figure 2.7: Acquisition geometry used for source signature estimation from a separate signature streamer (ministreamer) placed 10-20 m below the air gun array. The weight is used to adjust the streamer to the desired depth.

The main motivation for using mid-field hydrophone (the signature streamer is placed 10-20 m away from the air guns) measurements instead of near-field measurements was to avoid the non-linear region close to the guns. Especially for compact air gun arrays, it is expected that the principle of superposing the pressure wave field from each gun will not be correct, due to non-linear effects taking place in the near-field of the air guns. Another motivation was to reduce the inaccuracies coupled to the bubble hydrophone distances. If the distances between the air gun bubbles and the hydrophones are of the order of 10-20 m rather than 1 m, an uncertainty in the positioning of 10 cm is of less importance for a mid-field hydrophone recording.

The ministeamer inversion method is based upon physical modeling of the bubble created by each air gun in the array. A modified version of equation 2.9 is used to model the motion of the bubble wall

$$R\ddot{R}\left(1 - \frac{\dot{R}}{C}\right) + \frac{3}{2}\dot{R}^2\left(1 - \frac{\dot{R}}{3C}\right) = H\left(1 + \frac{\dot{R}}{C}\right) + R\frac{\dot{H}}{C}\left(1 - \frac{\dot{R}}{C}\right) - \alpha\dot{R} + \beta\dot{R}^2 \quad (2.12)$$

where α and β are inversion parameters influencing the damping and bubble time period, respectively. It turns out that the introduction of these two new parameters in equation 2.12 gives enough flexibility to fit the recorded ministreamer data to an accuracy of 1-2% measured in residual error energy. When a good fit between modeled and measured ministreamer data is achieved, the estimated notional-source signatures are used to extrapolate a far-field signature that can be compared with measured far-field signatures. In a recent field test the average residual error energy between estimated and measured far-field signatures were found to be 2.8%. This test included various array configurations (with and without clustered air guns) and various far-field angles, totally 20 comparisons.

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2.6 The source ghost spectrum

Consider an air gun deployed at a depth z_g . Let $p(t)$ denote the notional source signal of the air gun. Assume that we measure the far-field signal of the gun at a distance R vertically below the air gun, as shown in Figure 2.8. According to equation 2.11 the far-field signal is now given by (assuming that the sea surface reflection coefficient is equal to -1):

$$s(t) = \frac{1}{R}p\left(t - \frac{R}{c}\right) - \frac{1}{R_g}p\left(t - \frac{R_g}{c}\right) \quad (2.13)$$

where c is the sound velocity in water, and $R_g = R + 2z_g$. Fourier transformation of equation 2.13 and assuming that $R_g \approx R$ in the denominator of the last term (but not in the exponent!) yields:

$$S(\omega) = \frac{1}{R}P(\omega)e^{-i\omega R/c} \left(1 - e^{-i2\omega \frac{z_g}{c}}\right) \quad (2.14)$$

Letting the source ghost spectrum be the last part of this equation (within the parentheses) means that the source ghost spectrum ($H(\omega)$) is

$$H(\omega) = 1 - e^{-i2\omega \frac{z_g}{c}} \quad (2.15)$$

The norm of this ghost spectrum is given as ($\omega = 2\pi f$)

$$|H(f)| = \left|2 \sin\left(\frac{2\pi f z_g}{c}\right)\right| \quad (2.16)$$

A sketch of this function is shown in Figure 2.9. We see that the effect of the free surface is to attenuate low frequencies and high frequencies around the notch frequency $c/(2z_g)$. If the source depth is increased, we see that this notch frequency decreases, which means more attenuation of high frequencies. Therefore, it is often recommended to deploy the source as shallow as possible, a commonly used source depth is 5 to 6 m.

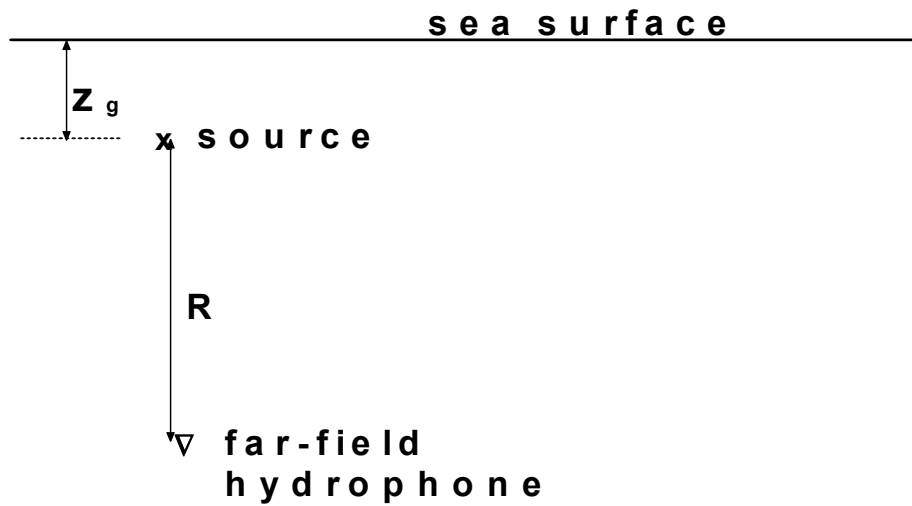


Figure 2.8: Geometry of a single air gun fired at a depth z_g and a far-field hydrophone.

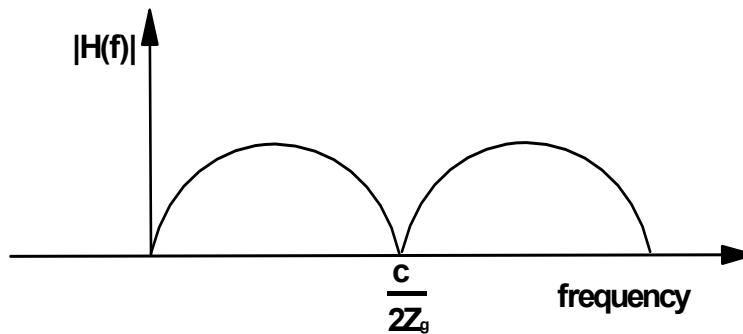


Figure 2.9: Amplitude spectrum of the source ghost response, assuming that the sea surface reflection coefficient is equal to -1 .

Figure 2.10 (top) shows a far-field signature of an air gun array deployed at 7.5 m together with the corresponding amplitude spectrum. From Figure 2.9 we see that the notch frequency for this case should be equal to (assuming a sound velocity of water of 1500m/s):

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$$f_{Notch} = \frac{c}{2z_g} = \frac{1500m/s}{15m} = 100Hz \quad (2.17)$$

which is exactly what we observe in Figure 2.10. If the far-field measurements is made an angle off the vertical, the traveltime distance between the direct arrivals and the ghost arrivals (surface reflected signals) decreases. This decreased traveltime separation corresponds to a shallower gun depth, and this is confirmed by the amplitude spectrum of the 30 degree far-field signature shown at the bottom of Figure 2.10. We see that the notch frequency has been shifted from 100 Hz for the vertical case to approximately 120 Hz for the 30 degree case.

2.7 Fighting the ghost

There has been many attempts to reduce or remove the effect of the source and receiver ghost on seismic data, both as a processing step (by applying filters that compensate for the ghost effect) and during acquisition. Fred Barr and Joe Sanders at Halliburton Geophysical Services developed an ocean bottom cable equipped with both a hydrophone sensor (measuring pressure) and a vertical geophone sensor (measuring particle velocity) in the 1980s. This ocean bottom dual sensor cable was tested in the Gulf of Mexico. The basic idea is that if the two sensors are calibrated

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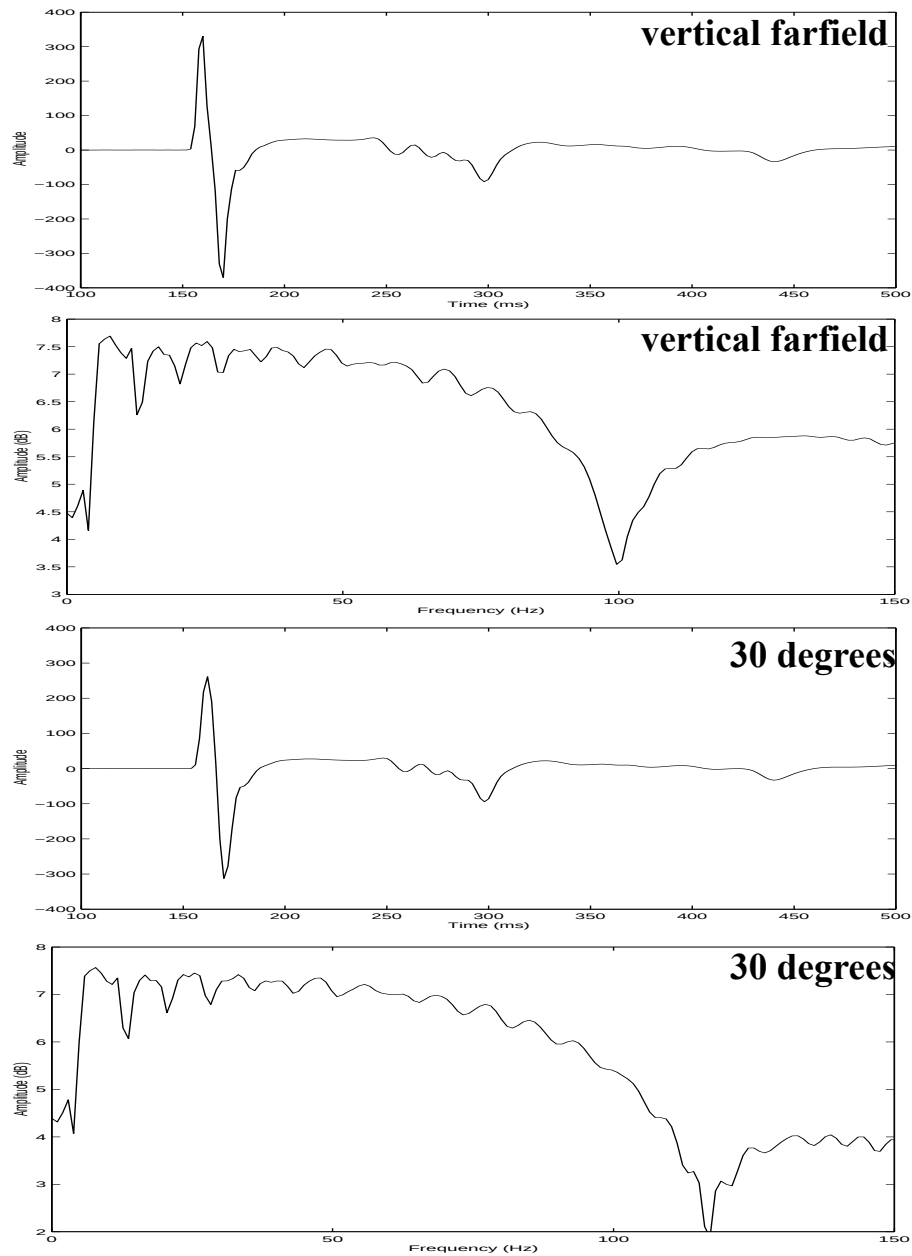


Figure 2.10: Vertical far-field signature of an airgun array deployed at 7.5 m and corresponding amplitude spectrum (top), and far-field signature measured at an angle of 30 degrees from the vertical axis and corresponding amplitude spectrum (bottom).

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relative to each other, the receiver ghost from the hydrophone and the geophone will exactly cancel each other, as shown in Figure 2.11.

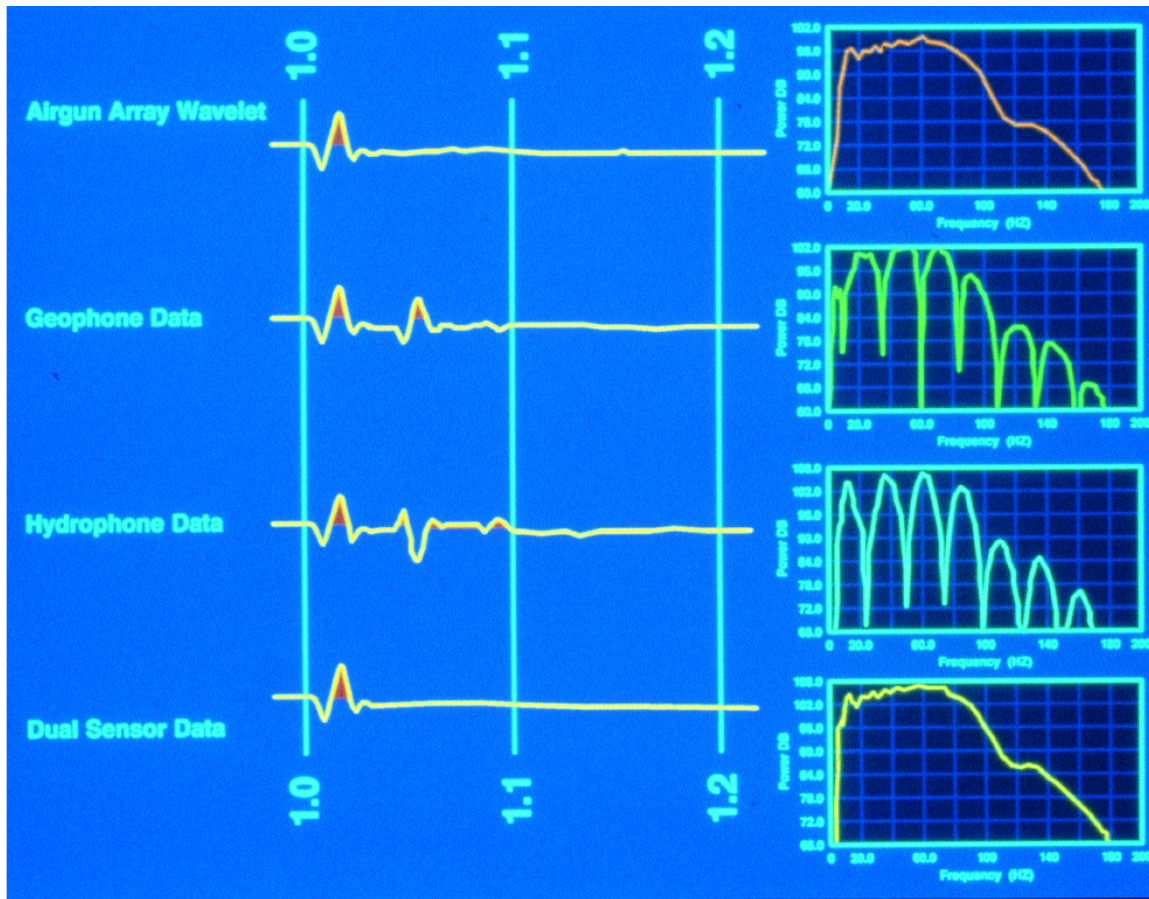


Figure 2.11: Principle of the dual-sensor OBC method: The airgun array wavelet (top) has no ghost reflections, while the geophone and hydrophone data both have a ghost reflection (opposite sign, however). When the two signals are added together, the dual sensor data has no ghost effects, and the spectrum is similar to the airgun array wavelet (bottom figure). (From Fred Barr, TLE 1997).

The main purpose of the dual-sensor technology was to attenuate water-column reverberations, and several examples showed that this method works. Several attempts were made to construct a marine streamer equipped with a vertical geophone to exploit the same technique as Barr and Saunders developed for the dual-sensor OBC cable. None of these attempts were successful until PGS launched their new streamer in 2007, and presented this at the EAGE meeting in London. Their field example shows encouraging results, and this year (2007) a 2D vessel will be

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equipped with the new streamer and tested at various locations. PGS plans to equip 3D vessels with the new streamer technology in 2008. The major challenge for geophone recordings in a marine streamer is to attenuate the noise caused by water motion close to the streamer, and especially to achieve high quality recordings of the vertical displacement field is difficult. Therefore, one should expect that these ghost-attenuating technologies work best for shallow seismic data, the deeper we want to image, the weaker the seismic signal will be, and hence the more difficult it will be to obtain a clean vertical geophone signal. The hydrophone signal is less influenced by local water movements close to the streamer, since it measures pressure variations and not particle velocity.

2.8 The receiver system

The main physical principle of a hydrophone is that a mechanical deformation caused by the acoustic pressure wave converts into an electric field within the hydrophone transducer. The transducer itself consist of a piezoelectric material that is sensitive to changes in the mechanical stress. Since the output from a hydrophone is measured in Volts the sensitivity for a hydrophone is often given in microVolts per Pascal. The sensitivity will vary with temperature. Since the transducer element is often shaped as a ring or as a plate, hydrophones might be directional, but only for very high frequencies (above 10 kHz). One basic assumption is therefore that the size of the hydrophone should be much smaller than the wavelength of the recorded signal, which means that the whole transducer is exposed to the same pressure. The hydrophone cable consists of hydrophone groups containing several single hydrophones. In order to reduce noise the pressure signal recorded by each hydrophone within a group is added together. The effect of this group summation can be expressed as a filter in the frequency-wavenumber domain. The array response of a group with N hydrophones and a distance Δx between the elements is simply given by the sum

$$H(k) = \sum_{j=0}^{N-1} e^{-2\pi i k j \Delta x} = \frac{1 - e^{-2\pi i k N \Delta x}}{1 - e^{-2\pi i k \Delta x}} \quad (2.18)$$

Rearranging this expression we obtain

$$H(k) = \frac{\sin \pi k N \Delta x}{\sin \pi k \Delta x} e^{-\pi i k (N-1) \Delta x} \quad (2.19)$$

A plot of the amplitude part (disregarding the exponential phase factor) of this function is shown in Figure 2.12, for $\Delta x = 0.5\text{m}$ and $N = 19$. We observe that the effect of group summation is a spatial low pass filter. Spatial frequencies above $1/(N\Delta x) = 0.11\text{m}^{-1}$ are attenuated. The function is periodic with period equal to two times the Nyquist frequency, that is $1/\Delta x$. An increase of the group length $N\Delta x$ thus leads to a decrease in the spatial frequency.

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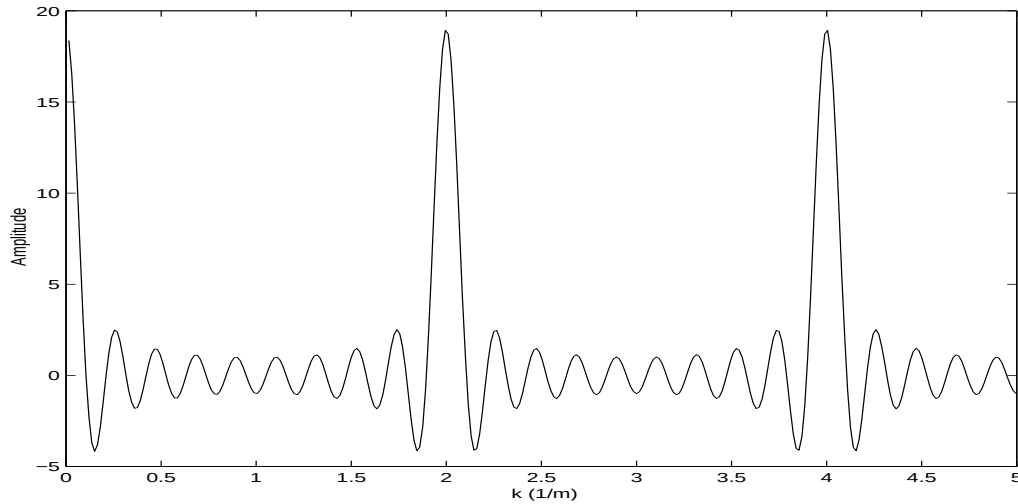


Figure 2.12: The array response function of a group consisting of 19 elements and with a group interval of 0.5m.

Since the seismic experiment is recording a transient signal, a wide dynamic range for the recording instrument is necessary. Today's recording instruments have sufficient dynamic range to cover signals between the noise level (1 microbar or even less) and a maximal signal above 1 bar. In order to avoid temporal aliasing an anti-alias acquisition filter is used. Typical high cut filter values are from 125 Hz to 250 Hz. In order to reduce swell noise and other low frequent noise a low cut filter is also often used in the acquisition, typically 3.5 or 5 Hz. This low cut filters often have a significant impact of the phase properties of the recorded signal.

In the last ten years the greatest improvement in seismic acquisition has probably been on the navigation side. The positioning of the source-receiver system is of great importance in 3D multi-streamer surveys, and this accuracy has definitely improved during the last decade. It is a challenge both in conventional 3D seismic, and especially for 4D seismic to exploit this increased accuracy also in the processing. Typical spec for a Teledyne T-2 hydrophone is ± 1.5 dB in sensitivity. After the group summation one expect that the typical variation in sensitivity is of the order of 2-4%. A temperature change of for instance 4 degrees C will lead to a corresponding change in sensitivity of approximately 1%.

Temporal aliasing is not a problem in today's acquisition, since 2ms sample rate is commonly used for conventional 3D seismic. Spatial aliasing is however more critical, and is governed by the distance between each receiver group, and the dip angle of the steepest reflector that we want to image. In order to avoid spatial aliasing the receiver group spacing should be chosen

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$$\Delta r < \frac{v}{2f\sin\theta} = \frac{v}{4f\sin\theta_{app}} \quad (2.20)$$

where v is the velocity, f the maximum frequency of interest, θ is the maximum dip angle and θ_{app} is the maximum apparent dip as seen on a stack section. This means for example that if one should image a reflector with a dip of 30 degrees and the maximum frequency of interest is 75 Hz, and assuming a velocity of 2000 m/s one should use a receiver group interval of 27 meters or smaller in order to avoid spatial aliasing. This is of course a simplified “rule of thumb” equation, for a complex structure with curved dipping interfaces, one should use seismic modeling techniques to obtain a good understanding of which reflectors that it is likely to image properly. When the data is sorted into CDP gathers, the shot spacing also enters into this equation and then the group spacing should be interpreted as the distance between two adjacent traces within a CDP gather.

To adjust the depth of a marine seismic streamer, so-called birds are attached to the streamer at some selected positions along the streamer, for instance every 250 m. By operating these birds, the actual depth of the streamer can be adjusted to the prespecified level. Since these birds create some additional flow patterns due to their size, it is expected that some noise might occur close to the birds, as shown in Figure 2.13.

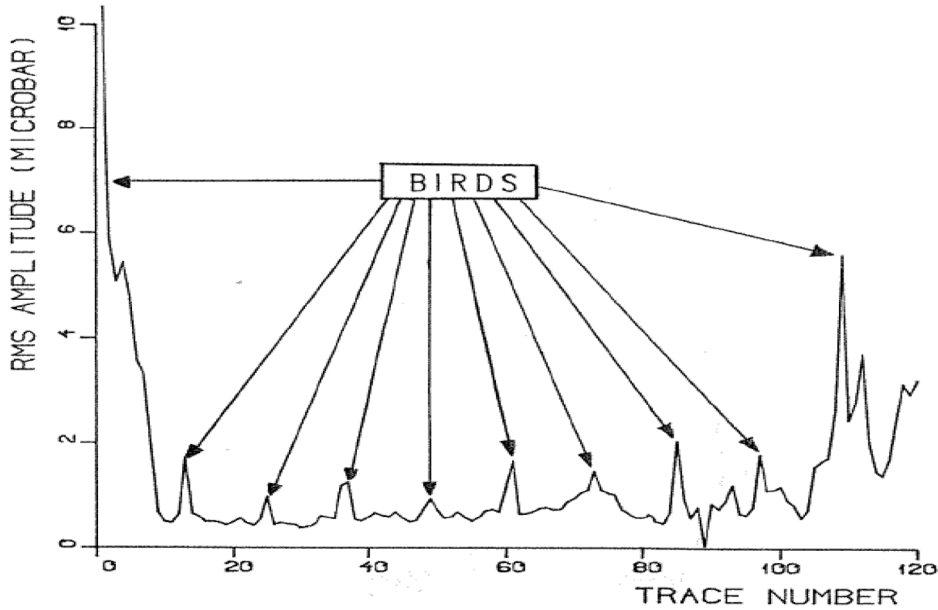


Figure 2.13: Recorded RMS (Root Mean Square) amplitude of the ambient noise for a 3000 m streamer. Notice the increased noise level close to the birds, and that the ambient noise level is higher for both ends of the steamer.

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The amplitude spectrum of the ambient noise recorded at 2600 m offset of a conventional marine seismic streamer is shown in Figure 2.14. The comparison for the full source strength

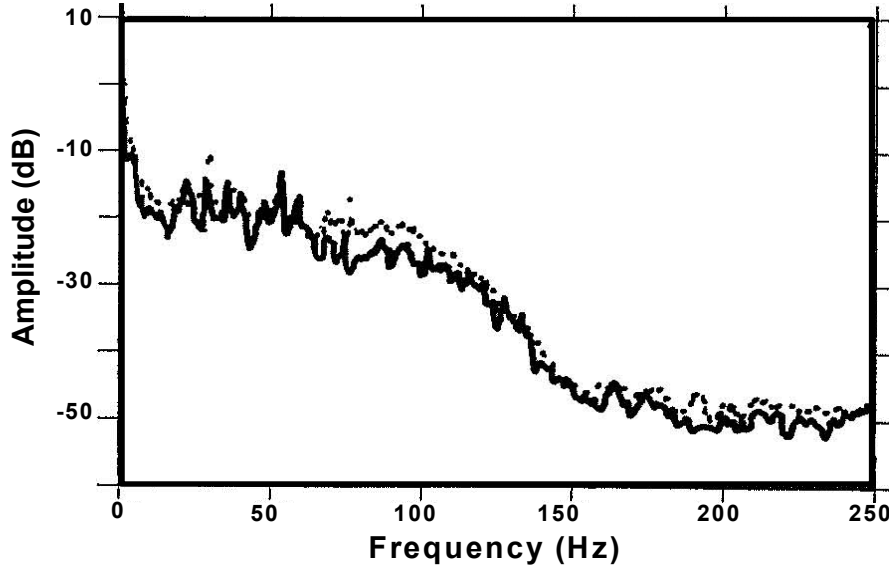


Figure 2.14: Amplitude spectra of the ambient noise recorded at 2600 m of a conventional marine streamer for a source array of 4832 cu. in. (dotted line) and a source array of 2416 cu. in. (solid line).

and the half source strength show that the major amplitude differences occur for frequencies between 50 and 100 Hz. We also observe that the amplitude spectrum is monotonically decreasing with increasing frequency. Larner et al., 1982, used a $1/f$ decay curve to represent a combination of ambient and streamer-related noise, and although the curve shown in Figure 2.14 does not match a $1/f$ decay curve, it is qualitatively in accordance with this simple assumption. From Figure 2.14 we observe that there is a strong low frequency component (0-5 Hz). This is often caused by swell noise, and often a low cut filter is used during acquisition to remove this noise. If the low cut filter is not used during acquisition, it is important that the dynamic range of the acquisition system is sufficient to record both the low frequency variations and the weaker seismic signals generated by reflected wave energy from subsurface interfaces. The dynamic range is given in dB and is a measure for the ratio between the largest and smallest number that can be measured by the recording instrument:

$$D = 20 \log \frac{A_{max}}{A_{min}} \quad . \quad (2.21)$$

For instance if the ratio between the largest and smallest number recorded in a seismic survey is 10^8 , the corresponding dynamic range is 160 dB. A 24 bit streamer has a dynamic range of 138 dB, since one bit is used for sign and $D = 20 \log 2^{23} = 138$ dB.

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2.9 The amplitude decay and the noise generated by the previous shot

The typical time between two adjacent seismic shots is around 10 seconds. This means that we assume that most of the energy generated by a marine seismic source is negligible after 10 seconds. Figure 2.15 shows how the RMS (Root Mean Square) amplitude level decreases as a function of recording time for a time interval from 0 to 30 seconds. It clearly shows that the ambient noise level is not reached even after 30 seconds. Therefore, it is important to be aware of this effect and relate it to practical noise level aimed for in the acquisition. The shot interval is also directly related to the cost of a seismic survey, so an increase of the shot time interval from 10 to 15 seconds might mean up to 50 % increase in the acquisition costs. Therefore one has to be practical and compare the desired noise level to the effect of the shot generated by the previous shot on the seismic data.

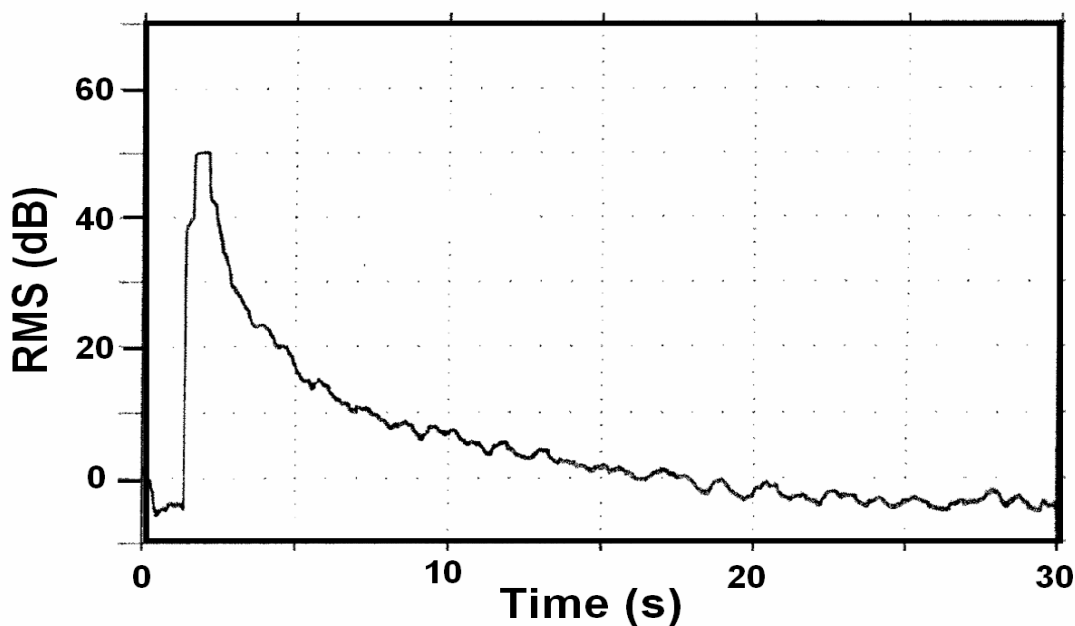


Figure 2.15: RMS amplitude for an offset of 2600 m (raw data) using a 500 ms sliding window for the RMS computation. The source volume is 4832 cubic inches. Zero dB corresponds to 1 microbar and 60 dB to 1000 microbar.

A simple, phenomenological model to describe the RMS-curve ($A(t)$) for seismic data recorded at a given offset might be given as

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$$A(t) = \frac{P_1 e^{-\pi f t / Q}}{t^2} + \frac{P_2}{t} R_0^{\alpha t} + n(t) \quad (2.22)$$

where P_1 and P_2 are scalars, Q is the quality factor, t is the recording time, f the dominant frequency, R_0 is a dominant reflection coefficient for the seabed, α is a fitting parameter representing the number of multiple reflections and $n(t)$ represents the background ambient noise level. A heuristic explanation for the three terms given in equation 2. is as follows: The first term represents primary reflected events, assuming that the typical geometrical spreading correction term is proportional to time squared, and a simple, constant Q -model term accounting for the absorption. The second term represents water layer multiples, assuming that the geometrical spreading correction for such events is proportional to the recording time, t . The $R_0^{\alpha t}$ -term is meant to account for the fact that higher order multiples will be attenuated by the dominant reflection coefficient to some exponent that is proportional to the recording time. Since the dominant part of the multiples propagate in the water layer, the absorption term is skipped for this term. A simple curve fitting using equation 2.22 is shown in Figure 2.16.

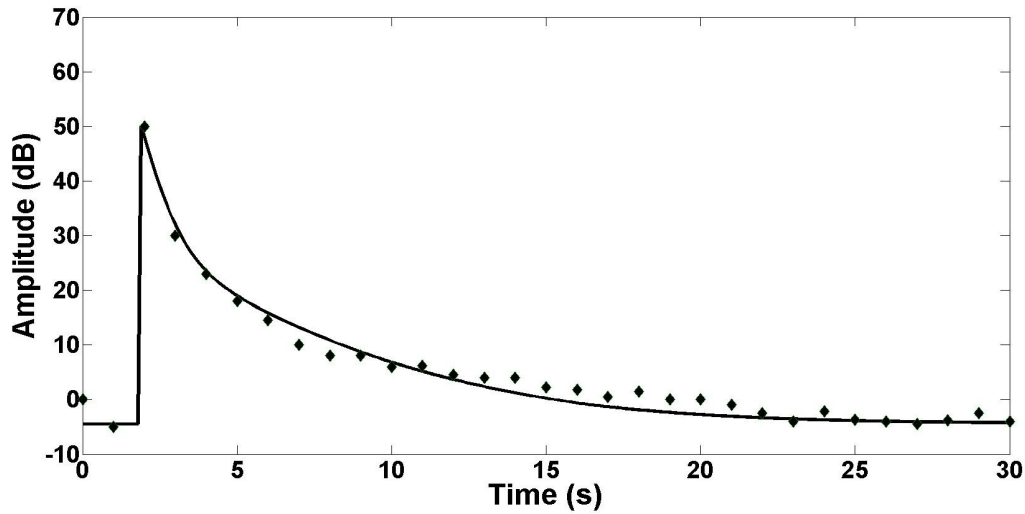


Figure 2.16: Curve fitting: Using equation 2. and selected parameters (solid line). The diamonds represent measured RMS-data.

For a given recording time t and a shot interval τ the signal to noise (from the previous shot) ratio is now given as

$$\frac{S}{N} = \frac{\frac{P_1 e^{-\pi f t / Q}}{t^2} + \frac{P_2}{t} R_0^{\alpha t}}{\frac{P_1 e^{-\pi f (t+\tau) / Q}}{(t+\tau)^2} + \frac{P_2}{(t+\tau)} R_0^{\alpha(t+\tau)} + n(t)} \quad (2.23)$$

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assuming that the noise consist of shot generated noise from the previous shot and the back-ground noise level only. This means that we neglect other types of noise, such as interference noise from other seismic vessels, ship traffic and so on. Figure 2.17 shows a comparison

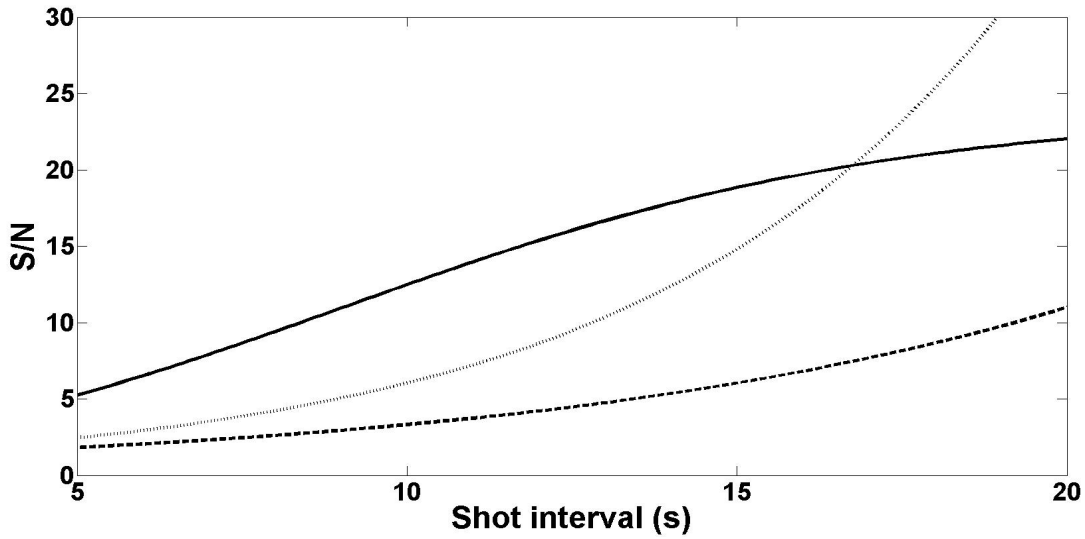


Figure 2.17: Comparison between S/N -estimation based on equation 2.23 (solid line) and two simple exponential decay models (dashed and dotted lines).

between equation 2.23 and two simple exponential decay models. The exponential decay model is based on the work by Herber et al., 1997, where the signal to shot ratio can be written as

$$\frac{S}{N} = \frac{C e^{-kt}}{C e^{-k(t+\tau)} + n(t)} \quad (2.24)$$

In Figure 2.17 a k -value of 0.12 is used for the dashed line while $k=0.18$ is the dotted line. For both decay models a C/n -value of 6660 is used. We notice a significant difference between the model given by equation 2.23 and equation 2.24.

We notice that the signal to noise ratio increases with increasing shot interval, as expected. One should be careful, however to use such curves without taking the other noise sources into account. There is no point in improving the signal to noise generated by the previous shot if there are other noise sources that are more significant. For most seismic surveys it is therefore not realistic to increase the shot interval significantly beyond 10 seconds, simply due to the extra acquisition costs associated with this. However, for permanent seismic recording (used for time lapse seismic data analysis) the extra costs associated with the shooting is limited, and for such surveys it might be considered as an option, in order to improve the signal to noise ratio and hence also to improve the repeatability between two or more seismic surveys.

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Using equation 2.23 it is possible to mimic the effect of increasing the source strength on the signal to noise ratio. Figure 2.18 shows the effect of doubling the source strength: Since the background ambient noise level is the same and reached after approximately 30 seconds, there is a potential for improved signal to noise ratio by increasing the source strength. Therefore, in most seismic surveys it is recommended to use a source above a certain size. Typical large air gun arrays have a total volume of more than 4000 cubic inches, and this size or above is therefore recommended for conventional seismic surveys.

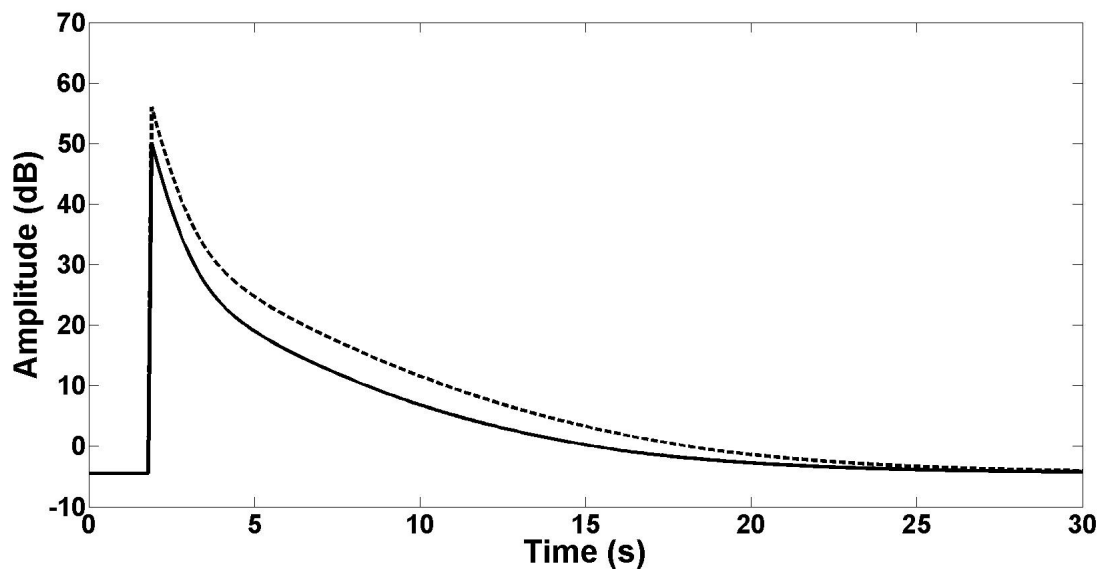


Figure 2.18: Modeled amplitude versus recording time for the calibrated model shown in Figure 2.16 (solid line) and a doubling of the source strength (dashed line).

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2.10 Acquisition of borehole seismic data

Borehole seismic data are often referred to as Vertical Seismic Profiles (VSPs). VSP surveys are acquired with standard sources (often smaller airgun arrays than in a conventional surface seismic survey) but with the receivers downhole. Typical VSP shot records show data as a function of receiver depth versus time instead of offset versus time. For fixed source acquisition we distinguish between the zero offset configuration and the offset configuration as sketched in Figure 2.19. For a moving source acquisition we have so-called walkaway VSPs where the sources walk away from the receiver array and normal incidence or walkabove VSPs where the source and receiver tool try to walk in the same direction so that the offset between the two are close to zero (this configuration requires deviated well paths).

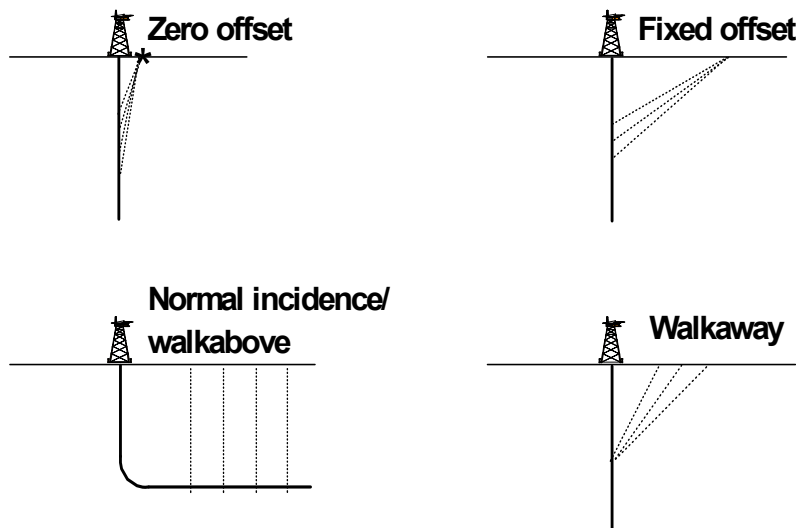


Figure 2.19: Four typical categories of VSP-acquisition geometries.

The receiver array consists of geophones that can measure (for most tools) particle velocities (or displacements) in x, y and z directions simultaneously. 5-level tools which can record the seismic signal at 5 levels simultaneously have been regularly used in the last decade, and today multilevel tools up to 12 and 20 receiver stations are being used. There are receiver tools designed for open holes as well as for cased holes (clamped devices) available. Good coupling with the formation is of course essential for acquiring high quality VSP data.

Another type of VSPs are so-called reverse VSPs, that is when the source is placed downhole, and receiver arrays are deployed at the surface. Real time reverse VSPs using the drillbit as a source have been tested, but is not very common to date.

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2.11 Acquisition of cross well seismic data

A typical cross-well acquisition geometry is shown in Figure 2.20. Seismic signals are recorded at several receiver locations for each shot fired in the neighbouring well. This type of acquisition is usually very time consuming since only a few shot and receiver locations can be occupied at the same time. Another limitation with the cross-well technique is the limited range of the downhole sources. In order to avoid damages on the borehole, the source strength must be limited, and this reduces the range of the seismic signal. Therefore, this technique has been frequently used onshore, on fields where the distances between neighbouring wells are of the order of 200-500 meters. For North Sea fields, this technique has not been frequently used. One advantage with cross-well seismic imaging, is that deeper sediments often have less acoustic absorption, and hence the received seismic signals are often more high frequent than corresponding signals for conventional reflection seismic. The most typical way of using cross-well data is to use the traveltimes for all the different raypaths recorded, and compute a traveltimes tomogram showing variations in P-wave velocity between the two wells.

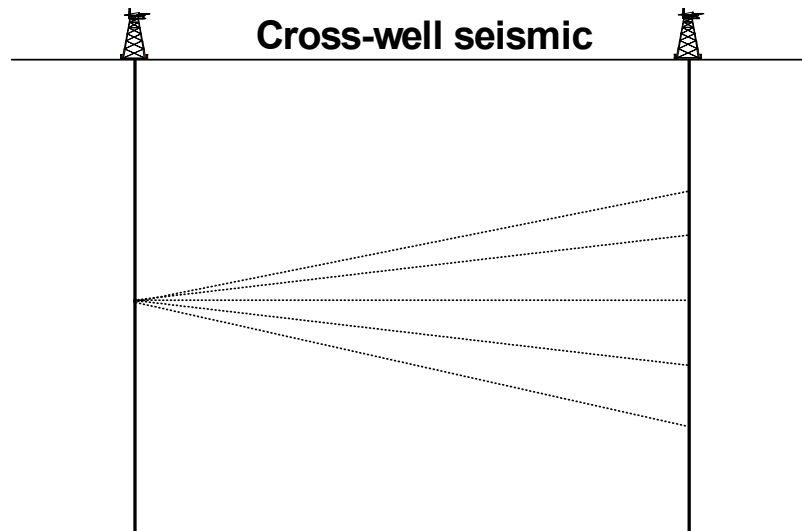


Figure 2.20: Typical geometry used for cross-well acquisition. The sources are located in one well and receivers in another well.

2.12 Acquisition of sea bed seismic data

Sea bed seismic acquisition has evolved during the last decades, and is perhaps the newest branch on the seismic acquisition tree. The first generation of seabed cables were equipped with one-component receivers (z-component). The next generation sea bed registration equipment

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also included three-component geophones in addition to a hydrophone. As for borehole seismic data, a good coupling between the receiver and the formation (ie. sea floor) is important. Various contractors have tested several different types of receivers, Figure 2.21 shows three typical sea bed receivers. Differences in coupling response between these three types of receivers are

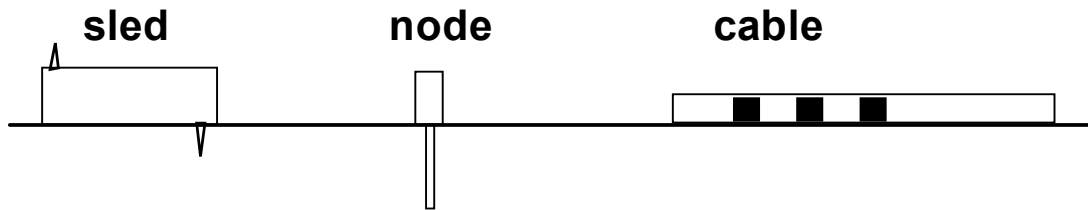


Figure 2.21: Three types of sea bed sensors, the sled (typically 50-60 kg), the stick, and the cable. All three types can be equipped with x,y and z-component geophones and a hydrophone.

still a reserach area, and we will probably see new devices in the years to come. Apart from coupling differences, there are also differences with respect to acquisition efficiency between the three types. A cable solution is more efficient than the other two devices, since typical sea bed cable lengths are of the order of kilometers, and thus offers a time efficient registration of sea bed data. For areas where the seabed is irregular, a cable solution might be impractical, and for such areas a dragged array solution (several sleds coupled together) or a node solution might be an appropriate choice. For cable solutions there are also practical limit with regard to water depth, today the cable solution can be used for water depths of maximum 1000 meters.

A nice feature of seabed seismic data is the multiple azimuth coverage: Since the shooting pattern is a regular grid with for instance 25 m shot point interval in both x and y direction, an almost continuous coverage of azimuthal angles are covered. This opens possibilities for improved imaging as well as more advanced anisotropic studies.

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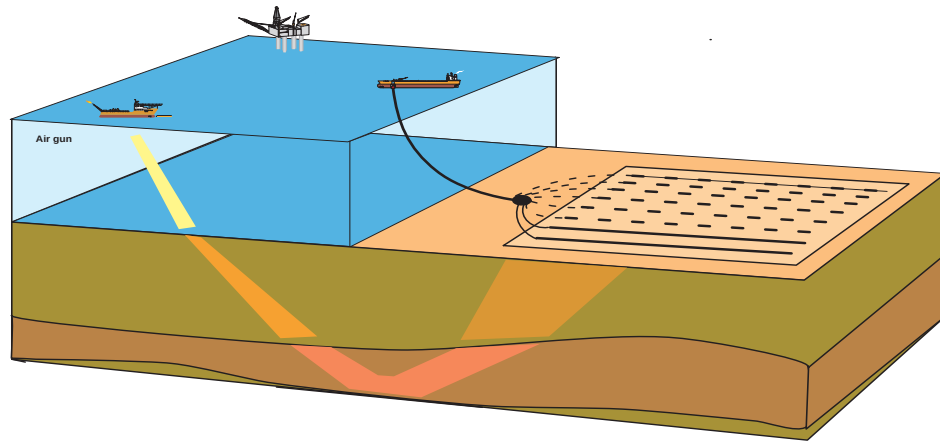


Figure 2.22: Sea bed seismic acquisition using one shooting vessel and one recording vessel. The recording vessel deploys a limited number of cables (in this case two) that are overshot, then the cables are moved into new positions, and overshot again.

2.13 Steerable streamers

In the 1990s WesternGeco developed a new generation of steerable streamers, the Q marine system. By using essentially the same technique as an aircraft, the streamer is equipped with wings that can be used to steer the streamer to the wanted position, both in depth and x-y coordinates. This might be desirable, especially if the cross-currents during acquisition is significant. The main motivation for developing this streamer came from success of time lapse seismic data in the 1990s. Experiments done for time lapse seismic (often referred to as 4D) experiments showed that it was extremely important to repeat the x-y positions of both sources and receivers from the base survey to the repeat survey. By developing a steerable streamer, it is possible to achieve this goal, given that the positioning of both sources and receivers are sufficient. Using today's GPS-systems, this positioning accuracy is adequate, and as a consequence, the normal-

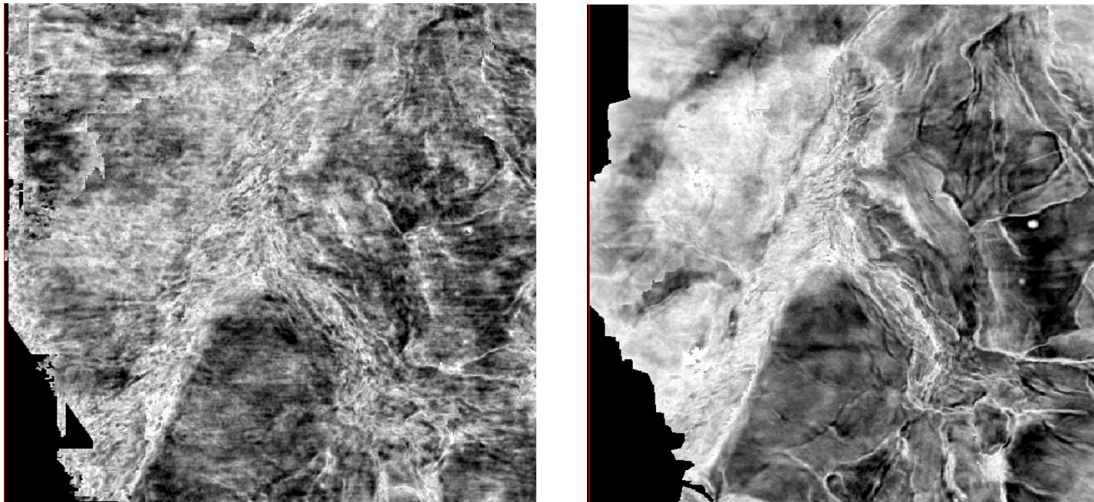
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ized RMS-error between two time lapse seismic surveys has decreased from 70-80 % to 20% today.

2.14 Acquisition of multi-azimuth streamer data

Recently there has been an increase in the number of multi-azimuth streamer data surveys. The main objective for this is improved seismic imaging of complex, 3-dimensional geological structures. Although the extra costs associated with several pass over a given structure is high, it has shown beneficial in some cases. The increased use of seabed seismic data has also contributed to this increased interest for multiazimuthal streamer surveys. Multiazimuth streamer surveys are used to improve imaging of complex structures, one example is shown in Figure 2.23 from the Varg Field in the North Sea. In simple words one might say that the advantage of multiazimuth surveys are similar to using a torch from various angles to illuminate a complex object: The more angles you illuminate your object from the better it is to reveal the structure of the object. For seabed seismic data (receivers located at the seabed) it is common to acquire data in a multi-azimuth set up, since the shooting pattern is normally a regular surface grid overshooting seabed receiver cables or nodes. An example of improved imaging of the Statfjord East flank is shown in Figure 2.24, where the complex structure of the top reservoir interface is better imaged on the multiazimuth OBC data than the conventional streamer data. By decimating the OBC data (using fewer azimuths) it is found that the image quality is reduced somewhat, indicating that multiazimuth seismic acquisition improved the image quality. However, multiazimuth processing often requires more work, since the velocity field might differ between various azimuth segments, requiring more work for velocity estimation. A future “dream” for multiazimuth streamer surveys is to cover a complex structure from different directions and continue seismic data acquisition until an acceptable image is obtained. This setup requires onboard, pre-stack depth imaging so that the image of the structure is continuously updated as the acquisition proceeds. Another use of multi-azimuth seismic data is for fracture detection. This is particularly important for carbonate fields, where the fracture orientation is important in order to understand the major flow directions within a reservoir, since high permeability is often correlated with high degree of fractures. By studying how the amplitude varies as a function of azimuth at the top reservoir, it is possible to estimate major changes in fracture orientation, since the reflection coefficient at top reservoir is dependent on the degree of fractures present in the reservoir. An example on how the amplitude variation with offset changes when the azimuth is changed by 90 degrees is shown in Figure 2.25 where a significant change is AVO (Amplitude Versus Offset) is observed between the N-S shooting direction and the E-W shooting direction.

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Ref.: Pre-Stack Depth Migration of Multi-Azimuth Towed Streamer Seismic: presented by Stian Hegna and Dorothee Gaus at the 2003 EAGE-meeting in Stavanger

Figure 2.23: Comparison between one-azimuth (left) and 3-azimuth shooting (right): Faults and noise content is much better on the right image compared to the left. (figure from PGS).

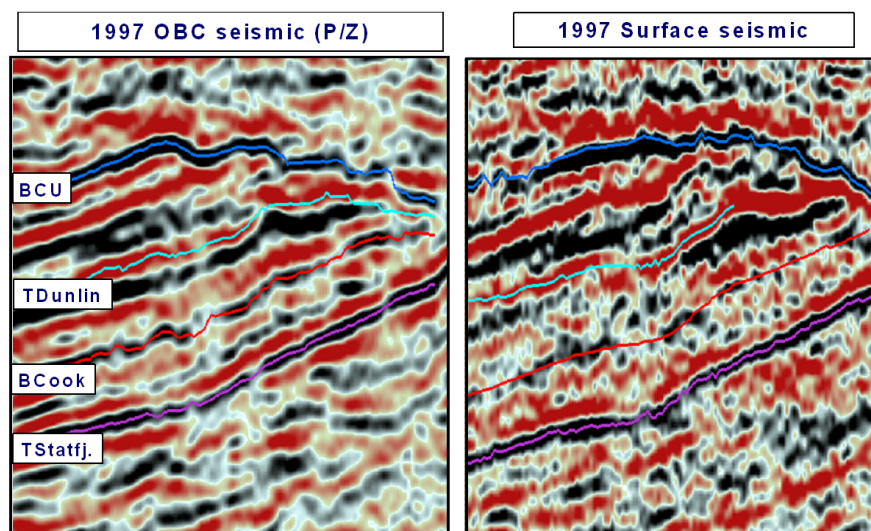


Figure 2.24: Example of multi-azimuth OBC acquisition from the East flank of the Statfjord Field, North Sea. Notice the improved imaging of the BCU marker defining the top reservoir at this field (Rognø, Kristensen and Amundsen, 1999).

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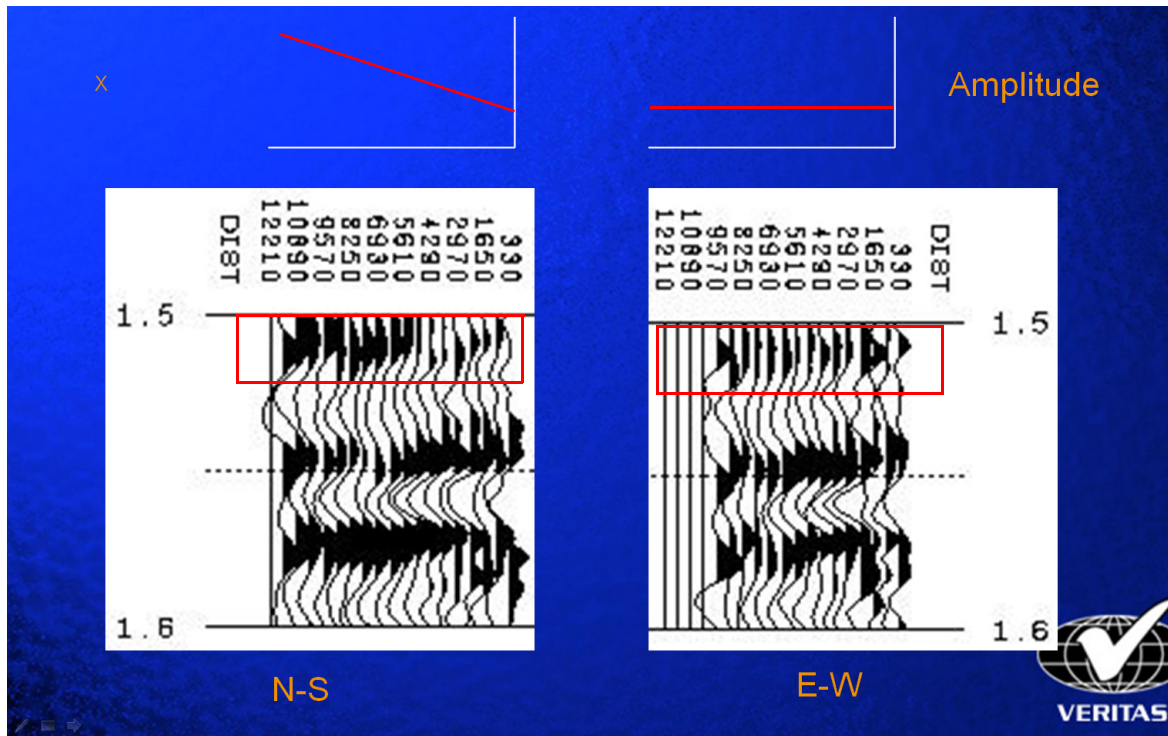


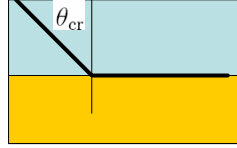
Figure 2.25: Variation in AVO with azimuth: Notice that the amplitude is increasing with offset (from right to left) on the N-S azimuth (left section, within the red rectangle), while there is no increase in amplitude for the E-W azimuth (right section). This is also illustrated by the amplitude versus offset cartoons shown above the seismic sections.

Another way of determining azimuthal anisotropy is to detect the variation in critical angle versus azimuth. This requires that there is a velocity increase at the interface under investigation. This is often the case for carbonate reservoirs, since the velocity of a carbonate rock is often higher than the overlaying cap rock. Figure 2.27 shows how the critical angle (derived from Snell's law) varies for anisotropic rocks. If the orientation of the cracks in a rock is vertical, there might be azimuthal changes in the anisotropic properties of the rock, and this will lead to changes in the anisotropic parameter epsilon. If we can characterize these azimuthal changes in anisotropy it might help us to better understand and map the major fracture orientation within a given field. Ultimately this knowledge will lead to a better reservoir management of the field, through improved positioning of producer as well as injector wells. There are however, several challenges related to the estimation of critical angle. For marine seismic data, the presence of the water column related noise (harmonic modes in the water layer) will often interfere with the signal close to critical offset for the top reservoir event. For land seismic data, the presence of the ground roll is the major obstacle for this type of analysis.

Critical angle for VTI and orthorhombic media

Snell's law:

$$\frac{\sin \theta_{\text{cr}}}{V_1(\theta_{\text{cr}})} = \frac{1}{V_{\text{hor},2}}$$



Isotropic/VTI interface:

$$V_{\text{hor}} = V_{P0} \sqrt{1 + 2\epsilon}$$

$$\sin \theta_{\text{cr}} = \frac{V_{P0,1}}{V_{P0,2} \sqrt{1 + 2\epsilon_2}}$$

Figure 2.26: The critical angle is a function of the vertical P-wave velocity and the anisotropic parameter epsilon (Thomsen-parameter) for the layer below the interface. By measuring the variation in critical angle with azimuth, it is possible to measure the variation in anisotropy.

2.15 High frequency signals from air-gun arrays

In recent years there has been an increasing interest to study the amount and characteristics of the high frequency signals emitted by an air-gun array. It is well known that marine mammals and fish have both hear and communicate in the frequency range from 10 Hz and up to 200 MHz. Therefore, it is interesting to investigate to what extent the high frequency signals produced by air gun arrays might disturb and harm marine life. Air gun arrays are designed to produce a strong signal in the seismic band, that is between 2-150 Hz. However, due to various mechanisms the source arrays will also produce a small amount of signals in the high frequency range, that is between 1 kHz to 100 kHz.

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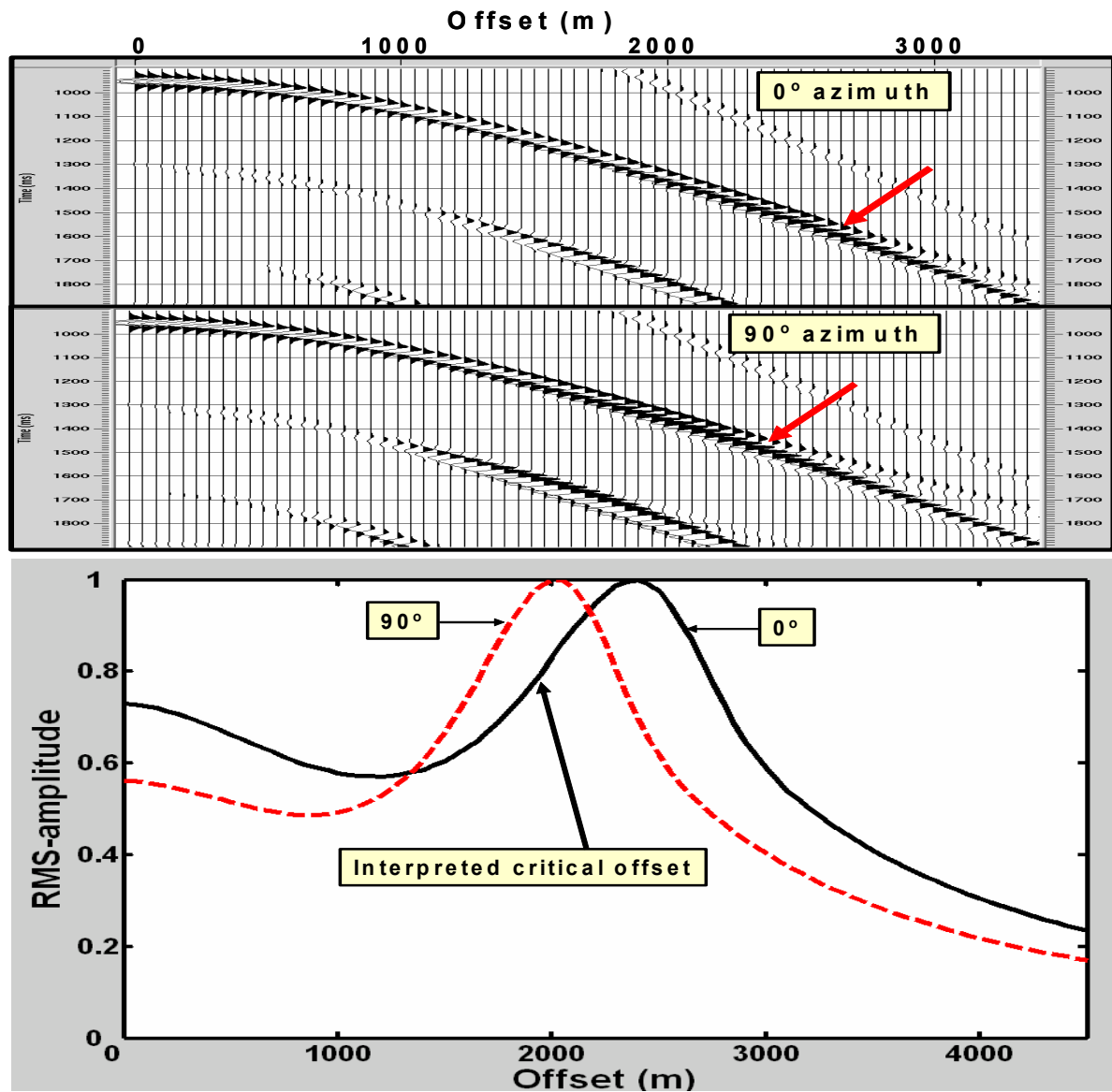


Figure 2.27: Comparison of synthetic seismograms for 0 and 90 degrees azimuth of an isotropic layer overlaying an orthorhombic (anisotropic) layer. Notice that the headwave split offs at a shorter offset for the 90 degree azimuth section than the 0 degree azimuth section. On the bottom plot we observe that the amplitude maximum is shifted by several hundred meters for the two azimuths. (From Landrø and Tsvankin, *Geophysics*, 72, D41-D50).

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The first and maybe the most obvious cause for high frequency signals from a single air gun is that the rapid movement of the air escaping through the air gun ports (Figure 2.3) creates cavities in the water close to the gun. This effect is similar to cavities created on propellers, or cavities caused by the water jet generated by the pistol shrimp, or cavities associated with turbulent flow in water. Figure 2.28 shows photographs taken immediately after the release of air from a small air gun, and on the second frame we notice small white bubbles that may be interpreted as small cavities being formed close to the sharp edge on the air gun above the gun ports. If these small white bubbles are cavities, they will collapse and produce high frequency signals. We don't know which of the effects mentioned above that actually could cause significant cavitation, or if all are present. Since the air gun has sharp edges, and that the water velocity close to these edges is high, we assume that cavities similar to propellers may be formed at edges close to the gun ports (Figure 2.3). Furthermore, it is likely that turbulent flow close to the expanding bubble creates cavitation. Since we believe that these cavities are created close to the gun, it is reasonable to assume that the amount and strength of the cavities are dependent on the design of the air gun. Hence, there might be differences in high frequency noise from air guns between various manufacturers.

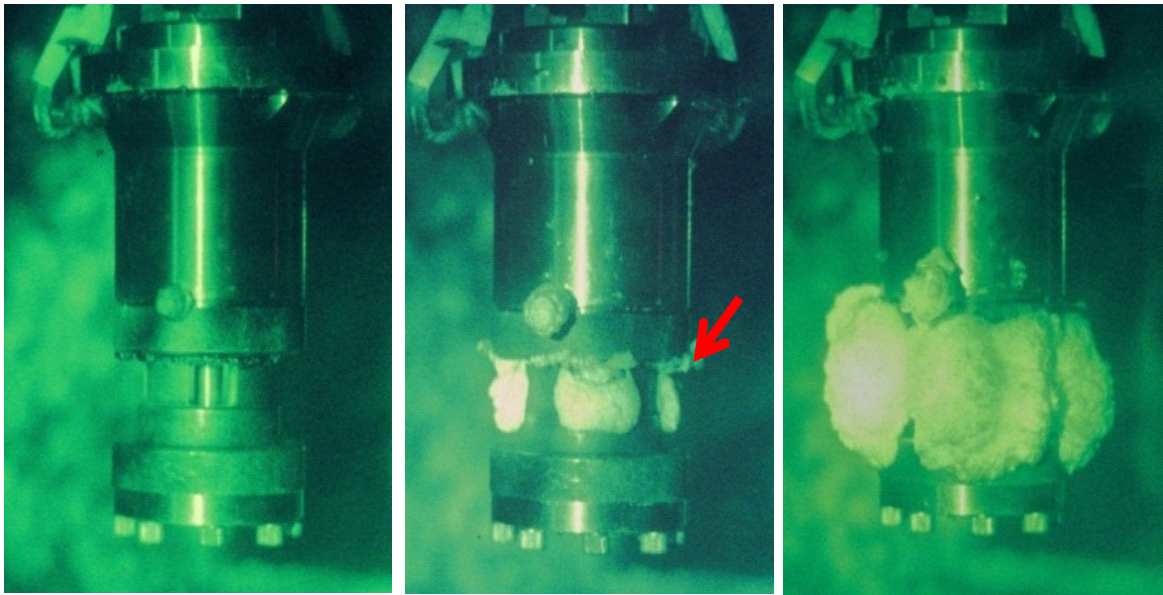


Figure 2.28: High speed photo of a small air gun for various times after the gun is fired. In the mid frame we observe small white bubbles (marked by red arrow) that are interpreted as vapor cavities. When these cavities collapse they generated a high frequency signal.

Another effect is the clicking in the solenoid (see Figure 2.3), which is the electric triggering unit sitting on top of the air gun. This will also create high frequency noise, especially at the early stage of an air gun signature. Furthermore, there is a possibility that the shuttle movement within the air gun might cause high frequency noise, for instance when the shuttle stops against

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the interior air gun steel. It is also well known that air guns jump immediately after they are fired, and this will cause shaking and introduce possibilities for mechanical high frequency noise. As far as we know, there has been limited research carried out to actually find out which of the suggested high frequency noise types that occur close to an air gun, and which mechanisms that dominate within various frequency bands.

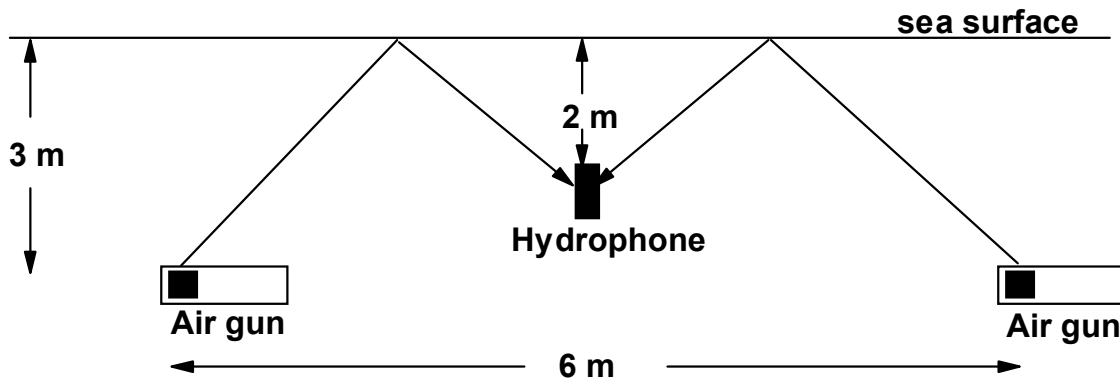


Figure 2.29: If the two air guns generate a peak pressure of 4 bars at 1 m, the reflected signal from the sea surface measured at the hydrophone between the guns is approximately -1.3 bars, assuming linear theory. The hydrostatic pressure at the hydrophone position is 1.2 bars, which means that the total pressure assuming that the pressure contribution from each source can be added linearly is negative. Figure from Landrø (2000).

Another, and far more sophisticated and maybe even speculative cause for high frequency emissions from air gun arrays is coupled to the effect of sound reflections from the sea surface. This effect was first mentioned by (Landrø, 2000), and in this paper we will elaborate further on this as a potential cause for generation of high frequency signals from compact air gun arrays. If two air guns as shown in Figure 2.29 emit a strong signal of say 4 bars, the pressure in the water between the two guns might approach zero since the reflected signals from the sea surface are negative. In this example, we have assumed that the signals from the two air guns can be linearly superimposed. However, as the water pressure approaches zero, non-linear effects will be more prominent, meaning that it is not straightforward to estimate exactly for which pressures cavitation will occur. In this context we will simply assume that when the linear theory predicts zero or even negative pressures in the water, cavitation might occur. The physical mechanism for this is that as the absolute pressure in the water approaches zero, then the water droplets will undergo a phase transition into water vapor, and a cavity filled with water vapor will be formed. We will refer to this phenomenon as “ghost cavitation”, since it is

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caused by close to simultaneous ghost arrivals from several air guns. This type of cavitation will be independent of type of air gun used, since it is simply a function of the geometry of the array. For a water gun, the cavities are formed by high speed water jets being formed outside the water gun, as described by (Landrø et al., 1993).

A synthetic modeling of this effect is shown in Figure 4, where we have put 8 identical air guns with a volume of 200 cubic inches 6 meters away (in a circle) from a hydrophone at 5 meters depth. Details on air gun modeling theory can be found in (Ziolkowski, 1970) and (Landrø, 1992) (for GI guns). Water gun modeling has been studied by (Landrø et al., 1993). The modeling we use in this paper is a calibrated version of the theory outlined by Ziolkowski. It also includes some extensions of effects that were not discussed in Ziolkowski's paper from 1970. However, in this context, these extensions are not considered important. We observe that for a short time, the absolute pressure is practically zero, which we interpret as a possibility for cavitation. In the source modeling code, there is no attempt to model the non linear effects discussed above. It should be stressed that the accuracy of this modeling for this low pressure region are strongly related to the width of the primary peak. In source modeling, there has not been a strong focus on modeling the first 3-5 milliseconds of the main peak with extreme accuracy, and therefore one should view these modeling examples as approximate and indicative.

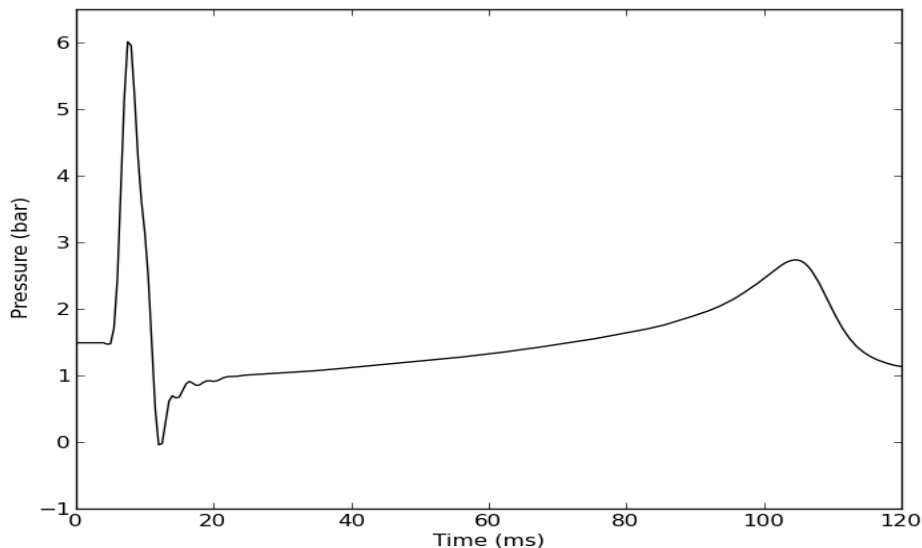


Figure 2.30: Source modeling of 8 200 cubic inch air guns fired at 6 m and 6 m away from a hydrophone located at 5 m depth. Notice that the absolute pressure in the water is close to zero for a short time interval around 12 ms.

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Despite of the fact that non-linear effects are crucial to accurately understand what happens when the negative-amplitude ghost reflections from several air guns hit an area between the air guns, as shown in Figure 2.29, we will claim that it is a possibility that cavities might be formed due to exceptionally low pressure that lasts for a few milliseconds.

And this is also the good news: If the major part of the high frequency signal generated by a compact air gun array is generated by ghost cavitation occurring between air guns, this effect can be reduced or even eliminated simply by increasing the distance between the guns. Lord Rayleigh showed in 1917 (Rayleigh, 1917) that the collapse time of a water vapor cavity is proportional to the cavity radius. The strength of the signal generated by the collapse of a cavity is also related to the size of the cavity (Landrø et al., 1993). Therefore, depending on the sizes of the cavities generated between the air guns, they will collapse at different delay times, and with various strengths. Generally, the strength and length of this high frequency signal will therefore increase with the size of the air gun array, and with the compactness of the array.

When cavitation occurs in water, the interior of the cavity is filled with water vapor. A cavity is formed as the water pressure is approaching zero, or more accurately as the pressure is equal to the vapor pressure. This pressure is given by the phase diagram for water, and it depends on the water temperature. For instance, for a water temperature of 17 Centigrades, the vapor pressure is 0.03 bar, practically zero.

Figure 2.31 shows an example of high frequency measurements of a small and a big (approximately double size) air gun array at 5 m depth, measured in the far field, vertically below the array. The distance between the three subarrays used for the full array configuration is 6 m. The small array consists of a single string, while the big array consists of the same string and two identical strings on each side, containing 3 pairs of clustered air guns. The peak-to-peak amplitude for the small and big array are 37 and 62 bar-m, respectively (corresponding to 251 and 256 dB re 1 microPascal).

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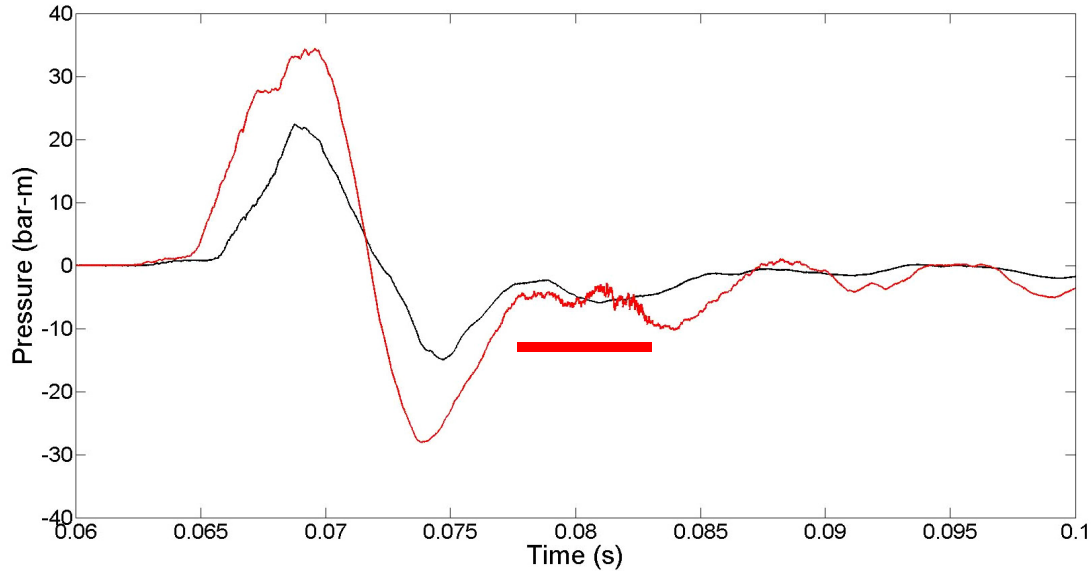


Figure 2.31: Measured source signatures from a single string array (black solid line) and a multi-string array (red solid line) (top). Notice the high frequency noise appearing at approximately 0.08 seconds (marked with a red bar) for the big source array. The amplitude of this high frequency noise is approximately 0.5 bar-m.

We notice that the amount of high frequency noise is not as strong for the small array compared to the full array configuration. The most prominent part of the high frequency noise is marked by a red bar in the figure. The amplitude level of the high frequency noise is approximately 0.5 bar-m (214 dB re 1 microPascal @ 1 m). Application of a 1 kHz high pass filter reveals the details of this high frequency noise, as shown in Figure 2.32. First of all, we notice that there is also a high frequency component associated with the small array, and more important: The high frequency signal starts at the very beginning of the air gun signal. We interpret this as associated with the noise generated by the solenoid. The signal generated at the same time as the signature reaches the maximum peak (marked on Figure 2.32) might be related to shuttle movement, jumping of the air guns, internal vibrations and cavities collapsing close to each individual gun. Furthermore, we believe that the fact that this signal is stronger for the big source, is simply because the big source contains more guns.

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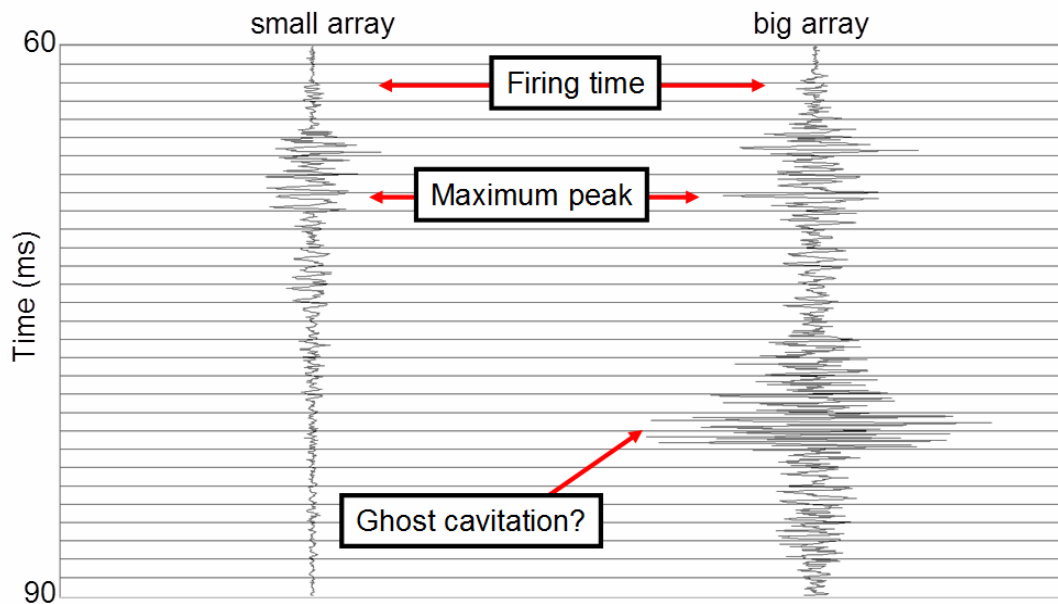


Figure 2.32: 1 kHz high pass filtered version of the small and big array far field signals. Notice that the high frequency noise is also present for the small array. The timing between each line is 1 ms. However, for times above 75 ms the big array shows significantly more high frequency signals.

2.16 Seismic acquisition and impacts on marine life

There has been several studies focusing on the effects of seismic shooting on various types of marine life. The general conclusion from such studies is that seismic activity does not kill fish or other species in the ocean. However, seismic activity influences the behavior of for instance fish, and therefore, it is not allowed to acquire seismic data during the spawning periods.

Marine mammals have a mammalian ear that through adaptation to the marine environment has developed broader hearing ranges than those common to land mammals. As a group they have functional hearing ranges from 7 Hz to 200 kHz. A comparison of hearing threshold for various types of marine animals is shown in Figure 2.33. We notice a huge variation in hearing threshold between various species. Normally the communication and hearing thresholds are in similar frequency bands, meaning that marine animals generate sound and listen within approximately the same frequency range. The clicks generated by sperm whales (Figure 2.34) are used both for echolocation and communication. From the mid-1960's, the clicks were described as multi-pulsed, long duration non-directional signals of moderate intensity levels with a spectrum peaking in the 2-8 kHz range.

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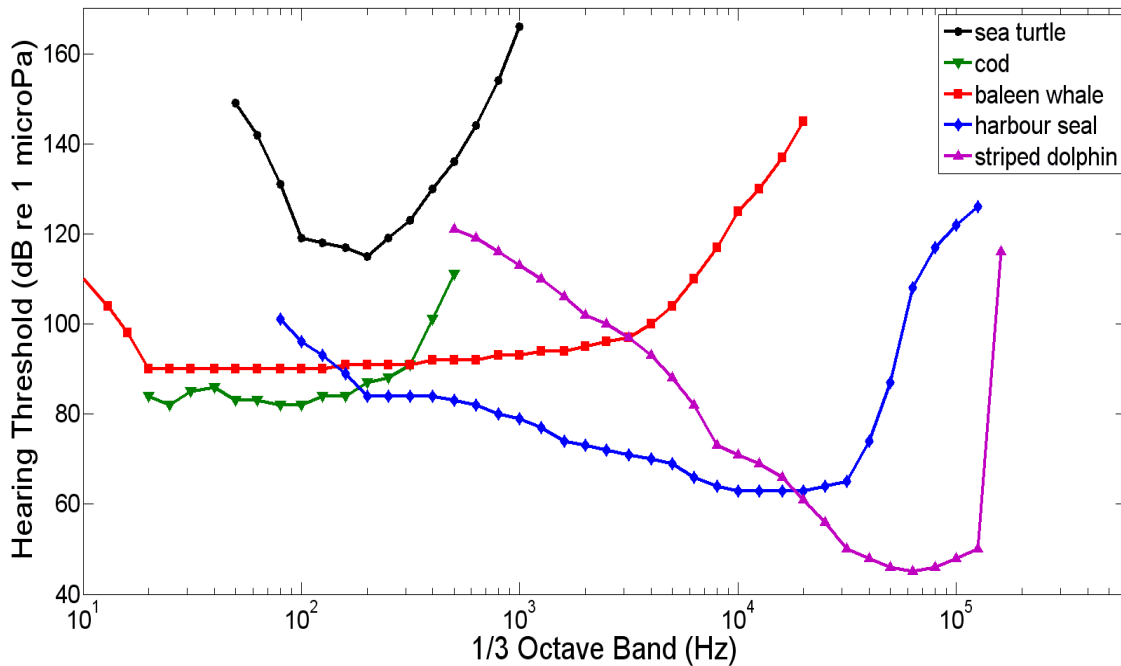


Figure 2.33: Hearing threshold for various marine animals. The data are taken from “Final EIR underwater noise analysis”, prepared by Tech Environment, Inc., 2006.

However, during the last decade studies by Møhl and co-authors, working in the Bleik Canyon offshore Norway, give new insights. The clicks recorded on large aperture hydrophone arrays are mono-pulsed, lasting 100 microseconds, and have pronounced directivity, with extremely intense (on-axis) source levels up to 236 dB (relative to 1 microPa), with spectral emphasis at 15 kHz. The measured sound levels make sperm whale clicks by far the loudest of sounds recorded from any biological source!

Figure 2.35 shows a comparison between the amplitude spectrum recorded 550 m away from an air gun array and the noise generated by the seismic vessel. For frequencies above 1-2 kHz, we notice that the vessel noise and the noise from the air gun array and the vessel are practically equal, indicating that the high frequency signal generated by the air gun array is drowning in the high frequency noise generated by the vessel. This is in agreement with Figure 2.31 where we observe that the high frequency signal is significantly lower in amplitude compared to the main seismic signal (the low frequency seismic signal constituting the main peak).



Figure 2.34: Sperm whale (photo: Discovery).

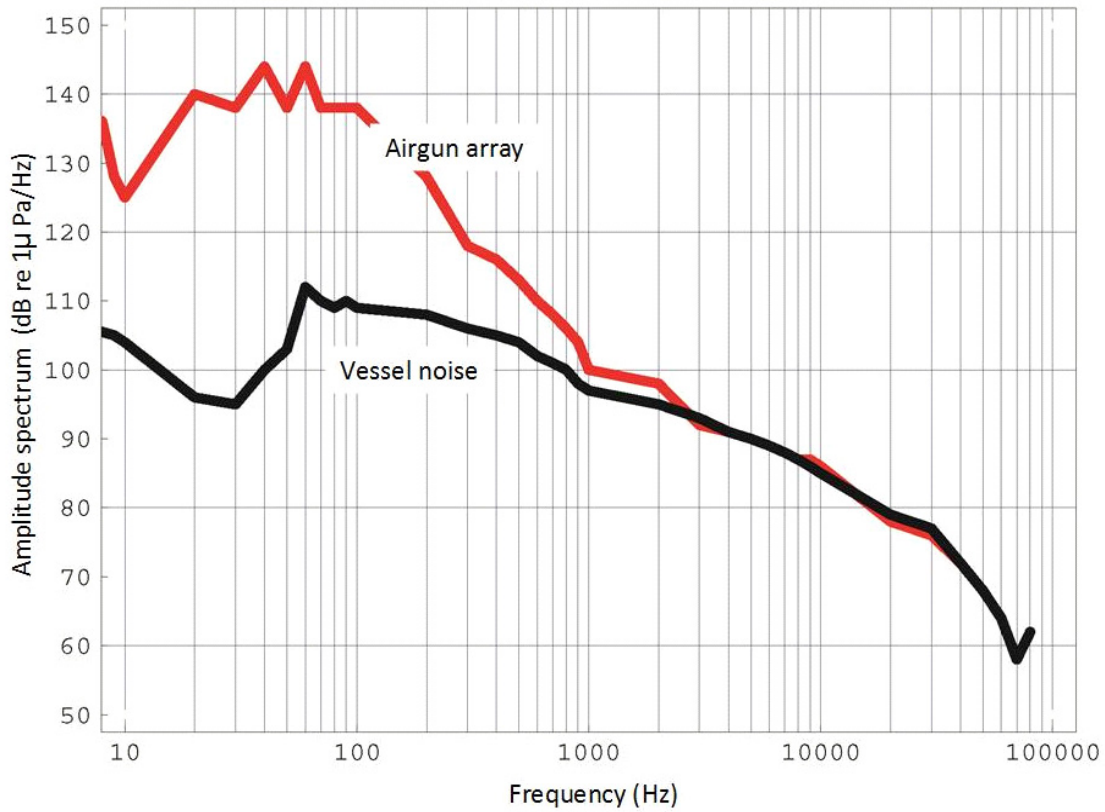


Figure 2.35: Broad-band smoothed amplitude spectra from 38.2 l gun array at 550 m range to a hydrophone at 35 m depth, and from source-vessels self-noise. (Modified from Breitzke et al, 2008). Below 1 kHz the amplitude spectrum of the air-gun signal differs significantly from the vessel's self-noise spectrum due to the low-frequency energy emitted by the air gun. Above 1 kHz the amplitude spectrum of the airgun signal agrees with the vessels self-noise almost completely. This indicates that the slow spectral level decay is mainly caused by ship-generated noise.

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3 Seismic Processing

3.1 Introduction

A typical processing flow consists of the following processes being applied in a sequential order:

- Read field tapes (both seismic data and navigation)
- Editing of data including removal of bad/noisy traces
- Setup of Field Geometry
- Field statics (land or seabed data only)
- Geometrical spreading correction
- Source signature deconvolution and trace balancing
- Multiple attenuation
- CMP (Common Mid Point) sorting
- Velocity analysis
- NMO and/or DMO correction
- Muting (typically near and far offset)
- Stacking
- Time-variant band pass filtering
- Migration (post stack)
- Apply appropriate gain

These are the most common processes, and the order they appear in various processing flows vary, dependent on the actual data set. For instance deconvolution can be applied to the data both before and after stack. Also seismic migration (the imaging step) can be applied both before (prestack migration) and after (poststack migration) stack. In the forthcoming subchap-

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ters these processes will be explained in more detail. For simplicity, most of the processes will be described assuming 2-dimensional seismic data, even though today they are applied to 3D seismic data.

3.2 Preprocessing

In the preprocessing step it is important to load the geometry of the seismic experiment. The navigation data giving the precise location of every shot and receiver for the whole seismic survey is loaded into the processing system. On the basis of notes given in the observers log, bad traces are removed from the data set. In addition to this editing of traces one can also apply an automatic detection and removal of bad traces, based upon preset noise criteria. It is important to remove traces containing high amplitude noise spikes, cause such spikes might lead to considerable problems for the forthcoming processing.

3.3 Field statics

The basic idea in field statics is to simulate a new data set, in such a way that the new data set is acquired along a horizontal interface. This means to replace the actual source and receiver locations down to a mutual horizontal plane, often referred to as the datum. This process is sketched in Figure 3.1.

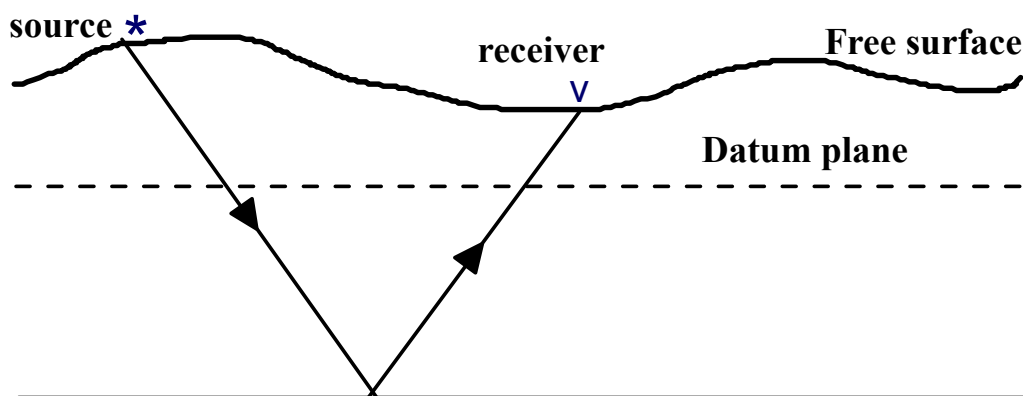


Figure 3.1: Illustration of the datum plane in field static corrections.

In order to move the sources and receivers down to the datum level, one need to know the P-wave velocity between the free surface and the datum level for P-waves, and correspondingly one need the S-wave velocity for doing the shear wave statics. In processing of land seismic data, the field statics is an important process, and its importance increases with the complexity

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of either the shape of the free surface, or strong lateral velocity variations in the medium between the free surface and the datum plane. For processing of shear wave events (both for land data and seabed data) this process is even more important, since the shear wave velocity is often very low in the uppermost layers, causing large static shifts, and corresponding uncertainties in the estimation of the statics. Field statics is not regularly used in processing of marine seismic data, apart from seabed seismic datasets.

3.4 Geometrical spreading correction

In an isotropic medium both the P- and S-wave amplitude decreases proportional to the inverse of the distance from the source location. This means that the recorded signal strength from reflectors at 3-4 km depth are attenuated by the reciprocal distance. In addition to this geometrical effect the amplitude of the seismic signal is attenuated due to transmission losses at each interface, and absorption effects. For interpretation purposes it is nice to have all the processed data within approximately the same amplitude level, independent of depth. Since the geometrical spreading effect scales as $\frac{1}{r}$ and r is proportional to the travelttime (assuming constant velocity) a correction factor being proportional to the travelttime or the travelttime raised into some arbitrary power is often being used. This process can be regarded as a programmed or deterministic gain control (PGC). At a later step in the processing one can choose to apply an automatic way of assuring that the amplitudes remains roughly constant with depth. This process is often based upon calculation of the RMS (Root Mean Square) value within fairly long time windows (400-1000 ms), and scaling of each time window by the inverse RMS-value. This process is often called AGC (Automatic Gain Control). For true amplitude processing, however, it is important to retain a deterministic control of the amplitude level in the processed seismic data, and then one often needs to take special precautions.

The square of the normalized geometrical spreading (g) can be expressed as (Ursin, 1990),

$$g^2 = \frac{\cos^2 \theta_1}{v_1^2} \sum_{k=1}^N \frac{D_k v_k}{\cos \theta_k} \sum_{n=1}^N \frac{D_n v_n}{\cos^3 \theta_n} \quad (3.25)$$

where D_k and v_k are the thickness and velocity of layer number k and θ_k is the angle the ray makes with the vertical in layer k . Based on the above equation Ursin derived the following approximation for geometrical spreading

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$$g^2 = \frac{T_0^2 v_{RMS}^4}{v_0^2} + \left[2 \left(\frac{v_{RMS}}{v_0} \right)^2 - 1 \right] x^2 + \frac{1}{T_0^2} \left[\frac{1}{v_0^2} - \frac{1}{v_{RMS}^2} \right] x^4 \quad (3.26)$$

where x is the offset, v_{RMS} is the RMS-velocity and T_0 is the normal-incidence traveltime. All the terms in the above given equation are estimated in a normal processing flow, which means that this equation is easy to implement as a more precise geometrical spreading correction than a conventional t^2 -scaling. The RMS-velocity is given by the Dix equation

$$v_{RMS}^2 = \frac{\sum_{k=1}^N \Delta t_k v_k^2}{\sum_{k=1}^N \Delta t_k} \quad (3.27)$$

where v_k and Δt_k are the interval velocity and vertical traveltime in layer k , respectively.

3.5 Deconvolution

A basic assumption in seismic processing is that the recorded seismic signal can be written as a convolution between a wavelet ($w(t)$) representing the input signal and the impulse response of the earth ($f(t)$)

$$y(t) = w(t) * f(t) \quad (3.28)$$

where the asterisk denotes convolution. From an exploration point of view, the interesting information about the subsurface is contained in the impulse response function, and therefore we want to extract the impulse response from the recorded seismic signal. Application of for instance the inverse filter (w^{-1}) of the wavelet defined by

$$w(t) * w^{-1}(t) = \delta(t) \quad (3.29)$$

will remove the effect of the wavelet on the seismic data. In principle one can design such inverse filters for all unwanted and deterministic effects that distorts the seismic data. For instance, the water layer reverberations can approximately be expressed as a periodic filter in the time domain, and therefore a corresponding inverse filter can be designed in order to remove the water layer reverberations. Wiener derived optimal filters that convert the input to any desired output. The typical form of the matrix equation derived by Wiener (see for instance Robinson and Treitel, 1980) is

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$$\begin{bmatrix} r_0 & r_1 & \cdots & r_{n-1} \\ r_1 & r_0 & \cdots & r_{n-2} \\ r_2 & r_1 & \cdots & r_{n-3} \\ \cdots & \cdots & \cdots & \cdots \\ r_{n-1} & r_{n-2} & \cdots & r_0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \cdots \\ a_{n-1} \end{bmatrix} = \begin{bmatrix} g_0 \\ g_1 \\ g_2 \\ \cdots \\ g_{n-1} \end{bmatrix} \quad (3.30)$$

where r , a and g are the autocorrelation of the input wavelet, the desired filter coefficients and the crosscorrelation of the desired output with the input wavelet, respectively.

3.5.1 Spiking deconvolution

If we choose g in equation 3.30 to be a zero-lag spike, the corresponding process is referred to as spiking deconvolution:

$$\begin{bmatrix} r_0 & r_1 & \cdots & r_{n-1} \\ r_1 & r_0 & \cdots & r_{n-2} \\ r_2 & r_1 & \cdots & r_{n-3} \\ \cdots & \cdots & \cdots & \cdots \\ r_{n-1} & r_{n-2} & \cdots & r_0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \cdots \\ a_{n-1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \cdots \\ 0 \\ 0 \end{bmatrix} \quad (3.31)$$

If the autocorrelation function matrix (r) is computed directly from a known source wavelet, that is

$$r_i = \sum w_{n-i} w_n \quad (3.32)$$

the process is denoted deterministic spiking deconvolution. If the autocorrelation function is simply computed as the autocorrelation of the input seismic signal (the basic assumption for this case is that the reflectivity series representing the earth is a purely random process), the process is called statistical spiking deconvolution.

3.5.2 Predictive deconvolution

By choosing g in equation 3.30 to be a time-advanced form of the input series we obtain a predictive deconvolution process. Given the input $x(t)$, we want to predict the value at a delayed time $x(t + \alpha)$, where α is the prediction lag. To obtain this we simply choose g as the autocorrelation function of the input data delayed with the lag α :

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$$\begin{bmatrix} r_0 & r_1 & \cdots & r_{n-1} \\ r_1 & r_0 & \cdots & r_{n-2} \\ r_2 & r_1 & \cdots & r_{n-3} \\ \cdots & \cdots & \cdots & \cdots \\ r_{n-1} & r_{n-2} & \cdots & r_0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \cdots \\ a_{n-1} \end{bmatrix} = \begin{bmatrix} r_\alpha \\ r_{\alpha+1} \\ r_{\alpha+2} \\ \cdots \\ r_{\alpha+n-1} \end{bmatrix} \quad (3.33)$$

This means that in order to design a predictive deconvolution operator we only need the auto-correlation of the input time series. In practice, one also has to specify the operator length (n) and the prediction lag (α). The main motivation for using predictive deconvolution in seismic processing is to attenuate multiples. In that case, a common procedure is to use a predictive deconvolution operator using a prediction lag equal to the twoway traveltime for the waterlayer, followed by a spiking deconvolution operator to compress the wavelet.

3.5.3 Directional deconvolution

Conventional marine seismic source arrays are directional, that is the amplitude and the phase of the emitted source signal varies as a function of emission angle with respect to the vertical direction. The directivity effects of the source array is strongly dependent of the areal distribution of the air guns, long and wide airgun arrays have more variations with angle than short and compact arrays. In order to avoid strong sidescatter effects, one can choose to acquire the seismic data with an extremely wide spread of the array, sometimes up to 100 metres. This will introduce strong directivity effects in the emitted signal, since the main seismic energy will be centered around the vertical axis, which of course also was the intention behind choosing a superwide array for such a survey.

In addition to source created directivity effects, absorbtion, transmission losses, and anisotropic effects within the subsurface might create additional directivity effects.

3.6 Multiple attenuation

In addition to predictive deconvolution (followed by spiking deconvolution) a number of different techniques for multiple removal has been derived.

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3.6.1 *F-k multiple removal*

F-k multiple removal utilizes that primaries and multiples have different velocities. If you apply a normal moveout correction (NMO) using a velocity inbetween the velocities for primaries and multiples, the multiples will be undercorrected, while the primaries will be overcorrected, as sketched in Figure 3.2. Transforming the x-t data shown in this Figure into f-k domain, yields a nice separation of the primaries and multiples, since the primaries will be mapped in the right quadrant, while multiples are mapped into the left quadrant in the f-k plot, as schetched in the figure. In the f-k domain, we can set all data with negative wavenumbers equal to zero, corresponding to removal of the multiples. After zeroing the left quadrant in the f-k domain, the data can be transformed back to the x-t domain again, but now the multiples are removed. In practice this technique for multiple removal is quite difficult, often because the deviation between multiple velocities and primary velocities is small, and therefore the two wavetypes do not separate well in the f-k domain.

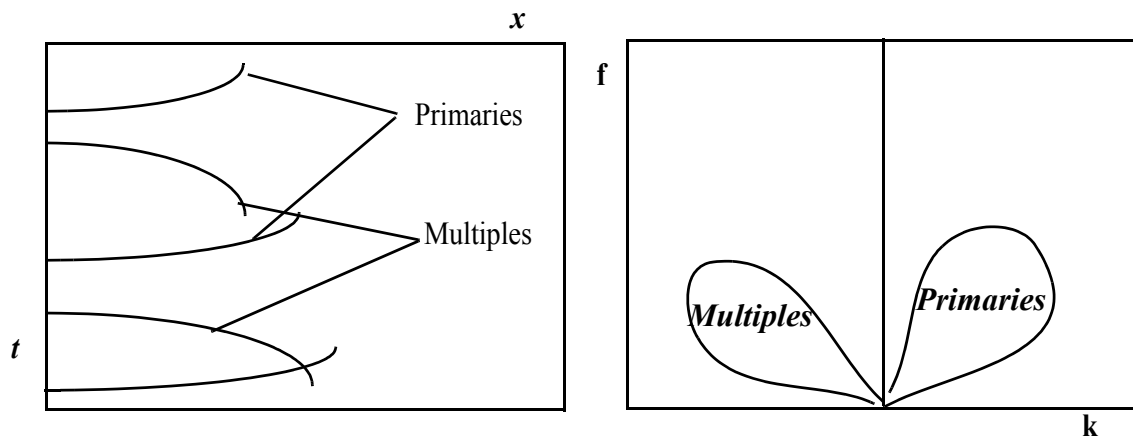


Figure 3.2: Schematic illustration of primaries and multiples in x-t domain after being NMO-corrected with a velocity function between multiple and primary velocities (left). The f-k transformed data is shown (schematically) to the right, where multiples and primaries map into different quadrants.

3.7 Band pass filtering

Normally a band pass filter is specified by 4 frequencies as shown in Figure 3.3. Band pass filter is a common way to attenuate both low and high frequency noise. Different band pass filter might be used for various time gates as well as offset gates.

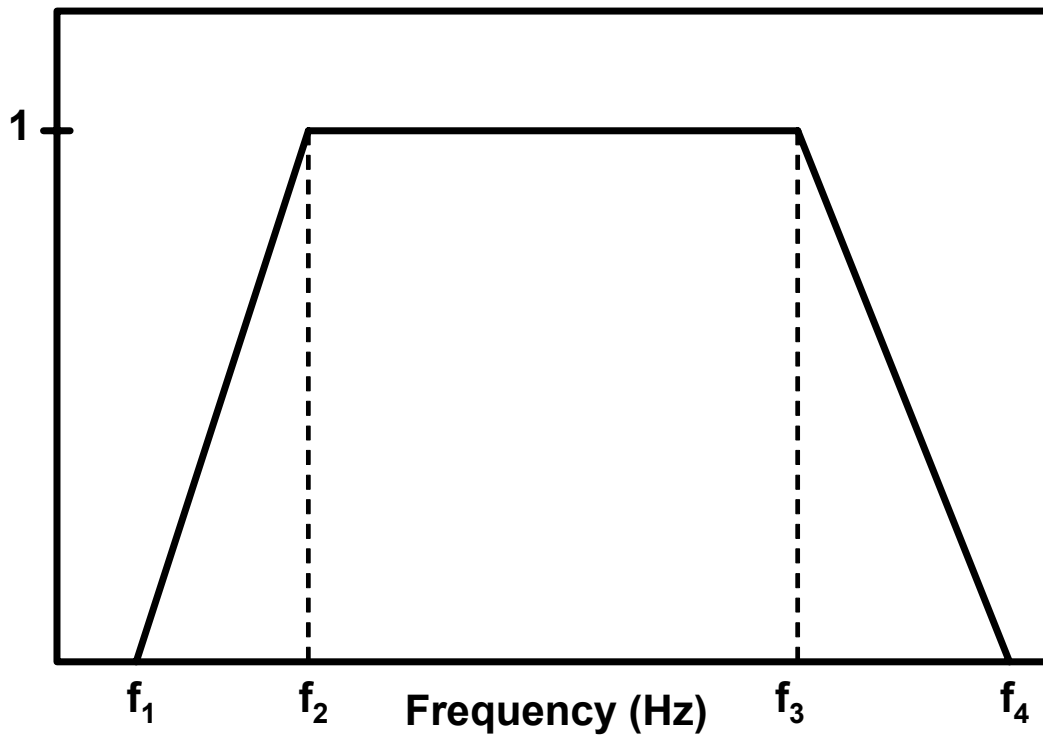


Figure 3.3: Example of a band pass filter specified by four frequencies.

3.8 Normal moveout correction

3.8.1 The traveltime equation

If we consider seismic reflections from a horizontal reflector at depth h (see Figure 3.4) the relationship between two-way traveltime (t) and offset (x) is given by:

$$t^2 = \frac{x^2 + 4h^2}{v^2} \quad (3.34)$$

where v is the velocity.

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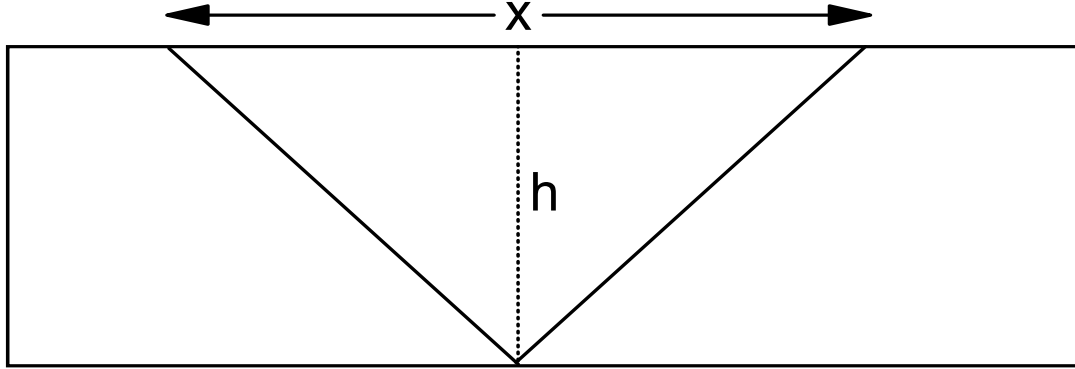


Figure 3.4: Geometry used for traveltime calculations for a horizontal seismic reflector.

If the two-way vertical traveltime ($t_0 = 2h/v$) is introduced in the above formula one obtains

$$t^2 = \frac{x^2}{v^2} + t_0^2 \quad (3.35)$$

If we assume that $2h \gg x$ the two-way traveltime can be expressed as

$$t = t_0 \left[1 + \frac{1}{2} \left(\frac{x}{vt_0} \right)^2 - \frac{1}{8} \left(\frac{x}{vt_0} \right)^4 + \dots \right] \quad (3.36)$$

From equation 3.36 we see that to the lowest order, the normal moveout (NMO) is given by

$$\Delta t_{NMO} = \frac{1}{2} \frac{x^2}{v^2 t_0} \quad (3.37)$$

There are various expressions and approximations for the normal moveout, and in recent years non-hyperbolic NMO-corrections have been used more frequently, especially in connection with studies of far offset seismic stacks.

3.8.2 NMO stretch

When the NMO correction formula is applied to seismic data (by for instance using equation 3.36) the seismic wavelet will be stretched due to the fact that different parts of the wavelet will

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be shifted slightly different. This effect is illustrated in Figure 3.5, where we observe that a seismic wavelet of length τ is stretched.

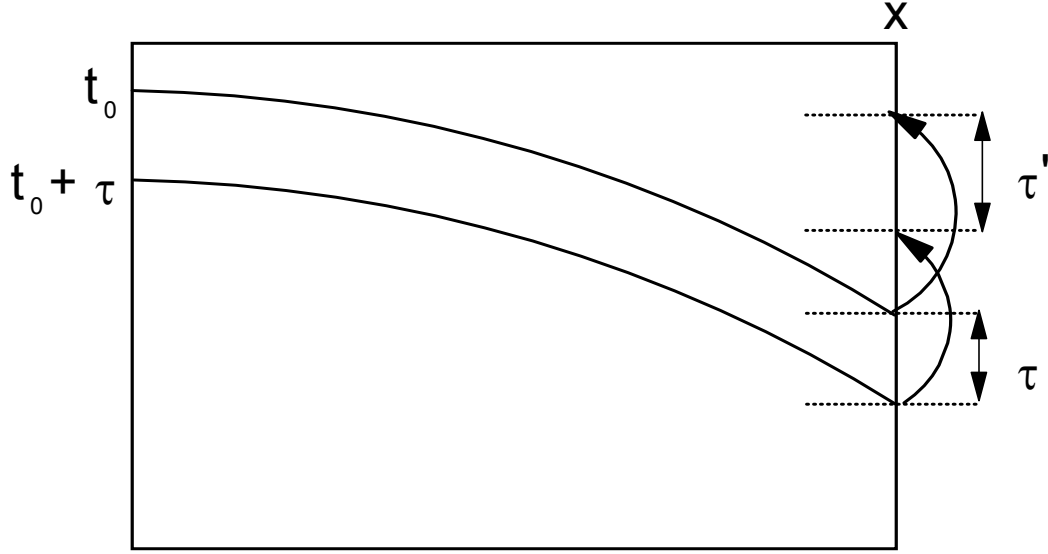


Figure 3.5: Illustration of NMO Stretch: A seismic wavelet becomes somewhat longer after NMO-correction has been applied.

Using the lowest order NMO-correction formula given in equation 3.37 and assuming the velocity to be constant, we obtain the following NMO-corrections for the start (onset) and end of the seismic wavelet for a given offset x :

$$\Delta t_{NMO}^1 = \frac{1}{2} \frac{x^2}{v^2 t_0} \quad (3.38)$$

$$\Delta t_{NMO}^2 = \frac{1}{2} \frac{x^2}{v^2 (t_0 + \tau)} \quad (3.39)$$

Assuming that the pulse width τ is much smaller than t_0 , we obtain an approximate expression for the pulse width after NMO-correction:

$$\tau' = \tau + \frac{1}{2} \frac{x^2 \tau}{v^2 t_0^2} \quad (3.40)$$

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The relative NMO-stretch can therefore be approximated by

$$\frac{\Delta\tau}{\tau} = \frac{\Delta t_{NMO}}{t_0} \quad (3.41)$$

A pulse-broadening, as given in equation 3.41 will of course also introduce a frequency shift towards lower frequencies. The relative frequency shift is given by

$$\frac{\Delta f}{f} = \frac{\Delta\tau}{\tau} \quad (3.42)$$

Since the NMO-correction increases with offset, and decreases with depth, the problem of NMO-stretch is most pronounced for shallow reflectors and far offsets.

3.9 Dip moveout correction

3.9.1 Traveltime equation for a dipping reflector

For a dipping reflector, the traveltime equation becomes somewhat more complicated, since the geometry of the raypaths become asymmetrical (Figure 3.6).

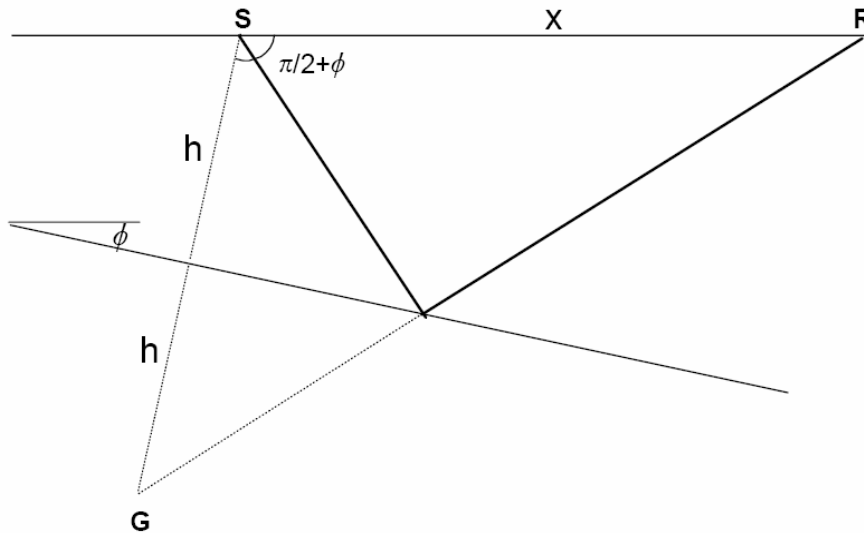


Figure 3.6: Geometry for a dipping reflector, including the source “ghost” point G.

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Using the cosine law to the triangle SGR (and $\cos\left(\frac{\pi}{2} + \varphi\right) = -\sin\varphi$), we obtain

$$v^2 t^2 = x^2 + 4h^2 + 4hx \sin\varphi \quad (3.43)$$

The traveltime equation can then be written

$$t = \frac{2h}{v} \sqrt{1 + \frac{x^2 + 4hx \sin\varphi}{4h^2}} \quad (3.44)$$

The traveltime difference between a downdip receiver with offset Δx and an updip receiver with offset $-\Delta x$ is approximately (after Taylor expansion of equation 3.44)

$$\Delta t_{dip} = \frac{2\Delta x}{v} \sin\varphi \quad (3.45)$$

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3.9.2 Shifted reflection point caused by dip

In addition to the traveltimes shift caused by the dip of the reflector, one also has to take into account that the reflection point is shifted with respect to the common midpoint between source and receiver, as shown in Figure 3.7.

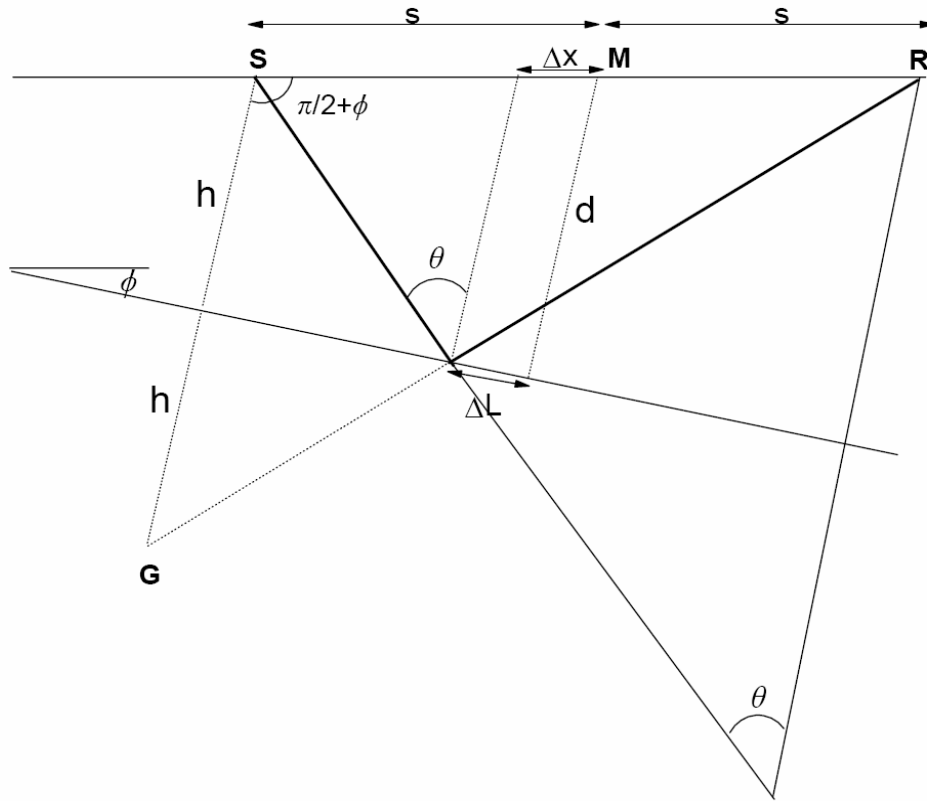


Figure 3.7: Shows how the actual reflection point is shifted when the seismic signal is reflected from a dipping reflector.

In order to find how much the true reflection point has moved updip compared to the common midpoint projection on the dipping layer, ΔL , one use the following intermediate steps:

$$h = d - s \sin \phi \quad (3.46)$$

$$\Delta L = s \cos \phi - h \tan \theta \quad (3.47)$$

After some tedious calculations, it is possible to obtain the following expression for $\tan \theta$:

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$$\tan \theta = \frac{s \cos \varphi}{d} \quad (3.48)$$

Inserting equations (3.46) and (3.48) into equation (3.47) yields

$$\Delta L = \frac{s^2}{d} \sin \varphi \cos \varphi \quad (3.49)$$

which is the well-known formula derived by Levin in 1971. In DMO-processing, common-reflection-point gathers are created, so one might say that DMO-processing involves traveltime corrections and a geometrical shifting of traces according to equation (3.49). Introducing the zero offset two way traveltime in the following way

$$T_0 = 2 \frac{d}{v} \quad (3.50)$$

we find that equation 3.43 can be simplified to (using that $d = h + s \sin \varphi$)

$$t^2 = T_0^2 + \frac{4s^2 \cos^2 \varphi}{v^2} \quad (3.51)$$

This equation can be used as an alternative to equation 3.44.

3.10 Velocity analysis

3.10.1 The velocity spectrum

The main objective in velocity analysis is to obtain stacking velocities that can be used in a normal moveout correction algorithm prior to stacking or summing the seismic traces belonging to a certain CMP-gather. The semblance function is commonly used as a velocity spectrum in order to ease the picking of velocities suited for stacking:

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$$SE(t) = \frac{\sum_t \left[\sum_{i=1}^M f_{i,t(i)} \right]^2}{\sum_t \sum_{i=1}^M f_{i,t(i)}^2} \quad (3.52)$$

where $f_{i,t(i)}$ represents the seismic amplitude of the i -th trace within a CMP-gather of totally M traces, for the two-way travelttime $t(i)$. The two-way travelttime is connected to the stacking velocity (v_{st}) in the usual way:

$$t(i) = \sqrt{t^2(0) + \frac{x_i^2}{v_{st}^2}} \quad (3.53)$$

where x_i is the offset for trace i . A typical semblance plot for a CMP-gather is shown in Figure 3.8.

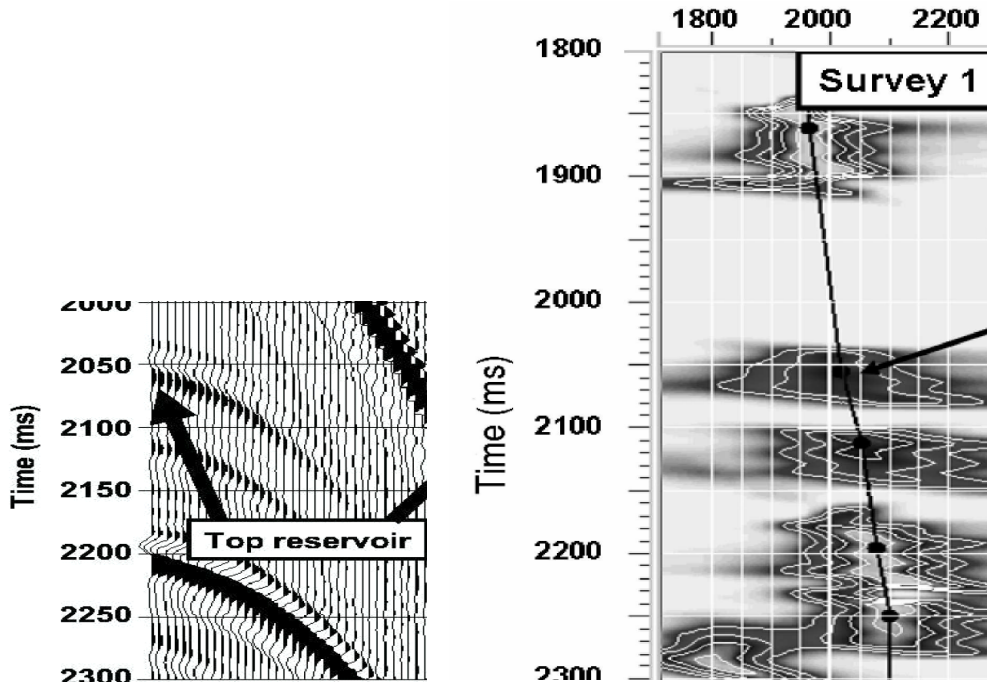


Figure 3.8 Seismic section (left) and corresponding semblance plot (right). The black solid line shows the picked velocities, and white lines represent constant-velocity levels.

3.10.2 Velocity picking

In practice other seismic events than primaries may contribute to high semblance values in the velocity spectrum. For instance, multiples will also give high semblance values. Therefore a certain part of the velocity picking cannot be done automatically, an interpretation procedure is

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needed in order to pick optimal stacking velocities. In areas where we know that the typical velocity trend is an increase with depth, one can choose to interpret high velocity values on the semblance contour map to be associated with primary events, while low velocity values can be assumed to be related to multiples. However, in other areas we might have velocity decreases with depth, and then it might be much more difficult to separate between primaries and multiples in the velocity picking.

In a normal 3D seismic processing flow, it is not normal to analyze and pick stacking velocities for every CMP-gather. A typical choice is to do velocity analysis every 500 metres. In complex areas one might of course choose a denser velocity grid. Typical guidelines used in velocity picking is that an increase in stacking velocity (with depth) is more probable than a decrease, interval velocities less than 1430 m/s and greater than 6700 m/s are unlikely. In reprocessing projects where the quality of the final stacked data is improved, one often finds that repicking of stacking velocities is an important contributor to the improved results. In interactive processing, it is straightforward to check the quality of the velocity picking by checking the flatness of the CMP-gather after NMO or DMO correction. If one is not satisfied with the alignment of certain events after NMO, one can go back and change the stacking velocity in order to optimize the flatness of the specific event.

3.11 Stacking, angle stacks

After NMO-correction of a CMP-gather, the primary events should ideally be horizontal well-aligned events, and a direct sum of the traces within a family should increase the signal to noise ratio considerably. This increased signal to noise ratio is indeed one of the major objectives for stacking. Another objective is multiple attenuation. Normally, water layer multiples have smaller stacking velocities than primaries, causing the multiples not to align very well on a NMO-corrected gather, and consequently reduced amplitude level on the final stacked (or summed) trace.

It is common to assume that each trace in a CMP-gather contain the same degree of information content about the subsurface, and thus each trace are given the same weight in the stacking process. There are, however, often reasons for applying a non-uniform weight function to the traces prior to stacking, as for instance putting more weight to the mid-streamer traces. This procedure is referred to as weighted stacking.

3.11.1 Muting

A real seismic CMP-gather often contains unwanted signals for near offsets and short recording times. This is shown in Figure X, where strong noisy signals distorts the reflected signals we ideally want to process. These noise signals are very often associated with strong mode conver-

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sions at the sea bottom, water bottom reverberations etc. Very often one also finds that the hydrophone groups nearest to the seismic vessel, and perhaps also those at the far end of the streamer are more noise contaminated than the other traces being recorded. Therefore, it is normal to zero out or mute such areas within a CMP-gather before the stacking process. Since these unwanted signals very often have high amplitudes, the effect of not applying muting prior to stacking is dramatic. It is essential for achieving good seismic data (especially for shallow events) to apply a proper mute prior to stacking.

3.12 Seismic Modelling

Seismic modelling is a process where we start with a given geological model of the subsurface, including geometry of subsurface layers and acoustic properties of the layers. Simulation is perhaps a more correct word than modelling for this discipline, since we talking about simulating (according to specific equations) how seismic waves propagate through the earth. In order to perform seismic modelling we have to simplify both the subsurface and the wave propagation procedure. It is impossible to create a geological model that includes all details and all properties of the real subsurface, we are forced into dividing the earth into layers or separated volumes that we can assign approximately the same properties to. If we specify only density and P-wave velocity for each volume (or layer) in our model, we have the acoustic case. In such a case we also assume that the earth is isotropic, that is, no changes in parameter values with respect to wave propagation direction. If the S-wave velocity is also specified for each volume of our geological model, we are talking about elastic modelling. Acoustic and elastic modelling are two major types of seismic modelling, and in seismic processing, acoustic modelling is most frequently used, as a basis for the seismic migration process. A model that includes anisotropy is referred to as for instance elastic anisotropic modelling, or similarly if absorption effects are taken into account the term visco-elastic modelling might be used.

Physical experiments made on small scaled synthetic models are also often referred to as seismic modelling, since a scaled version of a real seismic experiment is being modelled. Such experiments are useful both as a way to test different processing algorithms on a known earth model, and as a way to calibrate and check the numerical schemes used to simulate seismic wave propagation through the earth.

There are several motivations for performing seismic modelling:

- 1) To compare and distinguish between various possible/probable geological models. Both comparison between the different models, and direct comparison between each model and the seismic data acquired over the specific location.

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- 2) For well tie and well calibration purposes. It is important to understand how the seismic data relates to lithology and saturation changes within a well, in order to interpret the seismic data properly between the well locations.
- 3) Wave extrapolation methods are essential in seismic migration.
- 4) Most feasibility studies in connection with a time lapse seismic study involve seismic modeling of pre and post-production scenarios, often based on fluid flow simulations.

3.12.1 The wave equation

For an elastic, isotropic medium the wave equation can be written:

$$\rho(x) \frac{\partial^2 u_i}{\partial t^2} = \frac{\partial \sigma_{ij}}{\partial x_j} + f_i \quad (3.54)$$

where ρ is the density, u_i is the particle displacement in direction i , f_i is the external force (i.e. source), and σ_{ij} is the stress tensor. t and x refer to time and position, respectively. In addition to the wave equation, one needs a constitutive relation between the stress tensor and the particle displacement. This relation is often referred to as Hookes law, and using the Lamé parameters λ , μ , this law can be written (assuming elastic isotropic earth)

$$\sigma_{ij} = \lambda \delta_{ij} \frac{\partial u_k}{\partial x_k} + \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (3.55)$$

For the acoustic case ($\mu = 0$) and denoting the isotropic stress $P = -\lambda \frac{\partial u_k}{\partial x_k}$ these equations simplify to (skipping the force term in equation (3.54) and assuming constant density)

$$\sigma_{ij} = -P \delta_{ij} \quad (3.56)$$

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$$\frac{\partial^2 u_i}{\partial t^2} = \frac{1}{\rho} \frac{\partial \sigma_{ij}}{\partial x_j} \quad (3.57)$$

Performing spatial derivation ($\frac{\partial}{\partial x_i}$) on both sides of equation (3.57) and inserting for pressure instead of stress (according to equation (3.56)) yields:

$$\frac{\partial^2 P}{\partial t^2} = c^2 \nabla^2 P \quad (3.58)$$

which is the scalar acoustic wave equation, and where we have introduced the P-wave velocity $c = \sqrt{\frac{\lambda}{\rho}}$ instead of the Lamé parameters. There are several methods available for solving these wave equations, and we will briefly describe each method in the following sections.

3.12.2 Convolutional modelling

This is the simplest way of performing seismic modelling, and it is often used to model a seismic trace representing the response at a given well location. The basic assumption is that a stacked seismic trace for a given location can be written as a convolution between a representative wavelet and a series of reflection coefficients representing the major subsurface interfaces, see for instance equation (3.28). Such a convolution is computationally efficient, and it is straightforward to obtain a seismic trace this way. The shortcomings are of course the extremely simplifications built into this approach, a one-dimensional earth model, no multiples etc.

3.12.3 Frequency-wavenumber modelling

If our model of the subsurface can be described as a composition of plane, horizontal layers only, a solution of the wave equation can be performed in the frequency-wavenumber domain.

3.12.4 Raytracing

One solution of the acoustic wave equation (3.58) can be written as

$$p(t, x) = \frac{1}{R} p_0 \left(t - \frac{R}{c} \right) \quad (3.59)$$

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where R is the distance from the source to the observation point and c is the P-wave velocity. For an inhomogeneous medium it is usual to assume that the solution is of the form

$$p(t, x) = A(x)p_0(t - \tau(x)) \quad (3.60)$$

where A represents amplitude variations and τ the traveltime. After temporal Fourier-transformation of the wave equation one obtains

$$\nabla^2 P + \left(\frac{\omega}{c}\right)^2 P = 0 \quad (3.61)$$

with the well known solution

$$P(\omega, x) = P_0(\omega)A(x)e^{i\omega\tau(x)} \quad (3.62)$$

In the above equations ω denotes the angular frequency. Using equation (3.62) and inserting into equation (3.61) one obtains the following equation

$$\nabla^2 A + 2i\omega \nabla A \nabla \tau + i\omega A \nabla^2 \tau - \omega^2 A (\nabla \tau)^2 + \left(\frac{\omega}{c}\right)^2 A = 0 \quad (3.63)$$

Treating real and imaginary parts of the above equation independently results in these two equations

$$2\nabla A \nabla \tau + A \nabla^2 \tau = 0 \quad (3.64)$$

$$\nabla^2 A - \omega^2 A (\nabla \tau)^2 + \left(\frac{\omega}{c}\right)^2 A = 0 \quad (3.65)$$

For high frequencies ($\omega \rightarrow \infty$) the last equation boils down to the Eikonal-equation:

$$(\nabla \tau)^2 = \frac{1}{c^2} \quad (3.66)$$

This equation is the basis for kinematic raytracing, and the solution for the traveltimes are calculated along raypaths (kinematic raytracing). The amplitudes (often referred to as dynamic ray tracing) are given by equation (3.64). At the interface between two layers, seismic waves will be reflected and refracted (transmitted). In the high frequency limit the angle of incidence and reflection/refraction angles are related by Snells law

$$\frac{\sin \vartheta_I}{c_I} = \frac{\sin \vartheta_R}{c_R} = \frac{\sin \vartheta_T}{c_T} \quad (3.67)$$

where I, R and T denote incidence ray, reflected and refracted rays, respectively. In a layered medium consisting of homogenous layers with constant velocities within each layer, the ray paths will be straight lines. Using Snells law to compute the ray bending at each interface, we are then able to follow a specific ray down to a specified interface, and back to the receiver. In this way it is possible to obtain a series of rays, representing various raypaths and reflection/transmission histories. A schematic illustration of the principles of kinematic raytracing is shown in Figure 3.9.

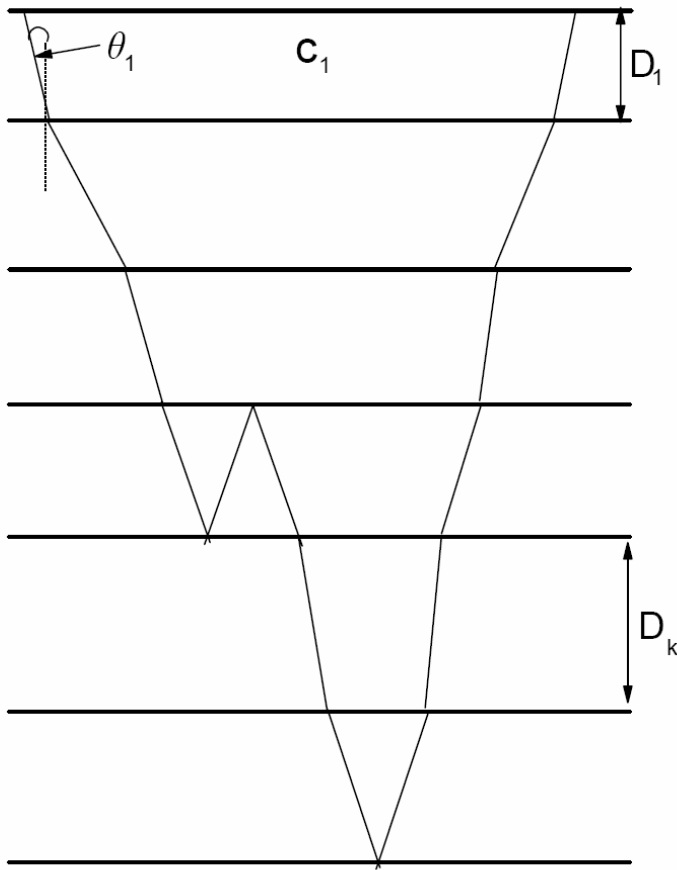


Figure 3.9: Schematic illustration of raytracing principles. At each interface the ray is either reflected or refracted (transmitted). All reflection and transmission angles can be calculated when the emission angle from the source is known, and the interval velocities within each layer. This example shows a ray that has been multiple-reflected.

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3.12.5 Finite difference modelling

For simplicity we study the 2-D acoustic wave equation (eq. 3.58) with a source term $s(t)$ included:

$$\frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2 P}{\partial t^2} + s(t) \delta(x - x_s) \delta(z - z_s) \quad (3.68)$$

In order to solve the above given equation we discretize in both time and space

$$t = n\Delta t \quad (3.69)$$

$$x = i\Delta x \quad (3.70)$$

$$z = j\Delta z \quad (3.71)$$

A Taylor series expansion of the second derivative of the pressure with respect to time yields:

$$\frac{\partial^2 P}{\partial t^2} \cong \frac{P^{n+1} - 2P^n + P^{n-1}}{(\Delta t)^2} \quad (3.72)$$

where $P^{n+1} = P((n+1)\Delta t)$ etc. If we choose to use a second order Taylor expansion for evaluating the spatial derivatives we obtain

$$\frac{\partial^2 P}{\partial x^2} \cong \frac{P^n(i+1, j) - 2P^n(i, j) + P^n(i-1, j)}{(\Delta x)^2} \quad (3.73)$$

for spatial differentiation in the x-direction, and similarly for the z-direction

$$\frac{\partial^2 P}{\partial z^2} \cong \frac{P^n(i, j+1) - 2P^n(i, j) + P^n(i, j-1)}{(\Delta z)^2} \quad (3.74)$$

Using these approximations, the wave equation can now be written

$$P^{n+1}(i, j) = 2P^n - P^{n-1} + (c(i, j)\Delta t)^2 \left\{ \frac{P^n(i+1, j) - 2P^n(i, j) + P^n(i-1, j)}{(\Delta x)^2} + \frac{P^n(i, j+1) - 2P^n(i, j) + P^n(i, j-1)}{(\Delta z)^2} - s^n(i, j) \right\} \quad (3.75)$$

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Given the source time function ($s(t)$) and its location (x_s, z_s), and the boundary conditions the above given equation can be solved if the velocity field ($c(i, j)$) also is known. The boundary conditions might for instance be that the pressure at the free surface is zero:

$$P(x, z=0, t) = 0 \quad (3.76)$$

and that the pressure is zero for time less than zero. When a seismic wave hits the edge of the finite difference grid, it will be reflected back into the grid again, and this will cause unwanted artefacts. In order to avoid this problem, one often surrounds the finite difference model grid with absorbers (not at the free surface of course, since reflections from the free surface are part of a real seismic experiment, and should therefore be included in the modelling). The effect of these absorbing layers surrounding the model is to attenuate the seismic energy so that the reflected waves are practically zero in amplitude when they are reflected at the edge of the finite difference grid.

Another problem to be aware of when using finite difference modelling is grid dispersion. Dispersion means that the phase speed of a wave is a function of the frequency, that is that for instance high frequencies travel faster than low frequencies. In order to analyze temporal dispersion we can use a one-dimensional wave equation:

$$\frac{\partial^2 P}{\partial t^2} = c_0^2 \frac{\partial^2 P}{\partial x^2} \quad (3.77)$$

and inserting an harmonic solution for the pressure

$$P \sim e^{i(\omega t - kx)} \quad (3.78)$$

into equation 3.77 yields

$$\omega^2 = c_0^2 k^2 \quad (3.79)$$

which means that the phase speed ($\frac{\omega}{k}$) equals the velocity of sound. However when we use a finite difference solution to equation 3.77

$$\frac{P^{n+1} - 2P^n + P^{n-1}}{(\Delta t)^2} = c_0^2 \frac{\partial^2 P}{\partial x^2} \quad (3.80)$$

and insert the harmonic solution we get

$$\frac{1}{(\Delta t)^2} [e^{i\omega\Delta t} - 2 + e^{-i\omega\Delta t}] = -c_0^2 k^2 \quad (3.81)$$

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The phase speed is equal to $\frac{\omega}{k}$, and after Taylor expansion of the left side of this equation we obtain

$$\frac{c^2}{c_0^2} = 1 + \frac{(\omega\Delta t)^2}{12} \quad (3.82)$$

which means that high frequencies travel faster than low frequencies, and that the amount of temporal grid dispersion is also proportional to the square of the time step used in the modeling. Similarly we can show that the spatial grid dispersion is given as

$$\frac{c^2}{c_0^2} = 1 - \frac{(k\Delta x)^2}{12} \quad (3.83)$$

Since a computer uses a limited number of digits to represent a certain number, there is always a danger for instabilities caused by such inaccuracies. In order to avoid numerical instabilities one should assure that the ratio between temporal sampling and spatial sampling is limited. For the 2nd order scheme discussed here it can be shown that one should choose this ratio in this way:

$$\frac{c\Delta t}{\Delta x} \leq \frac{1}{\sqrt{2}} \quad (3.84)$$

Finally, one should also be aware of that there is a lower number of how many grid points you should use per shortest wavelength. Typical values for fine grid methods is 20, while coarse grid methods (using higher order differential operators) can use only 4-5 gridpoints per shortest wavelength.

3.13 Migration

3.13.1 Introduction

There are several objectives for applying a seismic migration operator to the data, and perhaps the most important one is to move reflectors to their “true” position. All migration algorithms need a velocity model as input, and the quality of the migration velocities has a significant impact on the final migrated images. Another major reason for doing migration is to collapse and focus diffractions. It is also commonly recommended to apply prestack migration prior to AVO analysis or prestack inversion, in order to obtain a variation with offset that is more correct.

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In short one can say that migration is the art of reversing wave propagation effects in order to obtain clear images of the subsurface. The first step in most migration algorithms is therefore wave propagation. The wave propagation step or wavefield extrapolation can be approximated in several ways, and numerous schemes representing various approximations to a true wavefield extrapolation exists. The next and final step in migration is the imaging step which usually involves some kind of cross-correlation of two extrapolated wavefields, and is usually much simpler than the wavefield extrapolation step. It is, however, important to stress that the most challenging and difficult step in seismic migration is the determination of the migration velocities. The main four types of migration algorithms can be listed as follows:

- Post-stack time migration
- Post-stack depth migration
- Pre-stack time migration
- Pre-stack depth migration

The main difference between time and depth migration schemes is that after time migration the images are displayed in two-way traveltimes, and that the wavefield extrapolation is done in a time-stepping way. For a depth migration scheme, the wave stepping is done with the depth parameter (z) and the images can be presented in true vertical depth. In the following chapters these four types of seismic migration will be presented in further detail.

3.13.2 *Post-stack migration*

The most common model used in post-stack migration is the exploding reflector model (ERM) which is shown in Figure 3.10. In the exploding reflector model a reflection point is regarded as a source point, and the rays emerging from this point travels with half the velocity of the corresponding zero-offset experiment. The basis for post stack migration algorithms is the assumption that post-stack seismic data is equal to (or approximately equal to) zero-offset data. The first migration step is to extrapolate the wavefield measured at the recording surface downwards:

$$P(x, z_n, t) = f(P(x, z = 0, t)) \quad (3.85)$$

followed by an imaging condition

$$I(x, z) = P(x, z_n, t = 0) \quad (3.86)$$

which simply means that one assumes that reflectors explode at time zero. These two steps are then repeated for all depth points in the model.

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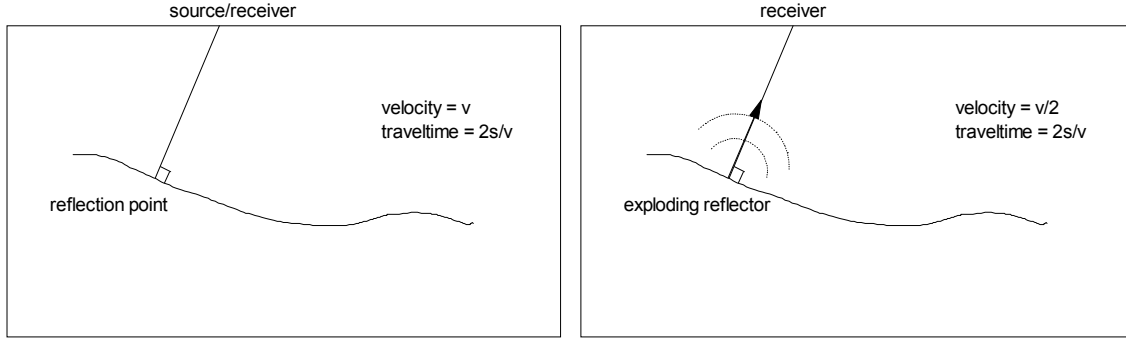


Figure 3.10: Typical zero-offset experiment (left) and the corresponding exploding reflector model (ERM) (right). Notice that the distance travelled by the ray is halvened in the ERM model, and this is compensated by choosing the velocity equal to the half of the true velocity .

3.13.3 The migration equation

For a constant velocity model as shown in Figure 3.11, we observe that a dipping interface is imaged with a wrong dip on an unmigrated section. From the figure we can derive the relation between the true dip angle θ and the apparant dip angle on the unmigrated section θ' , and the result is

$$\sin \theta = \tan \theta' \quad (3.87)$$

Since θ is larger than θ' this means that the migration process will move dipping events up-dip, so that the reflector becomes steeper. The above equation is often referred to as the migration equation (i.e. the zero offset migration equation). We also see from Figure 3.11 that the migration process changes the two way traveltime (t' denotes unmigrated traveltime)

$$t = \frac{t'}{\cos \theta'} \quad (3.88)$$

which means that the twoway traveltime increases after migration. For zero dip we see that the migration equation leaves the traveltime unchanged ($t = t'$). As an example we see that for an interface with a dip angle of 15 degrees, the change in twoway traveltime between unmigrated

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and migrated seismic profiles is 3.4%, which corresponds to several milliseconds for a reflector at a reasonable depth.

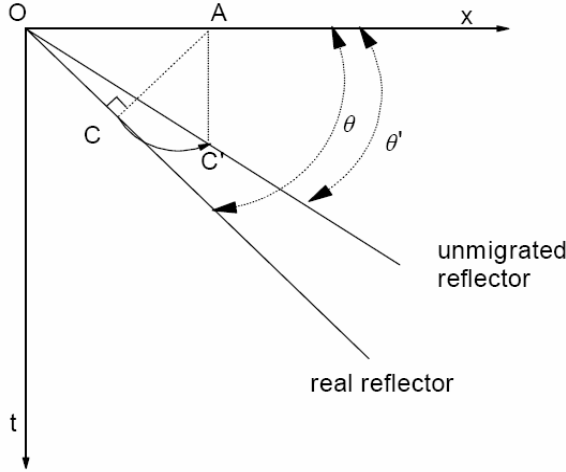


Figure 3.11: Migration in a constant velocity model. The real reflection point C appears to be at position C' on an unmigrated section. A correct migration operator will move the point C' back to the correct position C again. Note that $AC=AC'$.

3.13.4 Wavefield extrapolation

We consider the 2D acoustic wave equation (3.58), and assume for simplicity that the velocity is constant:

$$\frac{1}{c^2} \frac{\partial^2 P}{\partial t^2} = \frac{\partial^2 P}{\partial x^2} + \frac{\partial^2 P}{\partial z^2} \quad (3.89)$$

After Fourier transformation in x and t we obtain

$$\frac{\partial^2 P}{\partial z^2} + \left(\frac{\omega^2}{c^2} - k_x^2 \right) P(z, k_x, \omega) = 0 \quad (3.90)$$

where k_x is the horizontal wavenumber and ω is the circular frequency. Defining the vertical wavenumber as

$$k_z = \sqrt{\frac{\omega^2}{c^2} - k_x^2} \quad (3.91)$$

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the solution to the above equation can be written

$$P(z) = e^{\pm i k_z z} P(0) \quad (3.92)$$

where the + solution represents downgoing waves and the - solution represents upgoing waves. From equation 3.92 we obtain

$$\frac{\partial P}{\partial z} = \pm i k_z P(z) \quad (3.93)$$

which is often referred to as the one way wavefield extrapolation equation. If the velocity is not constant, but varies with depth, that is $c = c(z)$, it means that k_z is depth dependent, and that the solution can be written in the following way:

$$P(z) = e^{\pm i \int_0^z k_z(z) dz} P(0) \quad (3.94)$$

3.13.5 Phase shift migration

Phase shift migration was first developed by Gazdag in 1978, and the basic assumptions are plane layered earth and zero offset (or stacked) data. Furthermore, the exploding reflector model is used, and it is assumed that the data has been transformed to the (ω, k_x) domain. The wavefield extrapolation is done using the following equation

$$P((n+1)\Delta z, k_x, \omega) = e^{-i \sqrt{\frac{\omega^2}{c_{ERM}^2} - k_x^2} \Delta z} P(n\Delta z, k_x, \omega) \quad (3.95)$$

where c_{ERM} is the velocity used in the exploding reflector model (i.e. half of the true velocity). This extrapolation equation is denoted phase shift, because it is only a phase shift for each depth step. Notice that the velocity model may vary with depth ($c_{ERM} = c_{ERM}(z)$). The imaging condition is as for the ERM model:

$$I((n+1)\Delta z, k_x, \omega) = P((n+1)\Delta z, k_x, t=0) \quad (3.96)$$

Using the inverse Fourier transform to convert back to time again yields (and setting $t=0$)

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$$I((n+1)\Delta z, k_x, \omega) = \frac{1}{2\pi} \int d\omega P((n+1)\Delta z, k_x, \omega) \quad (3.97)$$

Extending this equation to the x-direction also, yields

$$I(x, z) = \frac{1}{(2\pi)^2} \int d\omega dk_x e^{ik_x x} P((n+1)\Delta z, k_x, \omega) \quad (3.98)$$

3.13.6 Stolt migration

Stolt migration (1978) is often called fk-migration. When c_{ERM} is constant, the phase shift operation (eq. 3.95) can be done in one step

$$P(z, k_x, \omega) = e^{-i \sqrt{\frac{\omega^2}{c_{ERM}^2} - k_x^2} z} P(z=0, k_x, \omega) \quad (3.99)$$

Using the dispersion relation in equation 3.91 we obtain

$$d\omega = c_{ERM} \frac{k_z dk_z}{\sqrt{k_x^2 + k_z^2}} \quad (3.100)$$

Using this relation in equation 3.98 yields the following relation

$$I(x, z) = \frac{1}{(2\pi)^2} \int dk_x dk_z e^{i(k_x x - k_z z)} \frac{c_{ERM} k_z}{\sqrt{k_x^2 + k_z^2}} P(z=0, k_x, c_{ERM} \sqrt{k_x^2 + k_z^2}) \quad (3.101)$$

Thus Stolt migration involves two operations in the fk-domain, first a mapping of ω into k_z and a scaling of the amplitudes with the factor $(c_{ERM} k_z) / (\sqrt{k_x^2 + k_z^2})$.

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3.13.7 Post stack Kirchhoff migration

The Helmholtz equation (3.61) including a source term and using the dispersion relation ($k = \omega/c$) reads

$$[\nabla^2 + k^2]P(x, \omega) = -S(x, \omega) \quad (3.102)$$

The corresponding Greens function (G) for the Helmholtz equation is defined by replacing the source term with Dirac's delta function

$$[\nabla^2 + k^2]G(x|x', \omega) = -\delta(x - x') \quad (3.103)$$

The Greens function can be regarded as the impulse response of the subsurface (since the source term is not included). Green's formula for a closed surface S

$$\int_V dV \nabla \underline{E} = \int_S dS \underline{n} \underline{E} \quad (3.104)$$

where underscore means vectors, and $\underline{n}$ is a unit vector pointing outwards from the surface S. Using Greens theorem and letting $\underline{E} = G \nabla P - P \nabla G$ yields

$$\int_V dV (G \nabla^2 P - P \nabla^2 G) = \int_S dS \underline{n} (G \nabla P - P \nabla G) \quad (3.105)$$

Using the definition of G and P (equations 3.102 and 3.103) yields

$$P(\underline{x}, \omega) = \int_V dV' G(\underline{x}|\underline{x}', \omega) S(\underline{x}', \omega) + \int_S dS' [G(\underline{x}|\underline{x}', \omega) \nabla' P(\underline{x}', \omega) - \nabla' G(\underline{x}|\underline{x}', \omega) P(\underline{x}', \omega)] \quad (3.106)$$

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which is known as the Kirchhoff integral. If we consider the case where we put the recording sur-

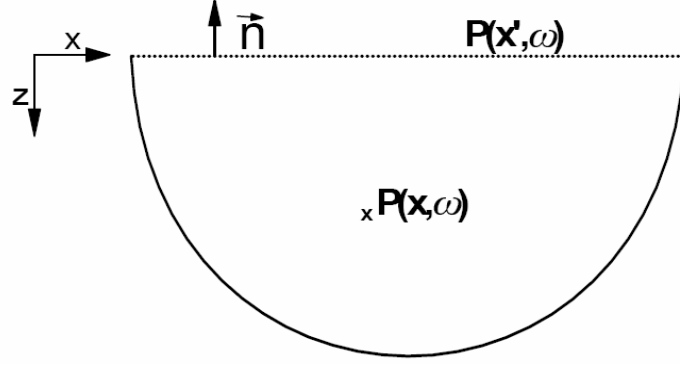


Figure 3.12: Shows configuration of receivers (dotted line) and how the surface S is chosen in order to compute the wavefield at position x for the two-dimensional case.

face outside the source, as sketched in Figure 3.12, we obtain the following expression for the pressure at the position x by using the Kirchhoff integral given in equation 3.106:

$$P(\underline{x}, \omega) = \int_S dS' [G(\underline{x}|\underline{x}', \omega) \nabla' P(\underline{x}', \omega) - \nabla' G(\underline{x}|\underline{x}', \omega) P(\underline{x}', \omega)] \quad (3.107)$$

Letting $\underline{x} = (x, y, z)$, and skip the y -direction for simplicity, we get (since the unit vector points in opposite direction to the z -direction

$$P(x, \omega) = - \int_{-\infty}^{\infty} dx' \left[G \frac{\partial P}{\partial z'} - \frac{\partial G}{\partial z'} P \right] \quad (3.108)$$

where we have assumed that the contribution along the half-circle vanishes when the radius approaches infinity. In the above expression, we know the Greens function, and the pressure is measured at or close to the free surface. The length of the recording surface in the x -direction is not infinity, so we have to assume that the contribution outside our recording surface is negligible. However, $\frac{\partial P}{\partial z'}$ is not known in general. For marine seismic acquisition one could in principle use two separate streamers and tow them over each other, so that we could compute the lowest order approximation to the vertical gradient of the pressure, but in general we must assume that such measurements are not available. For an upgoing wavefield it can be shown that

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$$G \frac{\partial P}{\partial z'} = -P \frac{\partial G}{\partial z'} \quad (3.109)$$

and hence

$$P(x, \omega) = 2 \int_{-\infty}^{\infty} dx' P(x', 0, \omega) \frac{\partial}{\partial z'} G(x, z | x', 0, \omega) \quad (3.110)$$

Choosing the free space Greens function for G

$$G(x, z | x', 0) = \frac{1}{4\pi R} e^{\pm i k R} \quad (3.111)$$

$$R = \sqrt{(x - x')^2 + z^2} \quad (3.112)$$

Choosing the minus sign in the exponential function for the Greens function, we find that for large R

$$\frac{\partial G}{\partial z'} = \frac{1}{4\pi} \frac{\cos \theta}{c R} (-i\omega) e^{-i\omega \frac{R}{c}} \quad (3.113)$$

where θ is the angle between the far-field point and the vertical axis, as shown in Figure 3.13.

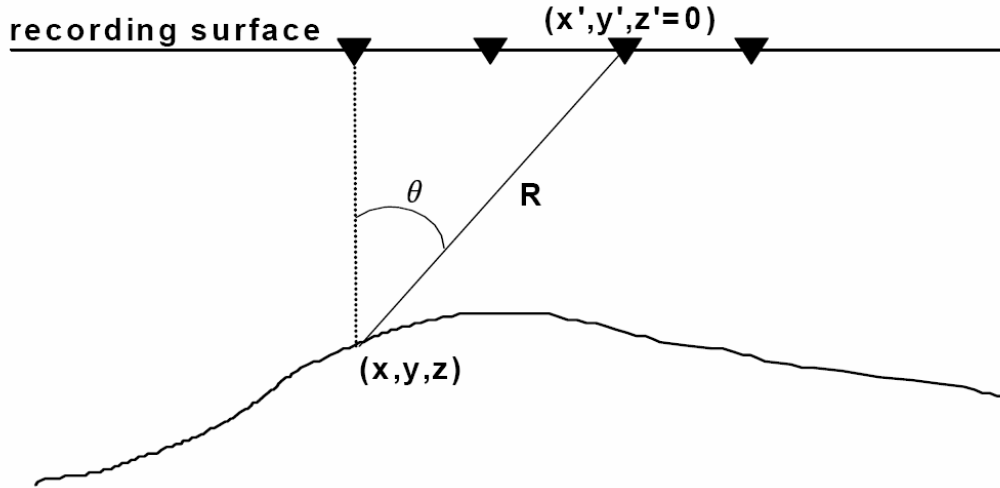


Figure 3.13: Geometry used in Kirchhoff migration: R is the distance between the receiver and the reflection point, and θ the angle between the

Inserting for the derivative of the Greens function yields

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$$P(x, \omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dx' P(x', 0) \frac{\cos \theta}{cR} (-i\omega) e^{-i\omega \frac{R}{c}} \quad (3.114)$$

Transforming the above equation back to time again, and inserting the y-coordinate that we have omitted for practical reasons yields:

$$P(x, y, z, t) = \iint_{-\infty}^{\infty} dx' dy' \frac{\cos \theta}{cR} \frac{\partial}{\partial t} P\left(x', y', 0, t + \frac{R}{c}\right) \quad (3.115)$$

which is usually referred to as the Kirchhoff integral equation for zero offset or post stack migration. In equation 3.115 the velocity $c = c_{ERM} = v_{true}/2$ according to the exploding reflector model. The migration is now done by performing the integration in equation 3.115 and setting $t=0$ for each output location. For post-stack Kirchhoff migration, the z-coordinate is transformed into two-way traveltimes again. Kirchhoff time migration is used when the velocities vary smoothly. It is also often used as a starting point for depth migration to obtain an initial structural model.

3.13.8 Finite difference migration

The basis for finite difference migration is the dispersion relation for the acoustic wave equation (assuming 2D)

$$\frac{\omega^2}{c^2} = k_x^2 + k_z^2 \quad (3.116)$$

For wavefield extrapolation we need an expression for k_z which is equal to

$$k_z = \pm \sqrt{\frac{\omega^2}{c^2} - k_x^2} \quad (3.117)$$

where the plus sign corresponds to downgoing waves and the minus sign corresponds to upgoing waves. Our interest is upgoing waves so we choose the minus sign. Displayed in the 2D wavenumber domain the solution to equation 3.117 is a half circle with radius ω/c , as sketched in Figure 3.14. Using a Taylor series expansion of the square root, assuming that $ck_x/\omega \ll 1$ gives

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$$k_z = -\left[\frac{\omega}{c} - \frac{ck_x^2}{2\omega} + \dots\right] \quad (3.118)$$

The first two terms in the Taylor expansion are also sketched in Figure 3.14, and we see that adding higher order terms in the Taylor expansion improves the fit with the exact solution (the half circle).

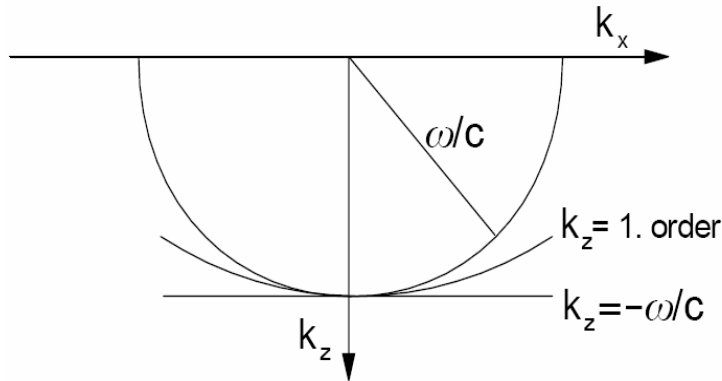


Figure 3.14: Solution of the acoustic dispersion relation in the 2D case is a half circle with radius equal to ω/c . The zero order and first order Taylor expansion solution to the exact relation are also shown.

The lowest order solution is often referred to as the 5 degrees equation, while the first order Taylor expansion is called the 15 degrees equation. If we choose to do the wavefield extrapolation with the 15 degrees equation, the following equation is used

$$P(x, z + \Delta z; \omega) = P(x, z; \omega) - i \left[\frac{\omega}{c} + \frac{c}{2\omega} \frac{\partial^2}{\partial x^2} \right] P(x, z; \omega) \Delta z \quad (3.119)$$

where $c = c_{ERM} = v_{true}/2$. The differentiation with respect to x is then written as a finite difference approximation in the same way as in for instance equation 3.73. Finite difference migration can handle varying velocity fields (both in x and z direction), but the algorithm is slow on a computer. Furthermore, as we have seen it is limited to certain dips, according to which degree of accuracy we choose for the wavefield extrapolation equation.

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3.14 3-D Seismic

3.14.1 Binning of seismic data

In 3-D seismic processing, common mid point gathers are created through a binning process. This means that the x-y surface is divided into cells or bins with a certain size (for instance 12.5 by 12.5 metres). For each source-receiver pair, the midpoint position is computed, and assigned to a specific bin. An example of source-receiver pairs and corresponding seismic data is shown in Figure 3.15. If one bin cell lacks a specific offset range, one way to circumvent the problem is to borrow a trace from the neighbouring bin cell. This is often referred to as flexi-binning. Another way to compensate for lack of traces within a bin cell is interpolation.

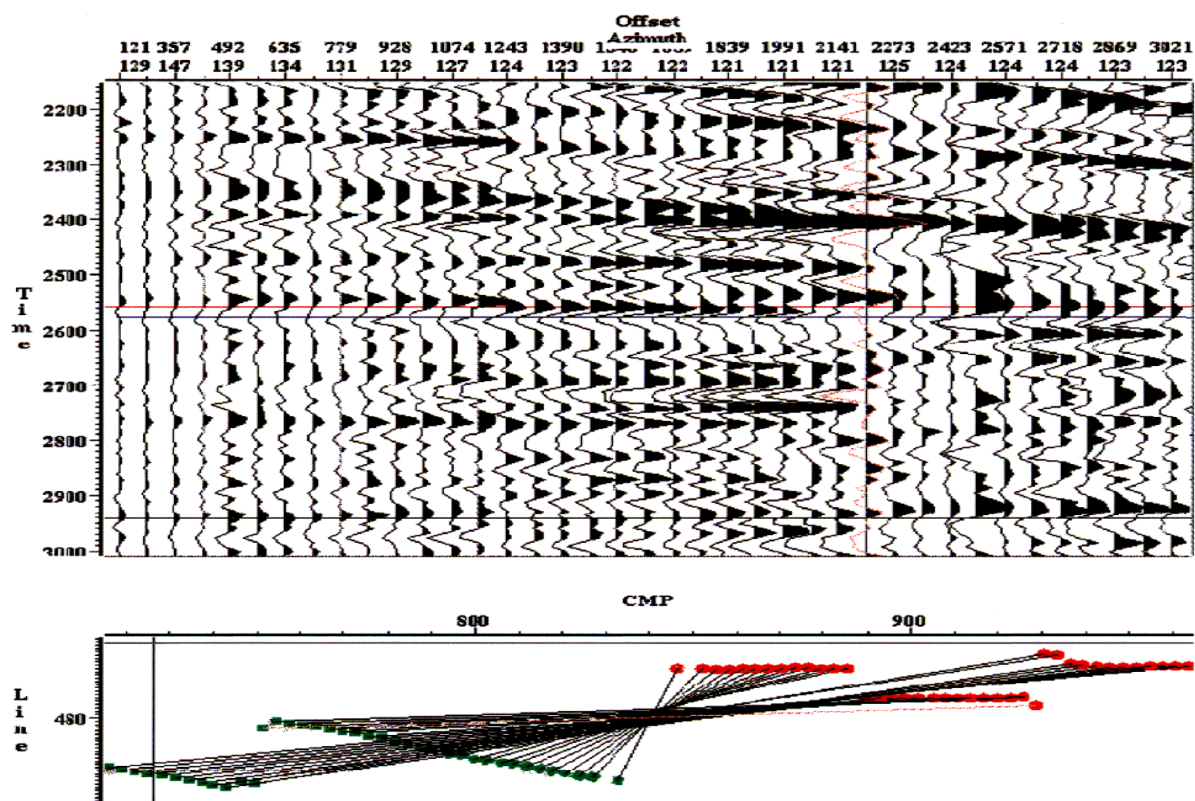


Figure 3.15: Bin gather (top) and corresponding source-receiver positions (bottom). The change in azimuth is approximately 8 degrees. Note the discontinuity in the reflector at 2.5 seconds corresponding to the change in azimuth.

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3.15 Processing of VSP data

A typical VSP processing sequence consists of the following elements:

- trace editing (remove bad and noisy traces)
- stacking of shots (if for instance each shot has been repeated 5 times this will increase signal to noise ratio.
- Band pass filtering
- Rotation of the geophone components so that x- y and z components are aligned with a “global” x-y-z coordinate system.
- Wavefield separation into up and downgoing waves (fk-filter or median filter)
- Deconvolution
- Imaging (migration). VSP-CDP transform, Kirchhoff migration or depth stepping

3.16 Processing of sea bed seismic data

Seabed seismic data can be divided into two different types: two-component data (hydrophone + vertical geophone) and four-component data (hydrophone + three geophone recordings). The first type has been used quite frequently in the Gulf of Mexico. In 1994 Berg, Svenning and Martin published the results of the first four-component survey acquired at the Tommeliten Field (North Sea). By exploiting that converted shear-waves are less attenuated by gas than P-waves, they demonstrated that it was possible to image the top reservoir event at Tommeliten. This event was very poorly imaged on conventional P-wave data. Figure 3.16 shows the typical wavemodes for a seabed seismic survey. We notice that the three wavemodes have different reflection/conversion points at the target reflector. In most four-component surveys acquired so far, it has been difficult to observe and utilize the SS-wave mode. The PS-converted wave mode generally has a higher signal to noise ratio, and on the x-component readings this wave type is easily observable. The subsurface location where the P to S conversion occurs, is often referred to as the conversion point. Therefore, for processing of the PS-data, one sorts the data into common conversion point gathers (CCP-gathers), instead of common midpoint gathers (CMP-gathers) as for P-wave data. The positions of the common conversion points are determined by the V_p/V_s -ratio. The position of a CMP-gather is determined by

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$$y = \frac{1}{2}(x_s + x_r) \quad (3.120)$$

where x_s and x_r are the source and receiver x-positions, respectively. For a Common Conversion Point gather (CCP-gather) the midpoint position is given by

$$x_{CCP} = x_s + \frac{x_r - x_s}{1 + \frac{v_S}{v_P}} \quad (3.121)$$

This means that in addition to conventional P-wave velocity analysis, one has to estimate the V_p/V_s -ratio in addition when processing PS-converted data.

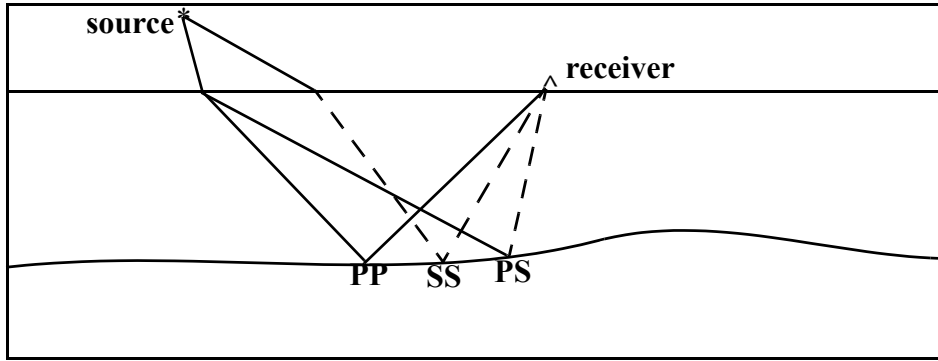


Figure 3.16: Wave modes in a four-component seabed seismic survey: PP-event, SS-event (converted from P to S at the seabed) and PS event (conversion at the interface).

Figure 3.17 shows a comparison between conventional streamer data, and a converted PS stack from the Tommeliten survey.

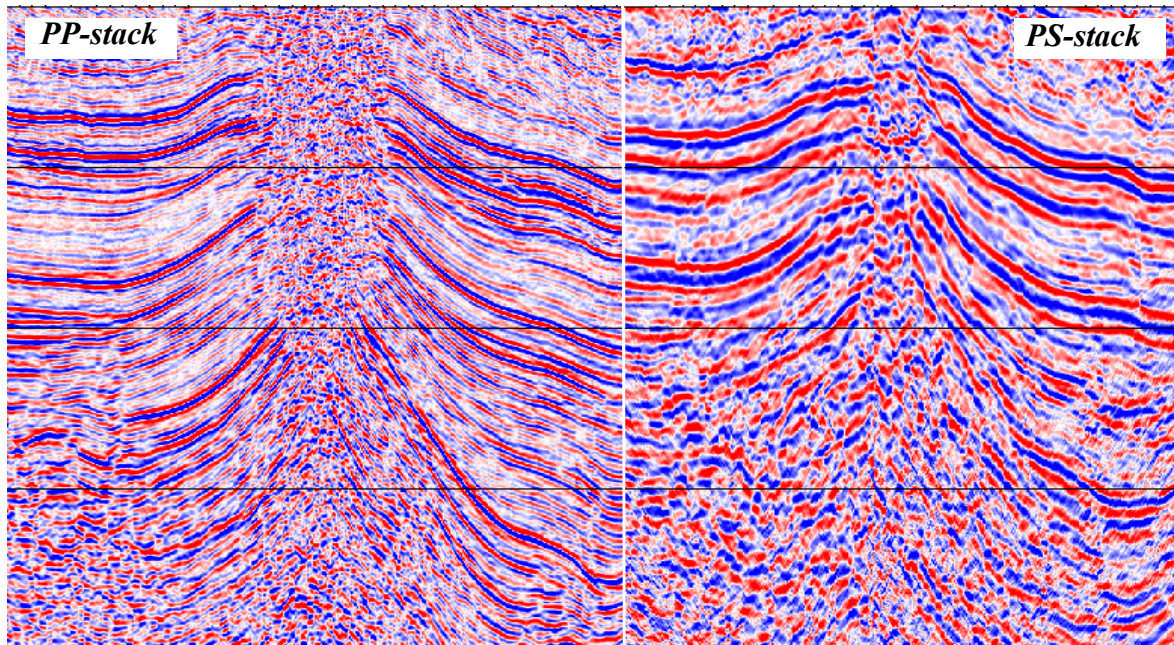


Figure 3.17: Comparison of conventional 3D seismic data (streamer survey) (left), and a PS stack generated from the x-component of a seabed seismic data (right). Notice better definition of the structure at the PS-image

4 Analysis of seismic data

4.1 Calibration with wells

A very important first step in the analysis of seismic data is to calibrate the seismic response observed on the seismic data with the response we expect based on observations in wells. Figure 4.1 shows an example of a well tie for two different incidence angles. The correspondance between synthetic traces based on the acoustic well logs and the real seismic data (stacked data) is quite good, and this example is representative for North Sea reservoirs. The synthetic trace are computed by a simple convolution between a wavelet and a reflectivity series. The reflectivity series are computed by using Zoepprits equations on the acoustic well logs. It is very important for all remaining analysis work that this calibration step is done properly, and that one focuses on obtaining a good understanding of which effects in the wells that really are observable on the seismic data.

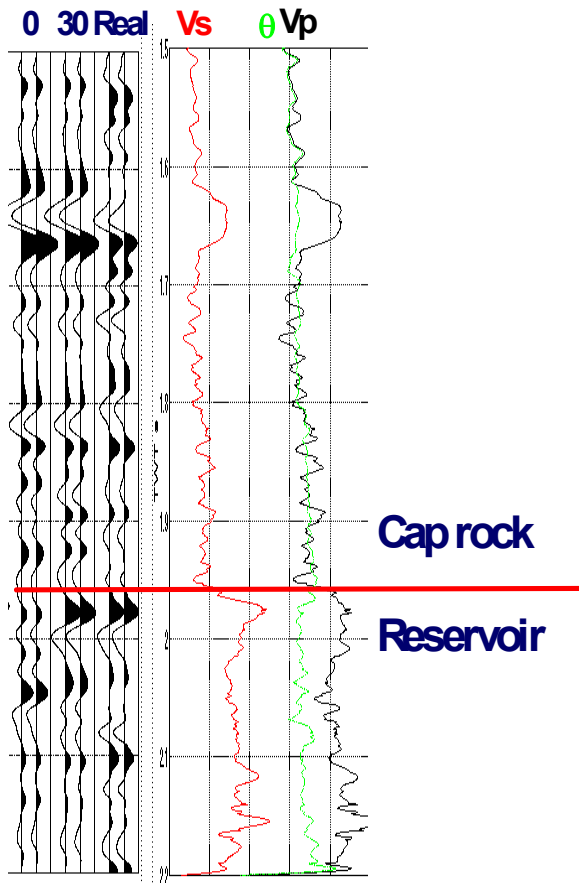


Figure 4.1: Example of well-tie. An input wavelet is convolved with a reflectivity series calculated for zero angle (left two traces) and for 30 degrees angle (second pair of traces). The reflectivity series are calculated using Zoeppritz equations with the three acoustic well logs (shown to the right) as input. Notice that the 30 degrees trace (corresponding to far offset trace) matches the stacked data (denoted Real) best at the top reservoir interface.

4.2 AVO analysis

AVO (Amplitude Versus Offset) analysis is used both in exploration and reservoir seismic. When a seismic wave is reflected at a given interface, the reflectivity will vary as a function of the incidence angle. If we consider a P-wave being reflected at an interface, as shown in Figure 4.2, we see that both a P and an S wave are reflected at the interface, as well as transmitted. The reflection and transmission coefficients are given by the Zoeppritz equations. Since the Zoeppritz equations are fairly complicated, several approximations have been derived.

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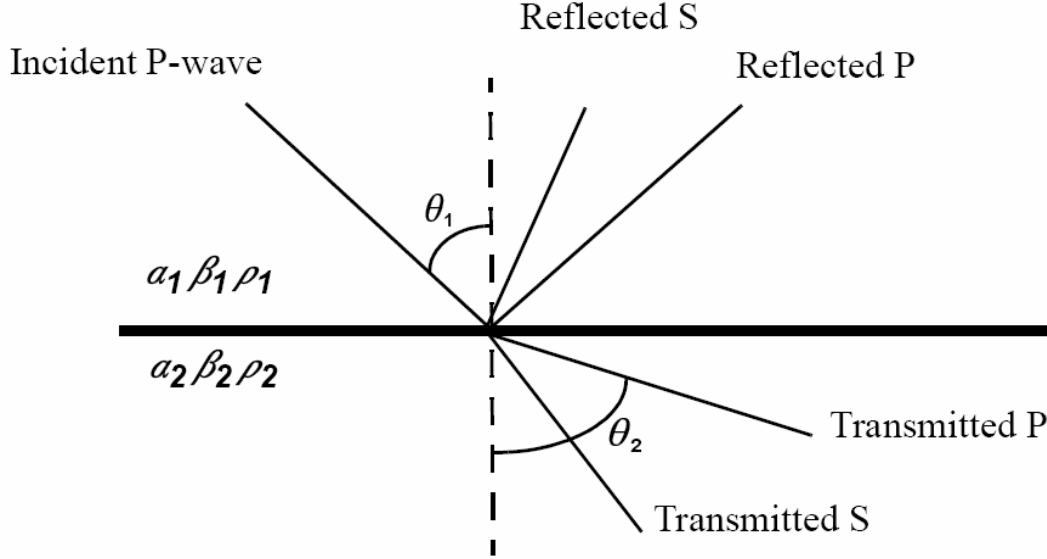


Figure 4.2: When a P-wave is reflected at an interface, both an S-wave and a P-wave is being reflected and transmitted.

One commonly used approximation for the PP reflection coefficient is the Smith and Gidlow approximation:

$$R(\theta) = \left(\frac{1}{2}\right)\left(\frac{\Delta\rho}{\rho} + \frac{\Delta\alpha}{\alpha}\right) - \frac{2\beta^2}{\alpha^2}\left(\frac{\Delta\rho}{\rho} + \frac{2\Delta\beta}{\beta}\right)\sin^2\theta + \frac{\Delta\alpha}{2\alpha}\tan^2\theta \quad (4.1)$$

where α and β denote P and S wave velocity, ρ density, θ incidence angle and $\Delta\alpha$ etc. denote changes in the seismic parameters across the interface. In AVO-analysis one often chooses to express the angle dependent reflectivity as a simple to term expression:

$$R(\theta) = R_0 + G\sin^2\theta \quad (4.2)$$

where R_0 is the intercept and G is the gradient term. A typical output from processing is to estimate 3D seismic cubes of the intercept and gradient as given above, either based on prestack seismic data, or simply based upon near and far offset stacked data.

It is not correct however, to assume that a prestack common midpoint gather, represents the reflectivity function as given by for instance equation 4.1. Source and receiver directivity effects will of course alter the seismic response with offset. Absorption effects will also lead to additive angle effects, since waves with longer travelpaths (high offset) will be more attentu-

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ated than waves with shorter travelpaths. Various focusing and defocusing effects caused by for instance curved interfaces will also cause additional directivity effects, and corresponding variation in amplitude with offsets. Finally, varying hydrophone sensitivities along the receiver cable will also lead to variations in the amplitude versus offset behaviour. Essentially there are two ways to attack this problem: One is to try to correct systematically for the above mentioned effects, that is in a deterministic way. Another commonly used approach is to use well logs to calibrate the AVO response at a given interface.

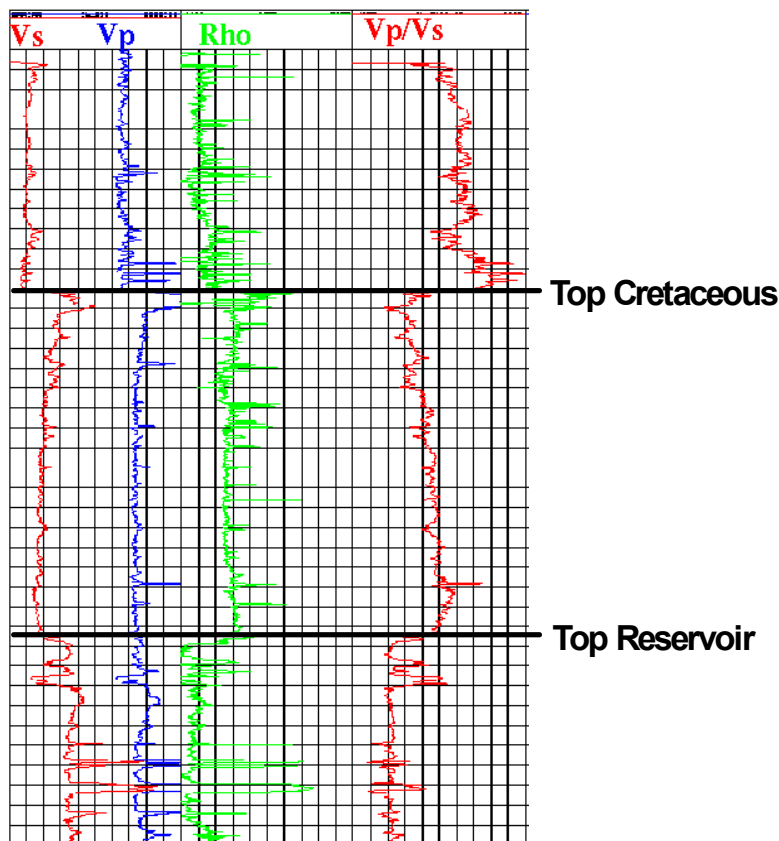


Figure 4.3: Acoustic well logs for well C-33 at Gullfaks: S-wave velocity, P-wave velocity, density and the V_p/V_s -ratio. Top Cretaceous and Top Brent (Reservoir) interfaces are marked.

From Figure 4.3 we notice that the shear wave velocity increases both at top Cretaceous and at

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the top Brent interface. According to the Smith and Gidlow equation (4.1) we estimate the following approximate expression (by picking approximate velocity and density contrasts directly from the log) for the P-wave reflection coefficient at the top Cretaceous interface:

$$R(\theta) = 0.2 - 0.2\sin^2\theta \quad (4.3)$$

A plot of this reflectivity function versus angle is shown in Figure 4.4.

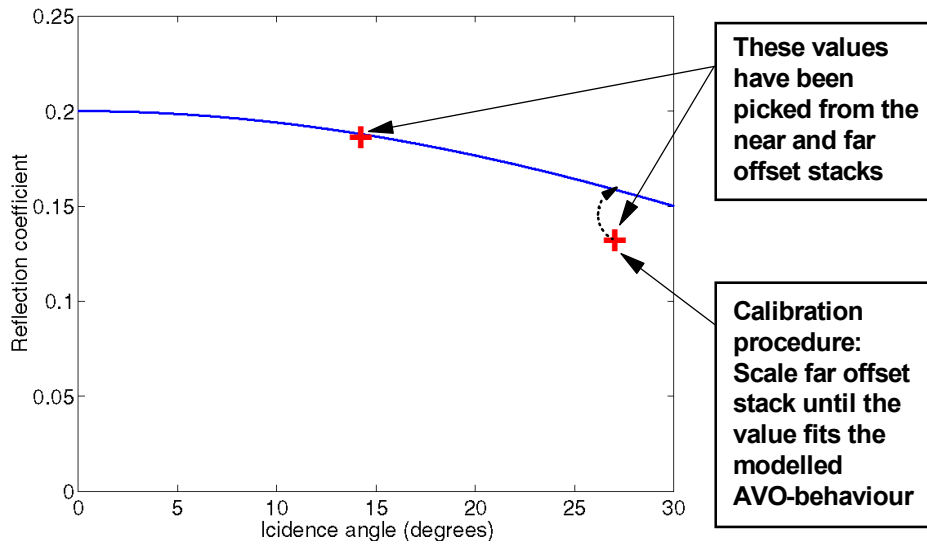


Figure 4.4: Modelled P-wave reflection coefficients for the top Cretaceous interface, based on the C-33 well logs. The crosses denotes values taken from the near and far offset seismic data. A calibration procedure will mean to scale the far offset amplitude so that the far offset cross matches the modelled reflectivity curve.

A plot of the near and far offset stacks in the neighbourhood of well C-33 is shown in Figure 4.5. We notice that the top Cretaceous amplitude increases somewhat when we go from near to far offset, while the top reservoir reflector (top Brent) shows an amplitude increase from near to far offset data. We also notice that the fit between modelled response based upon the acoustic well logs, and the seismic data improves when the calibration step described for the top Cretaceous interface is applied to the far offset stack.

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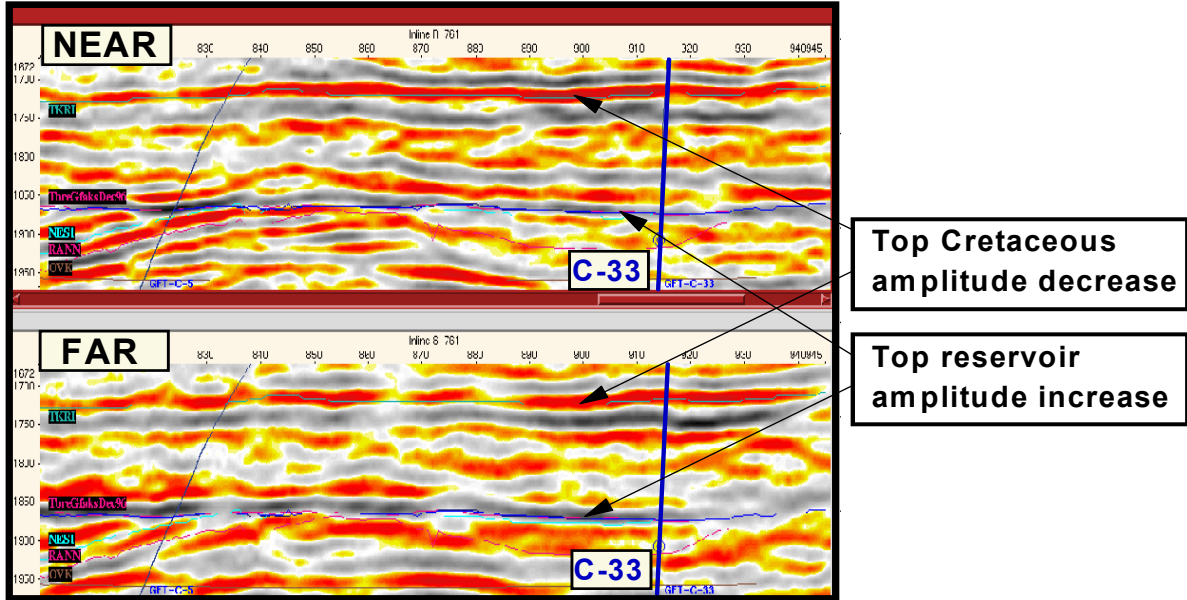


Figure 4.5: Near and far offset stacks for a seismic inline intersecting wellpath C-33 at Gullfaks. Notice that the top reservoir amplitude increases from near to far, while the top Cretaceous amplitude decreases.

Following the same procedure for the top Brent reflector, we find

$$R(\theta) = -0.05 - 0.37 \sin^2 \theta \quad (4.4)$$

A plot of this reflectivity curve is shown in Figure 4.6, together with amplitude values taken from the seismic near and far offset cubes. The AVO-effect observed in this figure is mainly caused by the strong increase in S-wave velocity at top reservoir. The reservoir rock is oil-filled in this area. If we model the AVO-response assuming that the reservoir rock is waterfilled instead, we observe that the AVO-curve is shifted towards weaker reflectivity. This means that the amplitude versus offset behaviour is lithology-dependent, but exchanging pore fluid from water to oil means an overall amplification of amplitudes at all offsets. At Gullfaks this means that high amplitudes on full stack data is often associated with presence of oil. Figure 4.7 shows an amplitude map of the top Brent interface at Gullfaks, clearly demonstrating the correspondence between oilfilled reservoir rocks and high seismic amplitudes.

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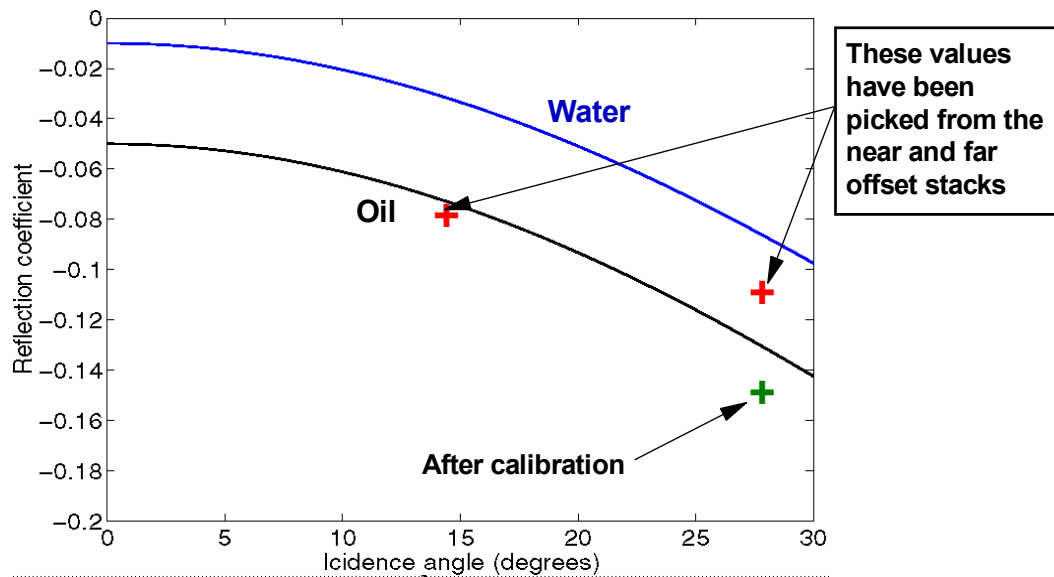


Figure 4.6: Modelled reflectivity as a function of incidence angle for oilfilled and waterfilled reservoir rocks. Notice that the fit between modelled response based on well logs and measured seismic data is improved after the calibration step described in Figure 4.4.

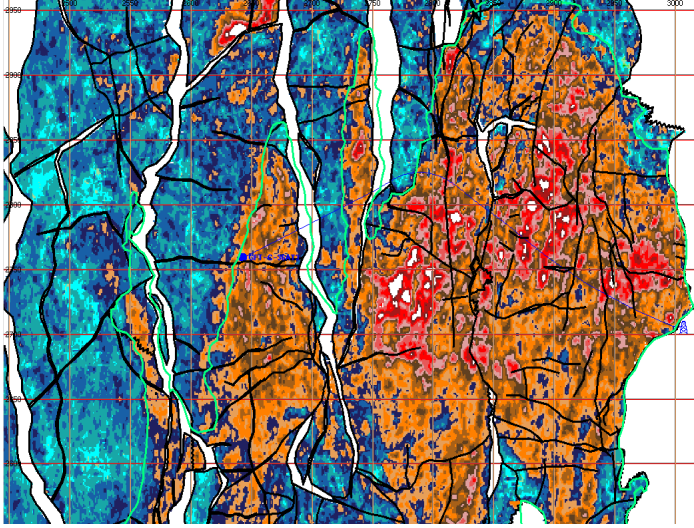


Figure 4.7: Amplitude map of top Brent interface at Gullfaks. Notice that strong amplitude coincides with presence of oil. Amplitudes decrease down-flanks.

4.3 Post stack inversion

Seismic data consists of reflections from the geological subsurface. The main objective for seismic inversion methods is to estimate physical parameters of the subsurface geological layers. These parameters are P- and S-wave velocities and the density. From post stack seismic data it is commonly used to estimate the P-wave acoustic impedance, assuming that the stacked seismic trace can be regarded as a zero-offset trace. The relation between the zero-offset reflection coefficient and the acoustic impedance at each side of an interface is given as:

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1} \quad (4.5)$$

where Z_1 and Z_2 are acoustic impedances in layer 1 and 2, respectively. The simplest way to estimate acoustic impedance from a stacked seismic trace is to apply an recursive inversion, often referred to as trace integration. From equation 4.5 we see that if the acoustic impedance in layer 1 is known the acoustic impedance in layer 2 (or sample number 2) is given by

$$Z_2 = Z_1 \frac{1 + R}{1 - R} \quad (4.6)$$

By using this equation recursively, we can obtain acoustic impedances as a function of two-way traveltime. Most commercial inversion algorithms uses more advanced ways to estimate the acoustic impedances (non-linear inversion, simulated annealing techniques etc.), often combined with use of additional information from for instance well logs.

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In this type of inversion it is assumed that the seismic trace does not contain multiple energy and that absorption effects have been removed from the trace.

4.4 Elastic impedance

One limitation with conventional post-stack inversion is that it is valid only for a zero-offset experiment. If we want to apply post-stack inversion to for instance a far offset stack, this assumption is not valid any more. It would therefore be nice if we could introduce a quantity playing the same role as acoustic impedance, that could be used for non-zero reflectivity angles. Connolly introduced a quantity called elastic impedance in 1998. The basic idea is that the reflection coefficient at a non-zero incident angle should be written in exactly the same way as for zero offset (equation 4.5):

$$R(\theta) = \frac{EI_2 - EI_1}{EI_2 + EI_1} \quad (4.7)$$

Since equation 4.7 is exactly equal to equation 4.5, it means that if we use conventional post-stack inversion on a far offset stack, the quantity we get out is elastic impedance. In conventional poststack inversion, a well tie is an important issue, since it links our well observations to the seismic data. It is therefore important to establish an explicit expression for the elastic impedance as a function of acoustic parameters that are measured in a well log. Based on the Smith and Gidlow equation (4.1) Connolly derived the following expression for elastic impedance:

$$EI = \alpha^{(1 + \tan^2 \theta)} \beta^{-8K \sin^2 \theta} \rho^{(1 - 4K \sin^2 \theta)} \quad (4.8)$$

where $K = \beta^2 / \alpha^2$. The basic assumption in Connolly's derivation of the elastic impedance is that the elastic impedance should play the same role for angle dependent reflectivity as the acoustic impedance for zero offset reflectivity. If that condition is fulfilled one can apply post stack inversion schemes (that are derived assuming that stacked seismic data is a convolution between a known wavelet and a zero offset reflection coefficient series) on far offset stacks. We notice that for an incidence angle equal to zero, the elastic impedance is equal to the acoustic impedance. Elastic impedance logs are now being used as a quick way of assessing AVO characteristics to be expected from a given well, and as a tool for calibration and inversion of offset seismic data. In the next session we will show how this concept can be extended to calibrate and invert sea bed seismic data.

4.5 Shear wave elastic impedance

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As an extension of elastic impedance, Landrø, Duffaut and Rognø derived a quantity denoted shear wave elastic impedance in 1999, that should play the same role for PS-converted data as elastic impedance does for conventional PP-data. An approximate equation (assuming weak contrasts and small angles) for P to S conversion as a function of incidence angle (θ) at a plane interface is given by (Aki and Richards, 1980):

$$R_{PS}(\theta) = -\frac{1}{2} \left[\left(1 + \frac{2\beta}{\alpha} \right) \frac{\Delta\rho}{\rho} + 4 \frac{\beta}{\alpha} \frac{\Delta\beta}{\beta} \right] \sin\theta \quad (4.9)$$

where α , β and ρ denote P wave, S wave velocity and density respectively, and $\Delta\beta$ etc refers to contrasts in the same parameters at the actual interface. Assuming that the shear wave elastic impedance (SEI) can be written as

$$SEI = \beta^B \rho^C \quad (4.10)$$

and requiring that the shear wave elastic impedance should relate to reflectivity in the same way as acoustic impedance for a zero offset reflection coefficient, we find that SEI can be written as:

$$SEI = \beta^{-4 \frac{\beta}{\alpha} \sin\theta} \rho^{-\left(1 + 2 \frac{\beta}{\alpha}\right) \sin\theta} \quad (4.11)$$

We observe that for $\theta=0$, the shear wave elastic impedance is equal to one (no contrasts), and this situation corresponds to no P to S conversion, as expected.

4.6 Examples of elastic and shear wave elastic impedances

Acoustic well logs (P-wave velocity, S-wave velocity and density) from a North Sea well are shown in Figure 4.8. We notice a strong increase in P- and S-wave velocities at the top Cretaceous interface. At the top reservoir interface (top of the Brent Group Formation) there is hardly any observable change in the P-wave velocity, while a significant increase in S-wave velocity is observed, and a small drop in the density. Figure 4.9 shows a comparison between the conventional acoustic impedance, the elastic impedance for an incidence angle of 30 degrees, and the corresponding shear wave elastic impedance for the same angle of incidence.

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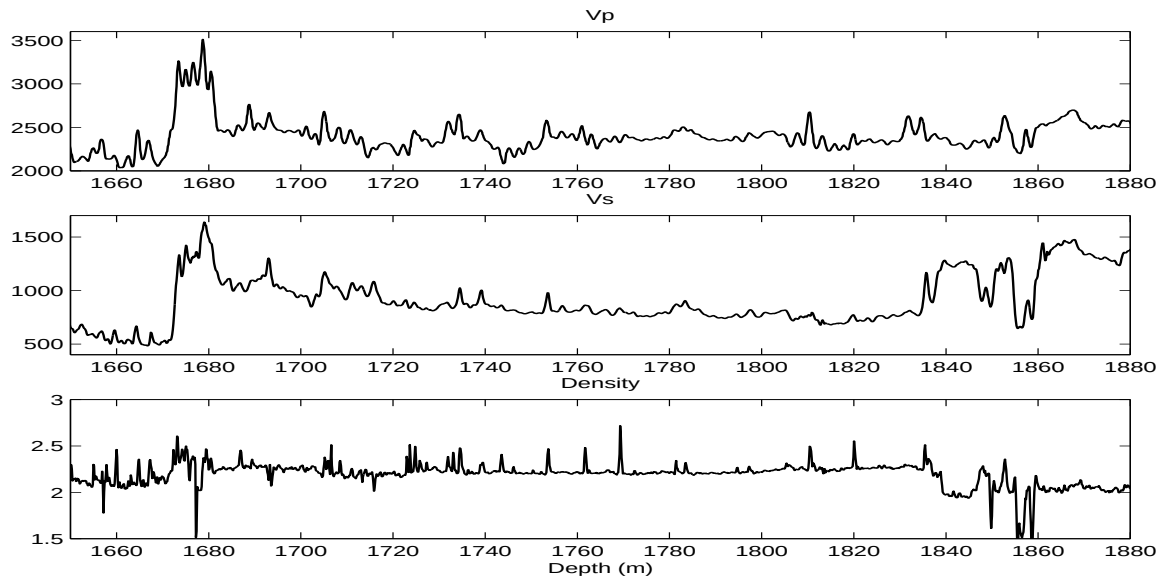


Figure 4.8: Acoustic well logs for a North Sea well: P-wave velocity (top), S-wave velocity (middle) and density (bottom).

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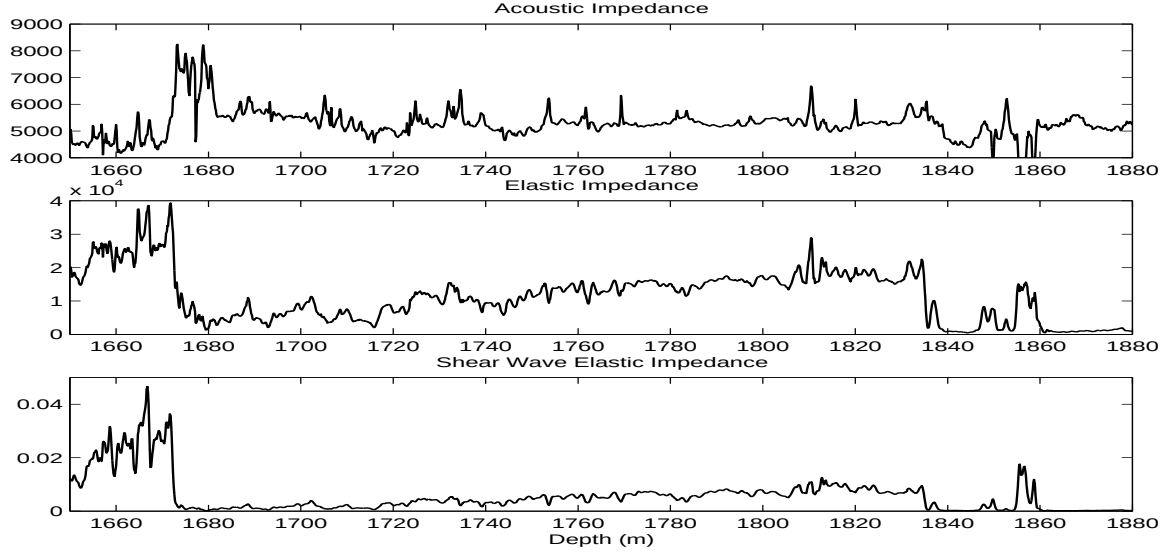


Figure 4.9: Comparison between acoustic impedance (top), elastic impedance (middle) and shear wave elastic impedance (bottom). Significant changes are observed for all three impedances for the top Cretaceous interface at approximately 1675 m. For the top Brent interface at approximately 1835 m, the elastic impedance and the shear wave elastic impedance show clearer changes than the acoustic impedance.

Assuming that we have inverted two cubes of seabed seismic data, one for acoustic impedance and one for shear wave elastic impedance, these cubes can be combined into one attribute cube representing for instance the Vp/Vs-ratio. In principle, we are then dealing with 3 unknowns and only two equations (one for AI and one for SEI). Assuming that for instance the density well log has smaller variations than the other logs (this can be observed in Figure 4.8, where we see that the relative changes in density are of the order of 10%, while the Vp and Vs logs have changes of the order of 50%). Using this assumption and choosing an incidence angle of 30 degrees leads to the following expressions for the two impedances:

$$AI = \alpha \rho_0 \quad (4.12)$$

$$SEI = \beta \begin{matrix} -2\frac{\beta}{\alpha} & -\left(\frac{1}{2} + \frac{\beta}{\alpha}\right) \\ \rho_0 & \end{matrix} \quad (4.13)$$

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Taking the logarithm on both sides of the last equation and inserting the first equation in the last yields

$$\log SEI = -2\frac{\beta}{\alpha}\log\frac{\beta}{\alpha} - 2\frac{\beta}{\alpha}\log\frac{AI}{\rho_0} - \left(\frac{1}{2} + \frac{\beta}{\alpha}\right)\log\rho_0 \quad (4.14)$$

Solving this equation with respect to the Vp/Vs-ratio (α/β) gives

$$\frac{\alpha}{\beta} = \frac{\log\rho_0 - 2\log AI + 2\log\frac{\alpha}{\beta}}{\log SEI + \frac{1}{2}\log\rho_0} \quad (4.15)$$

The dominant terms in the above equation is the acoustic impedance term and the shear wave elastic impedance term. Assuming that the Vp/Vs-ratio is approximately 2, and that the density is also approximately equal to 2, gives the following expression for the Vp/Vs-ratio

$$\frac{\alpha}{\beta} = \frac{3\log 2 - 2\log AI}{\log SEI + \frac{1}{2}\log 2} \quad (4.16)$$

Neglecting all terms except the impedance terms gives an even simpler equation

$$\frac{\alpha}{\beta} = -2\frac{\log AI}{\log SEI} \quad (4.17)$$

A recursive solution using the above given equation yields:

$$\frac{\alpha}{\beta} = \frac{\log 2 - 2\log AI + 2\log\left(-\frac{2\log AI}{\log SEI}\right)}{\log SEI + \frac{1}{2}\log 2} \quad (4.18)$$

As a test of the given equations, a comparison between the exact Vp/Vs-ratio and the three different approximations derived in equations is shown in Figure 4.10. We see that equations 4.16 and 4.17 give Vp/Vs-ratios that are somewhat shifted with respect to the exact solution, while equation 4.18 is quite accurate. This shows the potential for combined use of acoustic inverted data with shear wave impedance inverted data.

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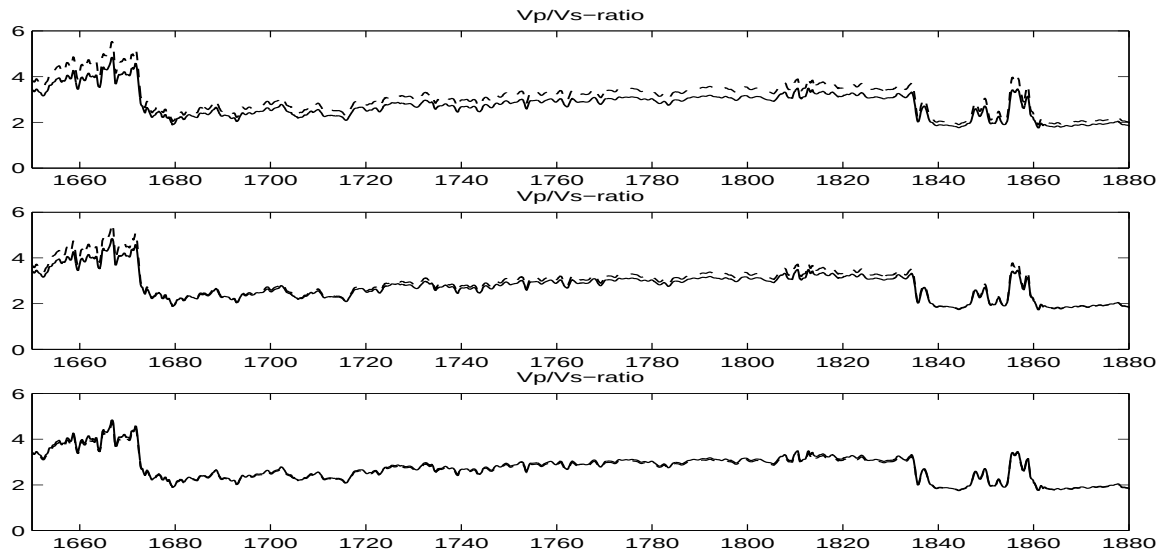


Figure 4.10: Comparison between the exact V_p/V_s -ratio (solid line) and the estimate based on acoustic and shear wave elastic impedance approximations (dashed line) for equation 4.17 (top), equation 4.16 (middle) and equation 4.18 (bottom).

For the top Cretaceous interface significant changes in acoustic impedance, elastic impedance and shear wave elastic impedance are observed. This means that this interface should be well defined on conventional stacked seismic data, far offset stack and on PS-converted seabed seismic data. Furthermore, we observe that the change for the top reservoir interface at approximately 1835 m depth, are most pronounced for the elastic impedance. The drop in shear wave elastic impedance is weaker for the top reservoir event, and therefore we might expect less conversion from P to S for this interface than for the top Cretaceous interface. Based on these well logs one should expect that far offset stacked data should image the top reservoir event clearly. Furthermore, the increase in elastic impedance and shear wave elastic impedance at approximately 1856 m, probably caused by the presence of coal in this layer, is similar in amplitude changes, indicating that this event should appear both on far offset stacks as well as PS-converted images. It is interesting to note that on a PS-converted stack, the top Cretaceous interface is more pronounced than the top reservoir interface. The main conclusion is that it is useful and meaningful to derive a shear equivalent to the elastic impedance introduced by Connolly (1998). Computing this shear wave elastic impedance in addition to more conventional impedances, enables us to predict at which interfaces we can expect strong P to S conversions, and to analyse the relative strength between P to S conversion and AVO-effects. Shear wave elastic impedance can serve

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as a calibration step for inversion of sea bed seismic data. It can also be used directly as input to a post stack inversion scheme of such data. Secondly, this impedance can be used to extract a wavelet that is representative for converted seabed data. The output from for instance a post stack inversion of a far offset angle stack of P to S converted sea bed data, could then be directly compared with the shear wave elastic impedance curve obtained from the acoustic well logs, and in that way it can be used as a guide in the interpretation of the inverted data. Finally, it has been demonstrated that a combined estimation of conventional acoustic impedance (from for instance post stack inversion of surface seismic data or the z-component of a seabed data set) and a shear wave impedance cube from PS-converted seabed data, gives us the opportunity to estimate an approximation for the V_p/V_s -ratio, based on the two inverted impedance cubes. This V_p/V_s estimate is based upon amplitude information, and may therefore serve as a complementary and more detailed information source to more conventional V_p/V_s estimates based upon for instance travel time correlations between PP and PS sections.

References

- Aki, K. and Richards, P., 1980, Quantitative seismology: W.H. Freeman & Co.
- Berg, E., Svenning, B., and Martin, J., 1994, Sumic - A new strategic tool for exploration and reservoir mapping: EAGE Expanded Abstracts 1994.
- Connolly, P., 1998, Calibration and Inversion of Non-Zero Offset Seismic: 68th Ann. Internat. Mtg., Soc. Expl. Geophys., Expanded Abstracts, 182-184.
- Thomsen, L., Barkved, O.I., Haggard, B., Kommedal, J.H., and Rossland, B., 1997, Converted-wave imaging of Valhall Reservoir: EAGE Expanded Abstracts Vol. 1, 1997.

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6 References

Buland, A., Landrø, M., Andersen, M. and Dahl, T., 1996. AVO inversion of Troll Field data. *Geophysics* **61**, 1589-1602.

Herber, R., Weigel, W., and Wong, H.K., 1997, Relationship between shot-induced noise and shot interval in seismic refraction experiments at sea: *Marine Geophysical Researches*, **19**, 257-265.

Larner, K., Hale, D., Zinkham, S.M. and Hewlitt, C., 1982, Desired seismic characteristics of an air gun source: *Geophysics*, **47**, 1273-1284.

Sheriff, R.E. and Geldart, L.P., 1995. *Exploration Seismology*. Cambridge University Press, Second Edition.

Beasley, C.J., Chambers, R.E., Workman, R.L., Craft, K. and Meister, L., 1997: A 3D ocean bottom cable repeatability experiment. *EAGE Expanded Abstracts Vol. 1*, Geneva 1997, B046.

Blanche, J.P., Capelle, P., Marraud, J., and Ducasse, M.L. 1994. Heimdal monitoring III: Seismic monitoring, methods, and results. *64th SEG Meeting Abstracts* pp. 335-337.

Sallas, J.J., 1984. Seismic vibrator control and the downgoing P-wave: *Geophysics* **49**, 732-740.

Ursin, B., 1990. Offset-dependent geometrical spreading in a layered medium: *Geophysics* **55**, 492-496.